SOME RESULTS ON INTERSECTIONS AND ENTROPY

by

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APPROVAL

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Abstract

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A central line of research in dynamical systems concerns the identification of geometric mechanisms necessary and/or sufficient for the production of entropy. In this thesis we prove several technical results directed (in some cases rather obliquely) towards this general project, motivated in particular by problems arising in attempts to prove Shub's entropy conjecture in finite regularity. We first prove a generalization of Poincaré's formula in integral geometry to arbitrary closed manifolds, where the transformation group is replaced by a compact family of C^1 diffeomorphisms satisfying two (essentially optimal) transversality conditions. We give two dynamical applications of this result, first a stochastic Artin-Mazur bound for arbitrary C^1 diffeomorphisms in the presence of sufficiently rich random noise, and second to the construction of uniformly expanding measures on $\mathrm{Diff}^r_{\mathrm{vol}}(M)$. Moving to the C^2 setting, we investigate domination assumptions necessary to guarantee the flatness of Bowen disks on subexponentially shrinking scales. Finally, we answer a question of Correa & Pujals by constructing a family of subshifts (and smooth realizations) with superpolynomial and subexponential generalized entropy. Returning to the entropy conjecture, we conclude by sketching some directions for future work.

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Preface

Chapter 2 is based on the preprint *An average intersection estimate for families of diffeomorphisms* [KS24], written in collaboration with Michael Shub.

Contents

1 Introduction **1**

1.1 Topological entropy and homology 1

1.2 The C^∞ solution 3

1.3 Stable volume growth 4

1.4 The intersections approach 7

1.5 Generalized entropy 10

1.6 Directions for future work 11

2 Average intersections for families of diffeomorphisms **13**

2.1 Introduction 13

2.1.1 Background and motivation 15

2.1.2 Assumptions 17

2.1.3 A digression on $\mathcal{H}$ 20

2.2 Preliminaries 22

2.2.1 Normal Jacobians 22

2.2.2 The solution manifold $\widetilde{\mathcal{V}}$ 23

2.2.3 The intersection manifolds $\mathcal{V}$ 24

2.2.4 The bundles $\mathcal{G}_k \to \widetilde{\mathcal{V}}$ 26

2.3 Proof of Main Theorem 33

2.3.1 The co-area formula . . . 34
2.3.2 Fiber integral bounds . . . 36
2.4 Generalization . . . 38
2.5 Construction of $(\mathcal{H}, \psi)$ . . . 44
2.5.1 Step 1: Local construction. . . . 47
2.5.2 Step 2: Patching together. . . . 51
2.6 Application 1: Stochastic Artin-Mazur . . . 53
2.7 Application 2: Uniformly expanding measures . . . 56

3 Domination and Bowen disks 64
3.1 Background . . . 64
3.2 Local version . . . 66
3.3 Main result . . . 68
3.4 Construction of example . . . 69
3.5 Pliss domination . . . 72
3.6 Characterizing uniform domination . . . 73

4 Density subshifts and generalized entropy 76
4.1 Density subshifts . . . 77
4.2 Smooth realization . . . 85

5 How (not) to prove the Entropy Conjecture 88
5.1 Stable volume growth . . . 89
5.2 Finding separated intersection sets . . . 91
5.2.1 A modified H_1 proof . . . 92
5.3 Nerves of dynamical covers . . . 94

A Submersions from manifolds with boundary 98

B Normal Jacobians formula **103**

Bibliography **108**

Chapter 1

Introduction

In 1972, Michael Shub posed the following problem, now known as the Entropy Conjecture:

Conjecture 1.0.1 ([Shu74])**.** *Let M be a closed manifold. For all $f \in \mathrm{Diff}^1(M)$ one has*

$$\log \mathrm{sp}(f_*) \leq h(f). \tag{EC}$$

Here $h(f)$ denotes the topological entropy of f, $f_* : H_*(M, \mathbb{R}) \to H_*(M, \mathbb{R})$ denotes the induced map on (total) real homology, and $\mathrm{sp}(f_*)$ its spectral radius.

In this thesis we will not prove this famous conjecture, and its relation to our actual, somewhat disparate results might appear oblique at best. Nevertheless, at least on the level of intellectual genealogy these results are unified primarily by their shared origin in years of thinking about the Entropy Conjecture. In this introduction we aim to sketch some of these connections.

1.1 Topological entropy and homology

The *topological entropy* of a continuous map $f : X \to X$ from a compact metric space to itself, denoted $h(f)$, measures the exponential growth rate of the number of orbit segments of

length n distinguishable up to arbitrary finite precision. In other words it provides a coarse measure of how the number of possible orbit behaviors in a dynamical system grows with time.

More precisely, we define a sequence of metrics d_n^f on X by

$$d_n^f(x,y) = \max_{0\le j\le n-1} d(f^j x, f^j y),$$

We write $B_f(x,n,\varepsilon)$ for the open ε-ball centered at x in the d_n^f metric, i.e. the (n,ε)-*Bowen ball* centered at x, and denote by $r_f(n,\varepsilon)$ the ε-covering number of the metric d_n^f, i.e.

$$r_f(n,\varepsilon) = \min\left\{ \#Y \ : \ Y\subset X \text{ s.t. } X = \bigcup_{y\in Y} B_f(y,n,\varepsilon)\right\}.$$

We also write $s_f(n,\varepsilon)$ for the maximal cardinality of an (n,ε)-separated subset, i.e.

$$s_f(n,\varepsilon) = \max\left\{ \#Y \ : \ Y\subset X \text{ s.t. } \forall x,y\in Y, x\neq y \implies d_n^f(x,y)>\varepsilon\right\}.$$

Then

$$\begin{aligned} h(f) &:= \lim_{\varepsilon\to 0}\limsup_{n\to\infty}\frac{1}{n}\log r_f(n,\varepsilon)\\ &= \lim_{\varepsilon\to 0}\limsup_{n\to\infty}\frac{1}{n}\log s_f(n,\varepsilon)\\ &= \lim_{\varepsilon\to 0}\liminf_{n\to\infty}\frac{1}{n}\log r_f(n,\varepsilon)\\ &= \lim_{\varepsilon\to 0}\liminf_{n\to\infty}\frac{1}{n}\log s_f(n,\varepsilon). \end{aligned}$$

The equivalence between these formulas is easy to check and it is often useful to have all options at hand. It is similarly easy to check that topological entropy is invariant under topological conjugacy, in particularly does not depend on the choice of metric generating the topology on X.

The spectral radius of f_* can also be seen as a dynamical growth rate, in terms of the action of f on the global topology of M. In particular, if $\mathrm{sp}(f_*) = \mathrm{sp}(f_{*,k}) = \lambda > 1$ then for

all $\alpha \in H_k(M, \mathbb{R})$ outside a positive codimension subspace we have

$$\|f_*^n \alpha\| \sim \lambda^n.$$

By integrating against a fixed basis for $H^k_{\mathrm{dR}}(M, \mathbb{R}) = [H_k(M, \mathbb{R})]^*$ it is easy to see that this implies the existence of some C^∞ k-simplex $\sigma : \Delta^k \to M$ such that

$$\int_{\Delta^k} \left\|\Lambda^k D_x(f^n \circ \sigma)\right\| \, dx =: \mathrm{vol}(f^n \circ \sigma) \sim \lambda^n.$$

In other words, *k-homology growth implies k-volume growth.*

1.2 The C^∞ solution

Many interesting partial results related to the EC have been proved over the years. Perhaps the most general is Yosef Yomdin's proof in 1987 that the conjecture holds in the smooth (i.e. infinitely differentiable) category:

Theorem 1 ([Yom87])**.** (EC) *holds for C^∞ maps.*

In Yomdin's proof this follows from a much stronger statement, namely that *volume growth lower bounds topological entropy* for C^∞ maps. This in turn follows from the fact that a C^∞ map f cannot have *local volume growth*, that is, exponential volume growth of a k-disk $D \subset M$ cannot be concentrated arbitrarily close to long orbit segments.[1]

A canonical class of examples due to Margulis shows that this stronger statement no longer holds in finite regularity, that is, for all $1 \leq r < \infty$ there exist C^r diffeomorphisms f with positive local volume growth. In particular, volume growth generally does *not* bound topological entropy from below in finite regularity.

[1] Finally this statement follows from a reparametrization estimate whose formulation and proof amounts to the primary technical substance of Yomdin's paper. Fortunately this most-difficult step is less relevant to the discussion here.

This fact immediately forecloses the possibility of proving (EC) in finite regularity from volume growth alone. On the other hand, it implies essentially nothing about the substance of the conjecture. The immediate reason is that homology growth produces not just volume growth but *stable volume growth*, of a relatively uniform kind, and moreover the stable growth of a number of other geometric parameters (e.g. intersections).

From this perspective, the proof of the C^∞ case remains conceptually unsatisfying, since the availability of the much stronger volume growth statement in the smooth setting obviates the need to identify any genuinely *homological* mechanism for the production of entropy.

1.3 Stable volume growth

One possible approach to finding such a mechanism focuses attention on the fact that volume growth from homology growth must persist over entire homology classes. In fact, for concreteness one can begin by considering volume growth stable under *homotopy*.

Namely, consider S a closed, immersed k-submanifold of M. (Formally this means that S is a closed k-manifold with $\varphi : S \to M$ some C^1 immersion, but we will frequently abuse notation and write S for its image in M as well.) Then we define the *homotopy volume growth of S under f* by

$$\mathrm{HVG}(S, f) = \liminf_{n\to\infty} \frac{1}{n} \log^+ \left(\inf_{\widehat{\varphi}\sim\varphi} \mathrm{vol}(f^n \circ \widehat{\varphi}) \right),$$

where the relation $\widehat{\varphi} \sim \varphi$ means that $\widehat{\varphi} \in C^1(S, M)$ with $\widehat{\varphi}$ homotopic to φ. Then the following gives a natural homotopy analogue to the Entropy Conjecture:

Conjecture 1.3.1. *If $f \in \mathrm{Diff}^r(M)$, then $\mathrm{HVG}(S, f) \leq h(f)$ for all immersed submanifolds S of M.*

Observation 1.3.2. *Conjecture 1.3.1 implies the C^r Entropy Conjecture.*

Proof. Consider $f \in \mathrm{Diff}^1(M)$. By Poincaré duality we have $\mathrm{sp}(f_{*,k}) = \mathrm{sp}(f^{-1}_{*,d-k})$ for all $1 \leq k \leq d = \dim(M)$. Since also $h(f) = h(f^{-1})$, up to replacing f by f^{-1} we can assume that

$$1 < \lambda = \mathrm{sp}(f_*) = \mathrm{sp}(f_{*,k})$$

for some $1 \leq k \leq d/2$. Now pick $\alpha \in H_k(M, \mathbb{Z})$ such that $\|f^n_*\alpha\| \sim \lambda^n$, i.e. for some $C > 0$ we have

$$\|f^n_*\alpha\| \geq C\lambda^n$$

for all $n \in \mathbb{N}$. By Thom's theorem, for some positive integer N, $N\alpha$ is representable, i.e. there is a closed k-manifold S and a continuous map $\varphi : S \to M$ such that $N\alpha = \varphi_*[S]$. Moreover, since $k \leq d/2$, by Whitney's embedding theorem we can perturb φ to make it a smooth immersion.[2] Finally for all $n \in \mathbb{N}$ and $\widehat{\varphi} \in C^1(S, M)$ homotopic to φ we get

$$\|(f^n \circ \widehat{\varphi})_*[S]\| = \|(f^n \circ \varphi)_*[S]\| = \|[f^n_*(N\alpha)]\| \geq \|[f^n_*\alpha]\| \geq C\lambda^n,$$

which implies that for some $\widehat{C} > 0$ we have

$$\inf_{\widehat{\varphi} \sim \varphi} \mathrm{vol}(f^n \circ \widehat{\varphi}) \geq \widehat{C}\lambda^n.$$

Thus $\mathrm{HVG}(S, f) \geq \log \lambda = \log \mathrm{sp}(f_*)$, so $\mathrm{HVG}(S, f) \leq h(f)$ forces $\log \mathrm{sp}(f_*) \leq h(f)$, as desired. □

For all immersed S it is easy to obtain a bound of the form

$$\mathrm{HVG}(S, f) \leq h(f) + \mathrm{HLV}(S, f),$$

where the error term $\mathrm{HLV}(S, f)$, which we define formally in Chapter 5, encodes exponential local volume growth which persists over the entire homotopy class of S. Thus to prove Conjecture 1.3.1 it would suffice to show that $\mathrm{HLV}(S, f)$ vanishes for all $f \in \mathrm{Diff}^r(M)$ and

[2] If $k < d/2$ we can obtain a smooth embedding.

immersed $S \subset M$.

The primary evidence for Conjecture 1.3.1 lies in the observation that the standard Margulis-Yomdin examples of local volume growth are not *stable*—the pathological volume growth phenomenon can be destroyed by a small perturbation—let alone persistent across the submanifolds entire homotopy class. In addition, our attempts to construct C^1 examples with $\mathrm{HLV}(S, f) > 0$ have so far been unsuccessful.

On the other hand, in general it is far from obvious how to actually *prove* that $\mathrm{HLV}(S, f) = 0$. This condition is equivalent to the following claim: For all $\delta > 0$, there exists $\varepsilon > 0$ and a sequence of $\widehat{\varphi}_n \sim \varphi$ such that

$$\sup_{x \in M} \mathrm{vol}\, f^n(\widehat{\varphi}_n(S) \cap B_f(x, n, \varepsilon)) \leq (1 + \delta)^n$$

for infinitely many $n \in \mathbb{N}$. Briefly this means that for infinitely many times n we can homotope S to $\widehat{\varphi}_n(S)$ such that the image by f^n of *any* (n, ε)-Bowen disk contained in $\widehat{\varphi}_n(S)$ has volume at most $(1 + \delta)^n$. The difficulty, of course, lies in "any": while it is easy to destroy (ε, n)-local volume growth at a specific point $x \in S$ by homotopy, it is much more delicate to do this in a way which avoids simply pushing the local volume growth elsewhere in the submanifold.

What is needed is some tangent-level property which allows control over the geometry of Bowen disks at time n and is moreover realizable at *all* points of $\widehat{\varphi}_n(S)$ for some $\widehat{\varphi}_n(S)$ homotopic to S. Classically, such geometric control is obtained via *hyperbolicity*, which implies (via contraction of the graph transform) that the local geometry of disks everywhere transverse to the stable bundle remains bounded. More generally one can obtain such control via a global *dominated splitting*, or at least locally at most points via *non-uniform hyperbolicity*. In the context of the Entropy Conjecture, the main problem with the first two properties is that they are rare. The problem with the last property, non-uniform hyperbolicity, is that its trademark non-uniformity makes it difficult to extract conclusions from the avail-

able information, namely global volume growth of a submanifold, since even with geometric control on an arbitrarily large measure subset of M it is difficult to rule out the possibility at any given time that the volume growth is produced entirely in the small regions where this control fails.

The desire for some finite-time derivative criterion less restrictive than uniform domination but still sufficient (at least in the C^2 setting) to give *some* geometric control over *some* Bowen disks (on a slowly decaying scale) is the main motivation underlying the discussion in Chapter 3. Though the main result of this chapter is negative—it amounts to constructing a simple counterexample to a desired claim—we hope it can help point towards future positive results in this direction.

1.4 The intersections approach

In addition to volume growth, there is another (essentially dual) approach to extracting entropy estimates from homology growth, which leverages the non-degenerate *intersection pairing*

$$\iota : H_k(M, \mathbb{R}) \times H_{d-k}(M, \mathbb{R}) \to \mathbb{R}.$$

As above we assume (without loss of generality) that $1 \leq k \leq d/2$ and $\operatorname{sp} f_* = \lambda > 1$ is realized on H_k. For simplicity and (in some sense) extra concreteness, we also assume (now in principle with some loss of generality, related to homology representation issues), that there are closed embedded submanifolds $V, W \subset M$, with $\dim(V) = k$ and $\dim(W) = d - k$, such that

$$\iota([f^n V], [W]) \sim \lambda^n.$$

With these assumptions, after a generic perturbation of V and/or W we can also assume that $f^n V \pitchfork W$ for all $n \in \mathbb{N}$. Under this transversality assumption the intersection form

asymptotics imply that for some $C > 0$ we have

$$\#(f^n V \cap W) \geq C\lambda^n.$$

Moreover this estimate holds for all $\widehat{V}$ and $\widehat{W}$, homotopic to V and W, respectively, such that $f^n\widehat{V} \pitchfork \widehat{W}$ for all $n \in \mathbb{N}$, therefore for C^1-generic $\widehat{V}$, $\widehat{W}$ in the homotopy classes of V and W.

From the perspective of the Entropy Conjecture, the basic, naive question becomes: Given submanifolds $V, W \subset M$ as above, does there exist $\varepsilon > 0$ such that for all $n \in \mathbb{N}$ sufficiently large we can find $\widehat{V}_n \sim V$ and $\widehat{W}_n \sim W$ such that $f^n\widehat{V}_n \pitchfork \widehat{W}_n$, and moreover $\widehat{V}_n \cap f^{-n}(\widehat{W}_n)$ (or at least most of it) is an (n, ε)-separated set? Analogously to the volume growth approach, the most natural route to a positive answer here would show that points in $\widehat{V}_n \cap f^{-n}(\widehat{W}_n)$ cannot generically (over C^1 homotopy classes) clump exponentially in (n, ε)-Bowen balls for ε arbitrarily small.

A somewhat paradoxical difficulty here is that the homotopy assumption, despite (but also because of) its strength, is difficult to work with geometrically. This suggests the need for some more restricted class of motions both large enough to rule out pathological intersection phenomenon (on average) and small enough to preserve some geometric control.

Motivated by this problem, in Chapter 2 we consider compact C^1 families of diffeomorphisms $\psi : \mathcal{H} \to \mathrm{Diff}^1(M)$ and provide close-to-optimal assumptions on such families sufficient to imply some natural such "regularizing on average" estimates.

Theorem A. *For any compact submersive family $\mathcal{H}$ there is a constant $C_U > 0$ such that $\mathbb{E}(\#(hV \cap W)) \leq C_U \operatorname{vol}(V) \operatorname{vol}(W)$ for all complementary dimensional submanifolds $V, W \subset M$. If the interior of $\mathcal{H}$ is also Grassmann-transitive, then there exists $C_L > 0$ such that $\mathbb{E}(\#(hV \cap W)) \geq C_L \operatorname{vol}(V) \operatorname{vol}(W)$ for all such $V, W \subset M$, as well.*

While the hoped-for application of this estimate to the Entropy Conjecture remains

incomplete, here we advertise its more general dynamical usefulness with two additional applications. Though somewhat far afield from our original motivation, both applications illustrate the broad utility of submersive (and/or Grassmann-transitive) families to dynamics.

In the first application we consider the classical Artin-Mazur theorem, which states that a dense subset of $C^r(M, M)$ has at most exponential growth of periodic point components. In its original form the theorem essentially follows from Nash approximation plus the observation that this estimate "obviously" (after some calculations) holds for polynomial maps (see e.g. [KH95, §7.4]). The apparent optimism of the density conclusion is deceptive in light of subsequent developments, by a number of authors, which cumulatively show (among other things) that the density conclusion cannot be improved, since maps with pathological (dominating any arbitrary sequence) periodic point growth can be locally generic.

Our result here provides a very modest counterpoint to this narrative by showing that such pathologies disappear in a natural stochastic setting, namely when one considers the iteration of any C^1 map f with submersive random noise.

Theorem B. *Let $(\mathcal{H}, \psi)$ be a compact, submersive C^1 family with normalized volume and write $\mu = \psi_* \operatorname{vol}_{\mathcal{H}}$. Then there exists $\lambda > 1$ such that for $\mu^{\mathbb{N}}$-a.e. $\mathbf{f} \in \operatorname{Diff}^1(M)^{\mathbb{N}}$ we have*

$$\#\operatorname{Per}_n \mathbf{f} \leq \lambda^n.$$

The assumption that $(\mathcal{H}, \psi)$ has *normalized volume* simply means that we have scaled the Riemannian metric on $\mathcal{H}$ so that $\operatorname{vol}(\mathcal{H}) = 1$.

Our second "application" of Theorem A is less an application of the theorem itself than of the class of C^1 families it considers. Namely, we again view a compact C^1 family $(\mathcal{H}, \psi)$ as inducing a probability measure μ on $\operatorname{Diff}^1(M)$, namely $\mu = \psi_* \operatorname{vol}_{\mathcal{H}}$ for $\operatorname{vol}_{\mathcal{H}}$ a normalized volume measure on $\mathcal{H}$. Such a measure defines a random walk, and a natural if rather naive first question about such a process is whether its stationary measures on M can be classified. For general probability measures on $\operatorname{Diff}^1(M)$ such a classification is too much

to ask; measures μ for which such a classification *is* possible are therefore interesting, and studied extensively under the general framework of *measure rigidity*.

Within this program, much attention has recently been devoted to the class of *uniformly expanding* measures. These measures are defined by requiring a strong form of uniform expansion-on-average for all tangent vectors in uniform finite time. This hypothesis has proved especially useful in extending measure rigidity results from the classical homogeneous context to more general smooth settings [Fil].

Perhaps surprisingly, such measures have been shown to exist in abundance on all manifolds [Pot21], [ES23]. In §2.7 we relate these existence results to our discussion by showing that the hypotheses on C^1 families in Theorem A are also sufficient (modulo a small, generically extraneous modification) to guarantee uniform expansion for their induced measures.

1.5 Generalized entropy

Classical topological entropy measures the *exponential* growth rate of number of separated orbits. This focus on exponential growth means that it is entirely unable to capture variations between systems with sub-exponential orbit complexity.

In [CP23], Correa & Pujals introduced the notion of *generalized entropy*. This is a topological conjugacy invariant of a dynamical system which takes values in the space of all *orders of growth*.

Since this invariant by definition involves no loss of information—it fully captures the asymptotics of the sequences $(r_f(n,\varepsilon))$—it is especially useful in the study of zero entropy systems. Relatedly, however, the space of orders of growth is extremely complicated. The formulation of this invariant thus immediately inspires a number of nontrivial realization questions. In general one would like more information about which orders of growth are actually attained as generalized entropies of (some class of) compact dynamical systems.

In Chapter 4 I answer one of their realization questions by constructing a class of subshifts f with $\sup \mathbb{P} < o(f) < \inf \mathbb{E}$. Using an elementary general technique one can then attain such growth rates within the class of smooth diffeomorphisms on $\mathbb{S}^3$.

1.6 Directions for future work

In the final chapter we return to the Entropy Conjecture, and in particular to the problem of finding approaches in low-ish regularity which more fully leverage the given homological (or at least homotopical) structure.

The main available paradigm for such an argument is Manning's homotopical argument that $\log \mathrm{sp}(f_{*,1}) \geq h(f)$. This argument relies on two claims. The first, Lemma 5.2.1, says that for sufficiently small $\varepsilon > 0$, any ε-small (i.e. diameter less than ε) curve with endpoints (n, ε)-close is homotopic to one whose image under f^n is a minimizing geodesic. This claim easily generalizes to higher dimensions. The second claim says that any sufficiently small curve admits a partition into segments with endpoints (n, ε)-close whose cardinality is bounded above by $r_f(n, \varepsilon/2)$. The (simple) proof of this latter claim appears to depend essentially on the order structure along a curve, thus resists higher-dimensional generalization.

Motivated by a desired to avoid the second claim (which appears a priori ungeneralizable to higher dimensions), in Chapter 5 we sketch an intersection-based proof of the H_1 result. While this proof still depends essentially on the first claim, it bypasses the second entirely. One hopes that this argument might point the way towards some generalization to higher dimensions.

Expanding on the first claim above, one could also naively ask e.g. whether it is possible to *triangulate* a submanifold $S \subset M$ such that the d_n^f diameter of any vertex set is $\leq 2\varepsilon$, but such that the total number of simplices is bounded by $C_\varepsilon(n) r_f(n, \varepsilon)$ for some constant $C_\varepsilon(n)$

growing subexponentially in n. For instance, one could take the Delaunay triangulation of S with respect to the d_n^f metric and some (n, ε)-separated set $X \subset S$ in the d_n^f. Leaving aside the immediate issue that such a triangulation is not guaranteed to exist, one expects the number of simplices in such a triangulation to be order $O(s_f(n, \varepsilon)^{\lceil \dim S/2 \rceil})$. This is worse than the desired estimate, but even such a bound would be interesting. In our final section we will make some remarks in this direction, circumventing the many technical issues around triangulations by instead looking at the *nerve* of a dynamical cover. Here the question becomes whether one can obtain estimates on the simplicial norm, in the nerve of a minimal (n, ε)-cover, of classes realizing homology growth. Such estimates would of course depend not only on n and ε but also in principle on the choice of cover and the represented class.

Chapter 2

Average intersections for families of diffeomorphisms

2.1 Introduction

Let M be a closed Riemannian manifold of dimension n, and let $\mathrm{Diff}^r(M)$ denote the space of C^r diffeomorphisms of M with the C^r topology. For $0 \leq k \leq n$, let $\mathrm{Gr}_k(TM)$ denote the k-Grassmannian bundle over M, i.e. the set of k-planes tangent to M.

In this chapter we study conditions on compact C^1 families in $\mathrm{Diff}^1(M)$ sufficient to guarantee a kind of "uniform mixing" on average of the (smooth) incidence geometry of M. By a C^r *family of diffeomorphisms of* M we will mean a pair $(\mathcal{H}, \psi)$, where $\mathcal{H}$ is a smooth manifold, possibly with boundary, and $\psi : \mathcal{H} \to \mathrm{Diff}^r(M)$ is such that the associated evaluation map $ev_\psi : \mathcal{H} \times M \to M$ is C^r. This implies in particular that the parametrizing map ψ is continuous. When a family $(\mathcal{H}, \psi)$ is fixed we will usually suppress the map ψ, using the same letter h to denote both an element in $\mathcal{H}$ and its image under ψ; by this mild abuse of notation we will often write, for instance, $ev_\psi(h, p) = \psi(h)(p) = h(p)$.

We will also assume that any such $\mathcal{H}$ comes equipped with a Riemannian metric, and

thus a Riemannian density which we denote by $d\mathcal{H}$. Given these assumptions, our main result is the following estimate:

Theorem 2.1.1. *Suppose $(\mathcal{H}, \psi)$ is compact C^1 family of diffeomorphisms of M such that*

(A1) *$ev : \mathcal{H} \times M \to M \times M$ defined by $ev(h, p) = (p, h(p))$ is a submersion;*

(A2) *$\mathring{ev}_k : \mathring{\mathcal{H}} \times \mathrm{Gr}_k(TM) \to \mathrm{Gr}_k(TM) \times \mathrm{Gr}_k(TM)$ defined by $ev_k(h, \sigma) = (\sigma, dh(\sigma))$ is surjective for all $0 \leq k \leq n$.*

Then there is a constant $C = C(\mathcal{H}, \psi) \geq 1$ such that

$$\frac{1}{C} \operatorname{vol}(V) \operatorname{vol}(W) \quad \leq \quad \int_{h \in \mathcal{H}} \#(h(V) \cap W) \; d\mathcal{H} \quad \leq \quad C \operatorname{vol}(V) \operatorname{vol}(W) \tag{2.1}$$

for all complementary dimensional submanifolds $V, W \subset M$.

Assumption (A1) says equivalently that for all $p \in M$, the map $ev_{\psi,p} : \mathcal{H} \to M$, $h \mapsto \psi(h)(p)$ is a submersion, or (again equivalently) that for all $(h, p) \in \mathcal{H} \times M$,

$$\partial_h (ev_\psi)_{(h,p)} : T_h \mathcal{H} \to T_{h(p)} M$$

is surjective. We call a family satisfying this assumption *submersive*.

Assumption (A2) says that for all $0 \leq k \leq n$ and $\sigma_1, \sigma_2 \in \mathrm{Gr}_k(TM)$, there is an $h \in \mathring{\mathcal{H}}$ such that $dh(\sigma_1) = \sigma_2$; in other words, the interior of $\mathcal{H}$ "acts transitively" on $\mathrm{Gr}_k(TM)$ for every k. We call a family satisfying this assumption *Grassmann-transitive*.

Here $\#(h(V) \cap W) \in \mathbb{N}_0 \cup \{\infty\}$ simply means the number of points in $h(V) \cap W$. Note that (2.1) implicitly includes the claim that the function

$$\mathcal{H} \ni h \longmapsto \#(h(V) \cap W) \in \mathbb{N}_{\geq 0} \cup \{\infty\}$$

is measurable, and integrable if both V and W have finite volume; in the latter case it implies that $\#(h(V) \cap W) < \infty$ for a.e. $h \in \mathcal{H}$.

By the volume of a submanifold $S \subset M$ we will always mean its intrinsic volume as a Riemannian manifold with metric inherited from M, i.e. $\mathrm{vol}(S) = \int_S dS$, where dS denotes the density induced by the restricted metric. Alternatively, if $\dim(S) = k$ we have $\mathrm{vol}(S) = \mathrm{Haus}^k_M(S)$, the k-dimensional Hausdorff measure of S in M.

Theorem 2.1.1 is actually a special case of a more general result, which applies whenever $\dim(V)+\dim(W) \geq \dim(M)$ and correspondingly replaces the cardinality of the intersection with its appropriate-dimensional Hausdorff measure. We focus on the complementary dimensional case for simplicity of exposition, but the proof of the general case works similarly. For the precise statement and some remarks on its proof, see §2.4.

By analogy with the (exact) kinematic formulas of integral geometry (see e.g. [Che66], [Cal20], [San04]), we call (2.1) a *kinematic inequality* for the family $(\mathcal{H}, \psi)$. Taken alone, Theorem 2.1.1 is a relatively straightforward adaptation of these classical formulas—which obtain when $\mathcal{H} = G$ is a compact Lie group acting isometrically on M and transitively on every $\mathrm{Gr}_k(TM)$—to a specific (a priori somewhat artificial and potentially rare) non-homogeneous context. Its possible utility derives from the following existence result, which we prove in §2.5:

Theorem 2.1.2. *Every closed manifold M admits an $\mathcal{H}$ and $\psi : \mathcal{H} \to \mathrm{Diff}^1(M)$ satisfying the hypotheses of Theorem 2.1.1.*

We emphasize that the $\mathcal{H}$ constructed here has non-empty boundary. This is the main motivation for allowing $\partial\mathcal{H} \neq \varnothing$ in the definition of C^r families, and in fact Theorem 2.1.2 fails if one requires $\partial\mathcal{H} = \varnothing$. For details see §2.1.3.

2.1.1 Background and motivation

Formulas of the type considered here have a long history in integral geometry and a well-developed theory in the setting of homogeneous spaces of compact Lie groups. From this

perspective the basic antecedent to our result is

Theorem 2.1.3 (Poincaré's formula for homogeneous spaces)**.** *Let M^n be a closed Riemannian manifold with G a compact Lie group acting smoothly and transitively on M by isometries. Suppose the action is also transitive on $\mathrm{Gr}_k(TM)$, for every $1 \leq k \leq n-1$. Then for every $k, \ell \geq 0$ satisfying $k + \ell \geq n$, there is a constant $C_{k,\ell} > 0$ such that for any submanifolds $V, W \subset M$ with $\dim(V) = k$ and $\dim(W) = \ell$, we have*

$$\int_{g \in G} \mathrm{vol}(gV \cap W)\ dG = C_{k,\ell}\, \mathrm{vol}(V)\, \mathrm{vol}(W), \tag{2.2}$$

where dG denotes the Haar measure on G.

In the form given here this result is due to Howard [How93]. The basic idea behind the proof is to view the left-hand side as a double integral over what we will later call the *intersection manifold* associated to V and W, i.e. the space of all intersections between $g(V)$ and W for $g \in G$. By changing the order of integration one then (essentially) equates this integral with the average volume of the spaces $G_{x,y} = \{g \in G \ : \ gx = y\}$ over all $(x, y) \in V \times W$. (In other words, one replaces the average volume of intersections with the average volume of the spaces of maps realizing each possible intersection.) Since all such spaces are isometric for any bi-invariant metric on G inducing the Haar measure, the result follows. For details, see [How93, §3] or [BC13, §A.4]. Our adaptation of this approach below is particularly indebted to [BCSS98, §13] and [SS94, §4]. For an exemplary presentation of the classical material, see [San04].

Observe that any action $G \curvearrowright M$ as in Theorem 2.1.3 can be viewed as a compact C^∞ family of diffeomorphisms of M, namely (G, ψ) where $\psi : G \ni g \mapsto (x \mapsto gx) \in \mathrm{Diff}^\infty(M)$. This family clearly satisfies (A1) and (A2), and thus Theorem 2.1.1 yields (2.2) *up to a uniform multiplicative error*. The fact that this error can be eliminated in the setting of Theorem 2.1.3 depends on the assumption that the family (G, ψ) consists of isometries. For

an analogous result applicable to general closed manifolds, however, this is too much to ask: if M is non-homogeneous—i.e. does not admit a transitive Lie group action—and a compact family $(\mathcal{H}, \psi)$ satisfies $\mathrm{im}(\psi) \subset \mathrm{Isom}(M)$ for some choice of Riemannian metric on M, then $\mathcal{H}$ must leave some compact, proper submanifolds invariant (namely the orbits of $\mathrm{Isom}(M)$, a compact Lie group acting non-transitively on M), and thus by choosing some such $\mathcal{H}$-invariant V—and then W some complementary-dimensional, arbitrarily small/large disk, intersecting/disjoint (respectively) from V, one sees that neither bound in (2.1) can hold uniformly for any measure on $\mathcal{H}$. More generally, this argument shows that for M non-homogeneous, no family of diffeomorphisms verifying (2.1) can be contained in a compact subgroup of $\mathrm{Diff}^1(M)$. Thus some geometric distortion is unavoidable. Consequently, an inequality incorporating a bounded multiplicative error seems to be the most one could expect from an extension of Poincaré's formula to non-homogeneous manifolds.

Our primary motivation for formulating such an extension comes from smooth dynamics.[1] Given a C^1 diffeomorphism $f : M \to M$ and a pair of complementary dimensional submanifolds $V, W \subset M$, one would like to understand how the growth of $\mathrm{vol}(f^n(V))$ affects the distribution of (a large compact piece of) the isotopy class of $f^n(V)$ with respect to W. Our results here imply that we can fix a compact family of diffeomorphisms $\mathcal{H}$ of M so that the average number of intersections between $h(f^n(V))$ and W has the same order of growth (in the sense of e.g. [CP23]) as $\mathrm{vol}(f^n(V))$.

2.1.2 Assumptions

Theorem 2.1.1 is a "backwards" theorem in the sense that we started with the estimate (2.1) and looked for reasonable (and in particular abundantly realized) assumptions that would make it true. Here we collect some comments on our eventual selections (A1) and (A2).

[1] For a similar generalization from a different direction, see the work of Fu in [Fu16], and especially his definition of an "admissible measured family" of diffeomorphisms in §6.

First, we emphasize that the statement of Theorem 2.1.1 is not quite optimal, in two main ways:

(i) The upper bound in (2.1) actually follows from (A1) alone; the surjectivity assumption (A2) is needed only for the lower bound. Note also that (A1), in contrast to (A2), is a purely local property of $(\mathcal{H}, \psi)$. Since the property (A1) is also easily seen to be invariant under (either left- or right-) translation of the family by a diffeomorphism g, this implies that an upper bound as in (2.1) can be obtained for finite-dimensional slices of $\mathrm{Diff}^1(M)$ concentrated arbitrarily close to any given diffeomorphism f.

(ii) In the other direction, it follows from the proof that (A2) is somewhat stronger than necessary for our result. What we really need is the following:

($\mathrm{A2}^*$) For any $\sigma_p \in \mathrm{Gr}_k(TM)$ and $\sigma_q \in \mathrm{Gr}_{n-k}(TM)$, there is an $h \in \mathring{\mathcal{H}}$ such that $h(p) = q$ and $dh(\sigma_p) \pitchfork \sigma_q$.

For further discussion of this point see the remark after Claim 2.2.9. We have opted to use (A2) in the statement of Theorem 2.1.1 nonetheless, primarily to emphasize the analogy with the transitivity assumption in Theorem 2.1.3, and also because it is no more difficult in practice to construct a family satisfying (A2) than one satisfying the weaker condition ($\mathrm{A2}^*$).

These qualifications in mind, the purpose of our assumptions is heuristically the following. The assumption that ev is a submersion implies that no particular intersection $h(p) = q$ between $h(V) \cap W$ can be locally persistent (since one can move $h(p)$ off itself by a small change in h). In particular, any non-transversal intersection between $h(V)$ and W (near which the cardinality of $h(V) \cap W$ could explode) can be destroyed by a small perturbation of h. This ensures that almost all intersections are isolated, and moreover that the total

number of such points (summed over $\mathcal{H}$) is never too large compared to $\mathrm{vol}(V) \cdot \mathrm{vol}(W)$. This yields the upper bound in Theorem 2.1.1 using (A1) alone.

To get a lower bound, one needs to show that $\mathcal{H}$ can always *produce* intersections between pairs of points in V and W. Moreover, since the quantity we are trying to bound is a sort of average of the number of intersections with respect to the volume measure on $\mathcal{H}$, these intersections should persist over open (thus positive measure) subsets of $\mathcal{H}$. In other words, $\mathcal{H}$ should be able to produce *transverse* intersections between any pair of points in V and W. This will certainly hold if $\mathcal{H}$ can move any tangential k-plane to any other, as guaranteed by (A2).

The following example cleanly illustrates the main ideas in the isometric case:

Example 2.1.4 ($\mathbb{T}^2 \curvearrowright \mathbb{T}^2$)**.** Let $M = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ be the standard flat 2−torus, and $\mathcal{H} = \mathbb{T}^2$ with $\psi : \mathbb{T}^2 \longrightarrow \mathrm{Diff}(\mathbb{T}^2)$ the smooth action of $\mathbb{T}^2$ on itself by translations; explicitly, $\psi : a \mapsto (\psi_a : x \mapsto x + a)$. Then the pair $(\mathcal{H}, \psi)$ satisfies assumption (A1), but not (A2), since $dh = \mathrm{id}$ for all $h \in \mathrm{im}(\psi)$ (using the canonical identifications $T_p\mathbb{T}^2 \cong \mathbb{R}^2$) and thus

$$\mathrm{im}(ev_1) = \{(\sigma_1, \sigma_2) \ : \ \sigma_1 \text{ and } \sigma_2 \text{ are parallel}\} \neq \mathrm{Gr}_1(T\mathbb{T}^2) \times \mathrm{Gr}_1(T\mathbb{T}^2).$$

More strongly, the fact that every element of $\mathcal{H}$ maps any element of $\mathrm{Gr}_1(T\mathbb{T}^2)$ to a parallel translate means that no element of $\mathrm{Gr}_1(T\mathbb{T}^2)$ can be made transverse to itself via $\mathcal{H}$.

Theorem 2.1.1 correspondingly fails. For instance, take I, J two parallel Euclidean line segments in $\mathbb{T}^2$. Then it is easy to see that $\psi_a(I) \cap J = \varnothing$ for a.e. $a \in \mathbb{T}^2$, so

$$\int_{a \in \mathbb{T}^2} \#(\psi_a(I) \cap J) \ d\mathbb{T}^2 = 0.$$

This implies that the lower bound in Theorem 2.1.1 cannot hold for any positive C.

If the angle between I and J is $\theta \neq 0$, however, it is similarly straightforward to show

that

$$\int_{a\in\mathbb{T}^2} \#(\psi_a(I)\cap J)\ d\mathbb{T}^2 = |\sin\theta|\,\mathrm{length}(I)\cdot\mathrm{length}(J).$$

In particular, we have

$$\int_{a\in\mathbb{T}^2} \#(\psi_a(I)\cap J)\ d\mathbb{T}^2 \leq \mathrm{length}(I)\cdot\mathrm{length}(J)$$

for any Euclidean line segments $I, J \subset \mathbb{T}^2$, and by a subdivision and approximation argument it is not difficult to see that this holds more generally for $I, J \subset \mathbb{T}^2$ any embedded C^1 curves. Thus the upper bound in Theorem 2.1.1 holds here with $C = 1$.

2.1.3 A digression on $\mathcal{H}$

One technical hurdle in proving Theorem 2.1.1 stems from our allowance of the possibility that the parameter space $\mathcal{H}$ has boundary. If one assumes instead that $\partial\mathcal{H} = \varnothing$, the availability of Ehresmann's fibration theorem (applied to the submersion ev) substantially streamlines the proof; see the remark concluding §2.2.4. However, here Ehresmann's theorem cuts both ways, imposing strong topological constraints on any boundaryless $\mathcal{H}$ satisfying (A1). These constraints imply in particular that for general M, the condition $\partial\mathcal{H} = \varnothing$ is too much to ask:

Proposition 2.1.5. *Suppose M is a closed manifold and $(\mathcal{H}, \psi)$ a compact, connected family of C^1 diffeomorphims of M with $\partial\mathcal{H} = \varnothing$, such that ev_p is a submersion for some $p \in M$. Then $\pi_1(M)$ is a finite extension of a quotient of a subgroup of $\pi_1(\mathrm{Diff}_0(M))$.*

Proof. Let $(\mathcal{H}, \psi)$ be such a family. After translating and perturbing we may assume that $\psi(\mathcal{H}) \subset \mathrm{Diff}_0(M)$ the identity component of $\mathrm{Diff}^\infty(M)$. Choose $p \in M$ such that ev_p a submersion.

By Ehresmann's theorem, $ev_p : \mathcal{H} \to M$ is a locally trivial fibration, say with fiber $\mathcal{F}$.

The end of the homotopy long exact sequence for this fibration reads

$$\pi_1(\mathcal{H}) \xrightarrow{(ev_p)_*} \pi_1(M) \longrightarrow \pi_0(\mathcal{F}) \longrightarrow 1.$$

Since $\mathcal{H}$ and (therefore) $\mathcal{F}$ are compact we must have $\#\pi_0(\mathcal{F}) < \infty$, so this sequence implies that $(ev_p)_*$ is virtually onto. On the other hand, $ev_p = EV_p \circ \psi$ where $EV_p : \mathrm{Diff}_0(M) \to M$, $f \mapsto f(p)$. Thus $(ev_p)_*$ factors through $\pi_1(\mathrm{Diff}_0(M))$, and the result follows. □

Corollary 2.1.6. *Suppose M is a closed hyperbolic manifold with* $\dim(M) \leq 3$. *Then there is no boundaryless compact family of diffeomorphisms of M satisfying* (A1).

Proof. By work of Earle-Eells (in dimension 2) and Gabai (in dimension 3), $\mathrm{Diff}_0(M)$ is contractible ([EE69], [Gab01]). In particular, $\pi_1(\mathrm{Diff}_0(M)) = 0$. Since (by hyperbolicity of M) $\pi_1(M)$ is infinite, by Proposition 2.1.5 we are done. □

The upshot here is that Theorem 2.1.2 fails if one requires that $\partial\mathcal{H} = \varnothing$. More strongly, these results (combined with the restrictions on higher homotopy groups obtained by continuing the above exact sequence) suggest that manifolds which admit a submersive compact family of diffeomorphisms without boundary may be quite rare. Clearly homogeneous spaces of compact Lie groups form the main class of examples. Are there others?

Question 2.1.7. *Which closed manifolds M admit a compact smooth family of diffeomorphisms $\mathcal{H} \longrightarrow \mathrm{Diff}(M)$ with $\partial\mathcal{H} = \varnothing$, such that the associated evaluation map $ev : \mathcal{H} \times M \longrightarrow M \times M$ is a submersion?*

We note that, on the question of existence, the choice of regularity is irrelevant. A C^r family of diffeomorphisms can be viewed as a map $\Phi \in C^r(\mathcal{H} \times M, M)$ such that $\Phi(h, \cdot) \in \mathrm{Diff}^r(M)$ for all $h \in \mathcal{H}$; with this notation, the submersion condition simply means that $\Phi(\cdot, p) : \mathcal{H} \to M$ is a submersion for all $p \in M$. When $\mathcal{H}$ and M are compact, these properties are open in $C^1(\mathcal{H} \times M, M)$, so if one obtains a boundaryless C^1 family satisfying (A1), by perturbing one can obtain a C^∞ family as well.

This intriguing diversion aside, the proof of Theorem 2.1.1 in the general, possibly-with-boundary setting turns out to work in much the same way as the Ehresmann-based proof for the boundaryless case, modulo some basic lemmas concerning submersions from compact manifolds with boundary which we relegate to Appendix A.[2] The content of these lemmas is to show that our required bounds (which follow trivially from compactness when the fibers of ev move C^1-continuously) still follow from our assumptions when we allow boundary, provided that the surjectivity requirement in (A2) is phrased in terms of the interior of $\mathcal{H}$.

2.2 Preliminaries

Let $(\mathcal{H}, \psi)$ satisfy the hypotheses of Theorem 2.1.1, and write $d := \dim(\mathcal{H})$. We begin by introducing the main objects used in our proof.

2.2.1 Normal Jacobians

Let E_1, E_2 be finite-dimensional real inner product spaces. Then for any linear map $A : E_1 \to E_2$, one defines the *normal Jacobian* by the formula

$$\mathrm{NJ}(A) := \det(AA^*)^{1/2} \in \mathbb{R}_{\geq 0}.$$

Note that $\mathrm{NJ}(A) > 0$ iff A is surjective. One can check that

$$\mathrm{NJ}(A) = |\det(A|_{\ker(A)^\perp})|.$$

For a C^1 map $f : M \to N$ between Riemannian manifolds, we define the *normal Jacobian of f* pointwise, by

$$\mathrm{NJ}(f)(x) := \mathrm{NJ}(df_x) = \det(df_x \cdot df_x^*)^{1/2}.$$

[2] Though these straightforward results are presumably well-known, we were unable to locate them in the literature.

Thus $\mathrm{NJ}(f) : M \to \mathbb{R}_{\geq 0}$, and since f is C^1 it is easy to see that $NJ(f)$ is continuous. By the above, $\mathrm{NJ}(f)(x) > 0$ iff x is a regular point of f.

2.2.2 The solution manifold $\widetilde{\mathcal{V}}$

The *solution manifold* $\widetilde{\mathcal{V}} \subset \mathcal{H} \times M \times M$ is the subspace consisting of all triples of the form $(h, p, h(p))$. Equivalently, $\widetilde{\mathcal{V}} = \Gamma(ev_\psi)$, where

$$\begin{aligned} ev_\psi \ &: \mathcal{H} \times M \longrightarrow M \\ &(h, p) \ \longmapsto \ h(p). \end{aligned}$$

Since ev_ψ is C^1, $\widetilde{\mathcal{V}}$ is a neat C^1 submanifold with boundary of $\mathcal{H} \times M \times M$, diffeomorphic to $\mathcal{H} \times M$.[3] In particular, it inherits C^1 projection maps $\widetilde{\pi}_1, \widetilde{\pi}_2$:

$$\begin{array}{ccc} & \widetilde{\mathcal{V}} & \\ {\scriptstyle \tilde{\pi}_1}\swarrow & & \searrow{\scriptstyle \tilde{\pi}_2} \\ \mathcal{H} & & M \times M. \end{array}$$

By (A1) and the $k = 0$ case of (A2), $\tilde{\pi}_2|_{\mathrm{int}(\widetilde{\mathcal{V}})} : \mathrm{int}(\widetilde{\mathcal{V}}) \to M \times M$ is a surjective submersion. For $(p, q) \in M \times M$ we write

$$\mathcal{V}_{p,q} := \tilde{\pi}_2^{-1}(p, q) \cap \mathrm{int}(\widetilde{\mathcal{V}}) = \left(\tilde{\pi}_2|_{\mathrm{int}(\widetilde{\mathcal{V}})}\right)^{-1}(p, q).$$

Then each $\mathcal{V}_{p,q}$ is an embedded $(d-n)$-dimensional submanifold of $\mathrm{int}(\widetilde{\mathcal{V}})$ with Riemannian metric inherited from $\mathcal{H} \times M \times M$ and induced Riemannian density which we will denote by $d\mathcal{V}_{p,q}$.

[3] By *neat* we mean that $\partial\widetilde{\mathcal{V}} \subset \partial(\mathcal{H} \times M \times M) = \partial\mathcal{H} \times M \times M$.

2.2.3 The intersection manifolds $\mathcal{V}$

Now let $V, W \subset M$ be complementary dimensional submanifolds of M, say $\dim(V) = k$ and $\dim(W) = n - k$. We define the *intersection manifold* $\mathcal{V}$ associated to $\mathcal{H}$, V, and W by

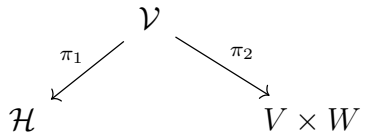
$$\begin{aligned}\mathcal{V} &:= \{(h, p, q) \in \mathring{\mathcal{H}} \times V \times W \ : \ h(p) = q\} \\ &= \left(\tilde{\pi}_2\big|_{\mathrm{int}(\widetilde{\mathcal{V}})}\right)^{-1}(V \times W) = \tilde{\pi}_2^{-1}(V \times W) \cap \mathrm{int}(\widetilde{\mathcal{V}}).\end{aligned}$$

Since $\tilde{\pi}_2\big|_{\mathrm{int}(\widetilde{\mathcal{V}})} : \mathrm{int}(\widetilde{\mathcal{V}}) \to M \times M$ is a surjective submersion, $\mathcal{V}$ is a C^1 submanifold of $\mathrm{int}(\widetilde{V})$ with $\mathrm{codim}_{\widetilde{\mathcal{V}}}(\mathcal{V}) = \mathrm{codim}_{M\times M}(V \times W) = n$, so

$$\dim(\mathcal{V}) = \dim(\widetilde{\mathcal{V}}) - n = \dim(\mathcal{H}) + n - n = \dim(\mathcal{H}).$$

Again note that $\mathcal{V}$ inherits a Riemannian metric from $\mathcal{H} \times M \times M$; as usual we will write $d\mathcal{V}$ for the induced Riemannian density.

For $i = 1, 2$, define $\pi_i := \tilde{\pi}_i|_{\mathcal{V}}$. Then we have the diagram



of C^1 maps, with π_2 again a surjective submersion.

In the first step of the proof we will apply the co-area formula to this diagram to convert our main integral

$$\int_{h\in\mathcal{H}} \#(h(V) \cap W)\ d\mathcal{H}$$

to an integral over $V \times W$ involving the normal Jacobians of π_1 and π_2. To estimate this latter integral, we will need a description of the tangent spaces

$$T_{(h,p,q)}\mathcal{V} \subset T_h\mathcal{H} \times T_pV \times T_qW.$$

This is given easily by

Claim 2.2.1. *For all $(h,p,q) \in \mathcal{V}$ we have*

$$T_{(h,p,q)}\mathcal{V} = \{(\dot{h},\dot{p},\dot{q}) \in T_h\mathcal{H} \times T_pV \times T_qW \ : \ J_{(h,T_pV,T_qW)}(\dot{p},\dot{q}) = d(ev_p)_h(\dot{h})\}$$

where $J_{(h,T_pV,T_qW)} : T_pV \times T_qW \to T_qM$ is defined by

$$J_{(h,T_pV,T_qW)}(v,w) = w - dh(v).$$

Proof. Differentiating the defining equation $h(p) = q$ yields

$$\dot{q} = d(ev_p)_h(\dot{h}) + dh(\dot{p}).$$

The claim follows. □

Notice that, since $d(ev_p)_h : T_p\mathcal{H} \to T_{h(p)}M$ is surjective for every $(h,p) \in \mathcal{H} \times M$, the projection map

$$\begin{aligned} d(\pi_1)_{(h,p,q)} \ : \ T_{(h,p,q)}\mathcal{V} &\longrightarrow T_h\mathcal{H} \\ (\dot{h},\dot{p},\dot{q}) &\longmapsto \dot{h} \end{aligned}$$

will be surjective if and only if $J_{(h,T_p,T_qW)}$ is surjective, i.e. iff $dh(T_pV) + T_qW = T_qM$. We summarize this observation in the following:

Proposition 2.2.2. *The regular points of the projection map $\pi_1 : \mathcal{V} \to \mathcal{H}$ are precisely those $(h,p,q) \in \mathcal{V}$ such that $h(V)$ intersects W transversally at q. Thus $h \in \mathring{\mathcal{H}}$ is a regular value of π_1 iff $h(V) \pitchfork W$.*

The restriction to the interior of $\mathcal{H}$ in the above statement is necessary because by definition $\mathcal{V} \subset \mathring{\mathcal{H}} \times V \times W$, and thus every $h \in \partial\mathcal{H}$ is trivially a regular value of π_1, even though $\mathcal{V}$ gives no (immediate) information about the intersections $h(V) \cap W$ for such h. However, since $\partial\mathcal{H}$ has measure zero in $\mathcal{H}$ and $\dim(\mathcal{H}) = \dim(\mathcal{V})$, by applying Sard's theorem to the C^1 map $\pi_1 : \mathcal{V} \to \mathring{\mathcal{H}}$ we get

Corollary 2.2.3. *$h(V) \pitchfork W$ for almost every $h \in \mathcal{H}$.*

Remark. The argument given here essentially follows the usual proof of the classical transversality theorem. In low regularity, however, its appeal to Sard's theorem requires constraints on the dimensions of V and W. In particular, in the C^1 category Corollary 2.2.3 fails when $\dim(V) + \dim(W) > \dim(M)$. This difficulty is the main reason for considering Hausdorff measures instead of volumes of intersections in the generalization to Theorem 2.1.1; see §2.4 for details.

An immediate consequence of the formula above is that the tangent space $T_{(h,p,q)}\mathcal{V}$ depends only on p, T_pV and T_qW. The same is therefore true of the linear maps $(d\pi_i)_{(h,p,q)}$ (themselves just restrictions of projections), and consequently their normal Jacobians.

We would like to show that more is true, namely that $\mathrm{NJ}(\pi_1)$ and $\mathrm{NJ}(\pi_2)$ are continuous when viewed as functions of h, T_pV, and T_qW. The next section introduces the objects necessary to make this statement precise.

2.2.4 The bundles $\mathcal{G}_k \to \widetilde{\mathcal{V}}$

For each $0 \leq k \leq n$ let

$$\mathrm{Gr}_k(TM) = \bigsqcup_{p \in M} \mathrm{Gr}_k(T_pM)$$

denote the k-Grassmannian bundle over M. This is a smooth fiber bundle over M, associated to the tangent bundle TM in the sense that it can be obtained by gluing local trivializations $U_\alpha \times \mathrm{Gr}_k(\mathbb{R}^n)$ according to the smooth clutching functions $\rho_{\alpha,\beta} : U_\alpha \cap U_\beta \to GL(n, \mathbb{R})$ for TM. For each k we let $p_k : \mathrm{Gr}_k(TM) \to M$ denote the standard (smooth) projection map. For $\sigma \in \mathrm{Gr}_k(TM)$ we will typically write $\sigma = \sigma_x$ for $x \in M$ to mean that $x = p_k(\sigma)$, i.e. $\sigma \in \mathrm{Gr}_k(T_xM)$. In words, $\sigma_x \in \mathrm{Gr}_k(TM)$ simply denotes a k-plane tangent to M at the point x.

Now define

$$\mathcal{G}_k := \{(h, \sigma_p, \sigma_q) \in \mathcal{H} \times \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM) \ : \ h(p) = q\}\,.$$

Clearly $\mathcal{G}_k$ is a C^1 submanifold of $\mathcal{H} \times \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$, and indeed the map $\mathrm{id} \times p_k \times p_{n-k}$ exhibits it as a C^1 $(\mathrm{Gr}_k(\mathbb{R}^n) \times \mathrm{Gr}_{n-k}(\mathbb{R}^n))$-bundle over $\widetilde{\mathcal{V}}$. (In particular, $\mathcal{G}_0 \cong \mathcal{G}_n \cong \widetilde{\mathcal{V}}$.)

Claim 2.2.4. *The map $\widehat{\pi}_2^k : \mathcal{G}_k \to \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$, $(h, \sigma_1, \sigma_2) \mapsto (\sigma_1, \sigma_2)$ is a C^1 submersion and surjective when restricted to $\mathring{\mathcal{G}}_k$.*

Proof. This follows from the observation that $\mathcal{G}_k$ can be obtained as the pullback of the smooth fiber bundle $\mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM) \to M \times M$ via the C^1 submersion $\tilde{\pi}_2 : \widetilde{\mathcal{V}} \to M \times M$. More precisely, we have the following diagram:

$$\begin{array}{ccc} \mathcal{G}_k & \xrightarrow{\widehat{\pi}_2^k} & \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM) \\ \downarrow{\scriptstyle \mathrm{id}\times p_k \times p_{n-k}} & & \downarrow{\scriptstyle p_k \times p_{n-k}} \\ \widetilde{\mathcal{V}} & \xrightarrow[\tilde{\pi}_2]{} & M \times M. \end{array}$$

In local coordinates (i.e. above any chart $U \subset \widetilde{\mathcal{V}}$), $\hat{\pi}_k$ has the form

$$\hat{\pi}_k \ \cong \ \tilde{\pi}_2 \times \mathrm{id} \ : \ U \times (\mathrm{Gr}_k(\mathbb{R}^n) \times \mathrm{Gr}_{n-k}(\mathbb{R}^n)) \ \longrightarrow \ \tilde{\pi}_2(U) \times (\mathrm{Gr}_k(\mathbb{R}^n) \times \mathrm{Gr}_{n-k}(\mathbb{R}^n))\,,$$

which is clearly a submersion because $\tilde{\pi}_2$ is. The surjectivity of $\hat{\pi}_k|_{\mathring{\mathcal{G}}_k}$ follows immediately from that of $ev\big|_{\mathring{\mathcal{H}}\times M}$, i.e. the $k = 0$ case of (A2). □

Now for each $(h, \sigma_p, \sigma_q) \in \mathcal{G}_k$ we can define

$$\begin{aligned} J_{(h,\sigma_p,\sigma_q)} : \sigma_p \times \sigma_q &\to T_qM \\ (v, w) &\mapsto w - dh(v), \end{aligned}$$

and

$$E_k(h, \sigma_p, \sigma_q) := \{(\dot{h}, v, w) \in T_h\mathcal{H} \times \sigma_p \times \sigma_q \ : \ J_{(h,\sigma_p,\sigma_q)}(v, w) = d(ev_p)_h(\dot{h})\}.$$

Finally, set

$$\begin{aligned}\Pi_1(h,\sigma_p,\sigma_q) &: E_k(h,\sigma_p,\sigma_q) \longrightarrow T_h\mathcal{H}\\ &\qquad (\dot{h},v,w) \longmapsto \dot{h},\\ \Pi_2(h,\sigma_p,\sigma_q) &: E_k(h,\sigma_p,\sigma_q) \longrightarrow \sigma_p\times\sigma_q\\ &\qquad (\dot{h},v,w) \longmapsto (v,w),\end{aligned}$$

and for $i=1,2$, define

$$\eta_i : \mathcal{G}_k \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$\eta_i(h,\sigma_p,\sigma_q) = \mathrm{NJ}(\Pi_i(h,\sigma_p,\sigma_q)).$$

The point of these objects, immediate from the definitions and Claim 2.2.1, is the following:

Lemma 2.2.5. *Let $V,W\subset M$ be complementary dimensional submanifolds with $\dim(V)=k$, and $\mathcal{V}$ their associated intersection manifold. Then for all $(h,p,q)\in\mathcal{V}$ and $i=1,2$ we have*

$$T_{(h,p,q)}\mathcal{V} = E_k(h,T_pV,T_qW);$$

$$d(\pi_i)_{(h,p,q)} = \Pi_i(h,T_pV,T_qW);$$

$$\mathrm{NJ}(\pi_i)_{(h,p,q)} = \eta_i(h,T_pV,T_qW).$$

We will also need the following properties:

Claim 2.2.6. *E_k is a continuous vector bundle over $\mathcal{G}_k$ of rank d.*

Proof. Let Ω_j denote the tautological vector bundle over $\mathrm{Gr}_j(TM)$, i.e. the smooth rank-j vector bundle $\Omega_j\to\mathrm{Gr}_j(TM)$ with fibers $\Omega_j(\sigma)=\sigma$. Observe that, since each $\sigma\in\mathrm{Gr}_j(TM)$ is a subspace of some tangent space of M, Ω_j carries a natural bundle metric obtained by

restricting the Riemannian metric on M.

Let $\mathcal{E}_k$ denote the restriction of the vector bundle $T\mathcal{H} \times \Omega_k \times \Omega_{n-k} \to \mathcal{H} \times \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$ to $\mathcal{G}_k$. Again this carries a natural bundle metric, namely the product metric induced from its factors.

We can now view J as a bundle map

$$J : \mathcal{E}_k \longrightarrow \gamma^* TM,$$

where $\gamma : \mathcal{G}_k \to M$ is given by $\gamma(h, \sigma_p, \sigma_q) = q$. Continuity of J follows from that of our parametrization $\psi : \mathcal{H} \to \mathrm{Diff}^1(M)$. Define

$$H : \mathcal{E}_k \longrightarrow \gamma^* TM$$

by

$$\begin{aligned} H_{(h,\sigma_p,\sigma_q)} : T_h\mathcal{H} \times \sigma_p \times \sigma_q &\longrightarrow T_q M \\ (\dot{h}, v, w) &\longmapsto d(ev_p)_h(\dot{h}), \end{aligned}$$

and note that H is a continuous bundle map because ev is C^1. Finally set $L := J - H$.

Since $d(ev_p)_h$ is surjective for all $(h, p) \in \mathcal{H} \times M$ and $J_{(h,\sigma_p,\sigma_q)}$ does not depend on $\dot{h}$, $L_{(h,\sigma_p,\sigma_q)}$ is surjective for all $(h, \sigma_p, \sigma_q) \in \mathcal{G}_k$. In particular, L is a constant rank vector bundle map, so $\ker(L)$ is a continuous vector subbundle of $\mathcal{E}_k$ (see e.g. [Lee13, Theorem 10.34]), with

$$\mathrm{rank}(\ker(L)) = \mathrm{rank}(\mathcal{E}_k) - \mathrm{rank}(TM) = (d + n) - n = d.$$

Since clearly $\ker(L) = E_k$, we are done. □

Claim 2.2.7. *η_1 and η_2 are continuous.*

Proof. Π_1 and Π_2 can be viewed as continuous bundle maps between metric vector bundles,

namely

$$\Pi_1 : E_k \to (\widehat{\pi}_1^k)^* T\mathcal{H};$$
$$\Pi_2 : E_k \to (\widehat{\pi}_2^k)^* (\Omega_k \times \Omega_{n-k}),$$

where $\widehat{\pi}_1^k : \mathcal{G}_k \to \mathcal{H}$ and $\widehat{\pi}_2^k : \mathcal{G}_k \to \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$ denote the natural projections. Thus their normal Jacobians are continuous functions of the base. □

Claim 2.2.8. *$\eta_2 > 0$ everywhere on $\mathcal{G}_k$.*

Proof. $\Pi_2(h, \sigma_p, \sigma_q)$ has full rank for all $(h, \sigma_p, \sigma_q) \in \mathcal{G}_k$. □

Now define

$$\widehat{\mathcal{F}}_{\sigma_p,\sigma_q} := \left(\widehat{\pi}_2^k|_{\mathring{\mathcal{G}}_k}\right)^{-1}(\sigma_p, \sigma_q) = (\widehat{\pi}_2^k)^{-1}(\sigma_p, \sigma_q) \cap \mathring{\mathcal{G}}_k.$$

Since $\widehat{\pi}_2^k|_{\mathring{\mathcal{G}}_k}$ is a surjective C^1 submersion, $\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}$ is an embedded $(d-n)$-dimensional submanifold of $\mathring{\mathcal{G}}_k$ for all $(\sigma_p, \sigma_q) \in \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$. As usual we let $d\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}$ denote the Riemannian density induced by the restriction of the metric on $\mathcal{G}_k$ to $\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}$.

Claim 2.2.9. *For all $(\sigma_p, \sigma_q) \in \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$, η_1 is positive somewhere on the fiber $\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}$.*

Proof. Let $(\sigma_p, \sigma_q) \in \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$. Clearly it suffices to find $h \in \mathring{\mathcal{H}}$ such that $h(p) = q$ and $\Pi_1(h, \sigma_p, \sigma_q)$ is surjective. Observe as in Proposition 2.2.2 that $\Pi_1(h, \sigma_p, \sigma_q)$ is surjective iff $dh(\sigma_p) \pitchfork \sigma_q$. But by (A2) we can find $h \in \mathring{\mathcal{H}}$ such that $dh(\sigma_p) = {\sigma_q}^\perp$, so we are done. □

Remark. Claim 2.2.9 is in fact the only place where (A2) is used, and its proof makes clear that this assumption is actually stronger than required for this result. What we really need is the following:

(A2*) For any $\sigma_p \in \mathrm{Gr}_k(TM)$ and $\sigma_q \in \mathrm{Gr}_{n-k}(TM)$, there is an $h \in \mathring{\mathcal{H}}$ such that $h(p) = q$ and $dh(\sigma_p) \pitchfork \sigma_q$.

So for instance (A2), which is equivalent to the claim that for all $\sigma \in \mathrm{Gr}_k(TM)$ and $q \in M$ we have

$$\mathrm{Gr}_k(T_qM) \subset \{dh(\sigma) \ : \ h \in \mathring{\mathcal{H}}\},$$

could be replaced by the requirement that for all $\sigma \in \mathrm{Gr}_k(TM)$ and $q \in M$, the set

$$\mathrm{Gr}_k(T_qM) \cap \{dh(\sigma) \ : \ h \in \mathring{\mathcal{H}}\}$$

has nonempty interior in $\mathrm{Gr}_k(T_qM)$.[4] We prefer (A2) here largely for simplicity, and because, in the general setting we consider, it is no more difficult to construct a family $\mathcal{H}$ satisfying this hypothesis than (A2*); see §2.5. The sufficiency of the weaker assumption may, however, be useful in applications where (A2*) is easy to verify for a given family $\mathcal{H}$, despite additional constraints which render (A2) unachievable.

The main result of this section is now an immediate consequence of the following general fact. For details of the proof, see Appendix A.

Proposition 2.2.10 (Appendix A, Lemma A.0.4)**.** *Let $f : M^m \to N^n$ be a C^1 submersion between compact Riemannian manifolds, with $\partial M \neq \varnothing$ and $\partial N = \varnothing$, and $\phi \in C^0(M, \mathbb{R}_{\geq 0})$. Write $\mathcal{F}_q := f^{-1}(q) \cap \mathring{M}$. Then there exists $C_U^\phi > 0$ such that*

$$\int_{\mathcal{F}_q} \phi \ d\mathcal{F}_q \leq C_U^\phi$$

for all $q \in N$.

Suppose in addition that $f\big|_{\mathring{M}}$ is surjective and ϕ is positive somewhere on each interior fiber $\mathcal{F}_q$. Then there also exists $C_L^\phi > 0$ such that

$$C_L^\phi \leq \int_{\mathcal{F}_q} \phi \ d\mathcal{F}_q$$

for all $q \in N$.

[4] This latter property is satisfied, in particular, if we assume that $\mathring{ev}_0$ is surjective (i.e. the interior of $\mathcal{H}$ acts transitively on points) and ev_k is a submersion for all k.

Sketch of proof. For the upper bound, it clearly suffices to show that $\mathrm{vol}_{m-n}(\mathcal{F}_q) \leq \widehat{C}_U$ for some uniform $\widehat{C}_U < +\infty$; then one can set $C_U^\phi := \widehat{C}_U \cdot \|\phi\|_\infty$. By attaching a collar to M and extending the metric we obtain an open Riemannian manifold $\widehat{M}$ isometrically containing M; by shrinking the collar we can assume that f extends to a submersion $F : \widehat{M} \to N$. Now choose a finite collection $\mathcal{B}$ of submersion charts for F which cover M, such that the plaques of elements of $\mathcal{B}$ have volumes bounded uniformly above by some $C_1 < +\infty$. Since every fiber $\mathcal{F}_q$ intersects each submersion chart in at most one plaque, we get $\mathrm{vol}(\mathcal{F}_q) \leq C_1 \cdot \#(\mathcal{B})$.

For the lower bound, one can use surjectivity of $f|_{\mathring{M}}$ and compactness of N to find finitely many submersion charts $B_1, \ldots, B_\ell \subset \mathring{M}$ for $f|_{\mathring{M}}$ whose images cover N, with fiber plaque volumes bounded below by some $\widehat{C}_L > 0$ and $\phi \geq \delta > 0$ on every B_j. Then each $\mathcal{F}_q$ contains at least one plaque of some B_j, so

$$\int_{\mathcal{F}_q} \phi d\mathcal{F}_q \geq \widehat{C}_L \delta > 0,$$

as desired. □

Corollary 2.2.11. *Let $\eta = \frac{\eta_1}{\eta_2} : \mathcal{G}_k \longrightarrow \mathbb{R}_{\geq 0}$. Then there exists $C_k > 1$ such that*

$$\frac{1}{C_k} \leq \int_{\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}} \eta \; d\widehat{\mathcal{F}}_{\sigma_p,\sigma_q} \leq C_k$$

for all $(\sigma_p, \sigma_q) \in \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$.

Proof. By Claims 2.2.4, 2.2.7, 2.2.8, and 2.2.9, this follows from Proposition 2.2.10 with $f = \widehat{\pi}_2^k$ and $\phi = \eta$. □

Remark. If we define

$$\Phi_k : \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM) \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$\Phi_k(\sigma_p, \sigma_q) = \int_{\widehat{\mathcal{F}}_{\sigma_p,\sigma_q}} \frac{\eta_1}{\eta_2} \; d\widehat{\mathcal{F}}_{\sigma_p,\sigma_q},$$

then Corollary 2.2.11 simply says that Φ_k is bounded uniformly above and away from zero. Since $\mathrm{Gr}_j(TM)$ is compact for all j and $\Phi_k > 0$ by Claim 2.2.9, this would be obvious if Φ_k were continuous, and indeed when $\partial\mathcal{H} = \varnothing$ this is the case by an easy application of Ehresmann's fibration theorem. When $\partial\mathcal{H} \neq \varnothing$, however, Ehresmann's theorem fails, and in fact Φ_k cannot be assumed continuous in general. We were led to Proposition 2.2.10 by the need to fill this gap, since it shows in particular that the (non-)continuity of Φ_k is inessential for the desired bounds.[5]

As a final aside, we note that Poincaré's formula (as stated in Theorem 2.1.3) reduces after the results of the next two sections to the claim that the functions Φ_k (and all the analogously defined functions $\Phi_{k,\ell}$ for $n \leq k + \ell \leq 2n$, where $\Phi_k =: \Phi_{k,n-k}$) are constant and positive in the homogeneous case, provided the induced action on each $\mathrm{Gr}_k(TM)$ is transitive. This is in turn a fairly straightforward consequence of some basic geometry of compact Lie groups.

2.3 Proof of Main Theorem

Let $V, W \subset M$ be complementary dimensional submanifolds of M, say $\dim(V) = k$ and $\dim(W) = n - k$. Our proof of Theorem 2.1.1 now consists of two steps. First, we convert the integral

$$\int_{\mathcal{H}} \#(h(V) \cap W)\, d\mathcal{H}$$

to an integral over $V \times W$. We then apply the results of §2.2.4 to bound the new integrand uniformly above and away from zero, proving the result.[6]

[5] An easy adaptation of the proof of the lower bound in Proposition 2.2.10 does, however, show that Φ_k is always lower semicontinuous.

[6] For a related use of this "double fibration" method, see [ACSS22].

2.3.1 The co-area formula

The key tool in the first step is the co-area formula, which we state here for convenience. Though we choose to give only the simplest form sufficient for our needs, note that this is a special case of a more general result; see e.g. [Fed69, §3.2], [Nic11], [Sim14].

Proposition 2.3.1 (The co-area formula, C^r case)**.** *Let $f : M^{n+k} \to N^n$ be a C^r map between smooth Riemannian manifolds (without boundary), with $r \geq k+1 \geq 1$. For $q \in N$, write $F_q = f^{-1}(q)$. Then for any Borel measurable function h on M we have*

$$\int_M h \cdot \mathrm{NJ}(f)\ dM = \int_{q \in N} \left(\int_{F_q} h\ dF_q \right) dN,$$

with both sides finite if one is. In particular, if h is (essentially) bounded and $\mathrm{NJ}(f) \in L^1(M)$, then

$$\left(\Phi : q \longmapsto \int_{F_q} h\ dF_q \right) \in L^1(N),$$

and

$$\int_M h \cdot \mathrm{NJ}(f)\ dM = \int_N \Phi\ dN.$$

In the special case when f is a submersion, the above holds for all $k \geq 0$ assuming only that f is C^1.

Remark. Note that the function

$$\Phi :\ N \ni q \longmapsto \int_{F_q} h\ dF_q \in \mathbb{R}_{\geq 0} \cup \{\infty\}$$

is well-defined only at regular values q of f, for which F_q is an embedded k-dimensional submanifold of M with induced Riemannian density dF_q. But then by Sard's theorem it is defined a.e. on N, so (provided it is measurable) its integral over N makes sense. This application of Sard's is the main reason for the required relation between the regularity of f and $k = \dim(M) - \dim(N)$ and explains why this regularity assumption is unnecessary

when f is a submersion, since in that case *every* point of N is a regular value for f. Indeed, in the submersive case the co-area formula is an easy consequence of Fubini's theorem.

More general versions of the co-area formula avoid these issues entirely by replacing dF_q with the k-dimensional Hausdorff measure associated to the metric space structure on M; see in particular Proposition 2.4.3. These measures agree when F_q is a k-dimensional submanifold of M.

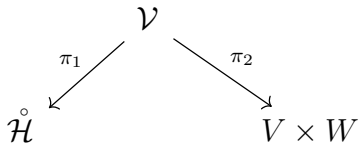
We now apply the co-area formula to prove the following, the main result of this section:

Proposition 2.3.2. *For $\mathcal{H}, V$, and W as in Theorem 2.1.1 we have*

$$\int_{h\in\mathcal{H}} \#(h(V)\cap W)\; d\mathcal{H} = \int_{(p,q)\in V\times W} \left(\int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\; d\mathcal{V}_{p,q} \right) dV\, dW,$$

with both sides finite if one is.

Proof. Recall the diagram



of Riemannian manifolds and C^1 maps. Since

$$\dim(\mathcal{V}) = \dim(\mathcal{H}) = d$$

and π_1 is C^1, we may apply Proposition 2.3.1 to π_1 (with $h \equiv 1$) to obtain

$$\int_{\mathcal{V}} \mathrm{NJ}(\pi_1)\; d\mathcal{V} = \int_{h\in\mathring{\mathcal{H}}} \left(\int_{\pi_1^{-1}(h)} d\pi_1^{-1}(h) \right) d\mathcal{H} = \int_{h\in\mathcal{H}} \#(h(V)\cap W)\; d\mathcal{H}. \tag{2.3}$$

The second equality follows from the fact that $\pi_1^{-1}(h)$ is a 0-dimensional submanifold of $\mathcal{V}$ for a.e. $h \in \mathcal{H}$, i.e. a discrete set of points, in which case we have

$$\int_{\pi_1^{-1}(h)} d\pi_1^{-1}(h) = \sum_{x\in\pi_1^{-1}(h)} 1 = \#(\pi_1^{-1}(h)) = \#(h(V)\cap W).$$

Note in particular that the a.e. defined function

$$h \longmapsto \int_{\pi_1^{-1}(h)} d\pi_1^{-1}(h) = \#(h(V) \cap W)$$

will be in $L^1(\mathring{\mathcal{H}}) = L^1(\mathcal{H})$ provided $\mathrm{NJ}(\pi_1) \in L^1(\mathcal{V})$.

We now apply the co-area formula a second time, this time to the map π_2 with $h = \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}$. Observe that this is well-defined because π_2 is a submersion, so $\mathrm{NJ}(\pi_2) > 0$ everywhere on $\mathcal{V}$. We get

$$\begin{aligned}\int_{\mathcal{V}} \mathrm{NJ}(\pi_1)\, d\mathcal{V} &= \int_{\mathcal{V}} \left(\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\right) \cdot \mathrm{NJ}(\pi_2)\, d\mathcal{V} \\ &= \int_{(p,q)\in V\times W} \left(\int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\, d\mathcal{V}_{p,q} \right) dV dW.\end{aligned}$$

Putting this together with (2.3) gives the desired result. □

2.3.2 Fiber integral bounds

By the preceding section, it now suffices to give uniform bounds for the integrals

$$\int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\, d\mathcal{V}_{p,q}.$$

Most of the necessary work has been done in §2.2.4. First recall Lemma 2.2.5, which gives

$$T_{(h,p,q)}\mathcal{V} = E_k(h, T_pV, T_qW),$$

$$d(\pi_i)_{(h,p,q)} = \Pi_i(h, T_pV, T_qW),$$

$$\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h,p,q) = \frac{\eta_1}{\eta_2}(h, T_pV, T_qW) = \eta(h, T_pV, T_qW).$$

Here $i = 1, 2$ and E_k, Π_i, η_i are as defined in §2.2.4. As an immediate consequence we obtain

Lemma 2.3.3. *For all $(p,q) \in V \times W$, the fiber $\mathcal{V}_{p,q}$ is isometric to $\widehat{\mathcal{F}}_{T_pV,T_qW}$. Moreover, we have*

$$\int_{\widehat{\mathcal{F}}_{T_pV,T_qW}} \eta \; d\widehat{\mathcal{F}}_{T_pV,T_qW} = \int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)} \; d\mathcal{V}_{p,q}.$$

Proof. Fix $(p,q) \in V \times W$. Using the fact that the Riemannian metrics on $\mathcal{G}_k$ and $\mathcal{V}$ are simply the restrictions of the product metrics on $\mathcal{H} \times \mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$ and $\mathcal{H} \times M \times M$, respectively, it is easy to see that the map

$$\widehat{\mathcal{F}}_{T_pV,T_qW} \longrightarrow \mathcal{V}_{p,q}, \quad (h, T_pV, T_qW) \longmapsto (h,p,q)$$

is an isometry.[7] The fact that

$$\int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h,p,q) \; d\mathcal{V}_{p,q} = \int_{\widehat{\mathcal{F}}_{T_pV,T_qW}} \eta(h, T_pV, T_qW) \; d\widehat{\mathcal{F}}_{T_pV,T_qW}$$

then follows by change of variables. □

Corollary 2.3.4. *If $C_k \geq 1$ is the constant given by Corollary 2.2.11, then*

$$\frac{1}{C_k} \leq \int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)} \; d\mathcal{V}_{p,q} \leq C_k$$

for all $p, q \in V \times W$.

Combining this with Proposition 2.3.2 and setting $C = \max\limits_{0 \leq k \leq n} C_k$ completes the proof of Theorem 2.1.1.

Remark. Our use of the rather abstract results of §2.2.4 and Appendix A to obtain the constants C_k leaves their geometric meaning somewhat opaque. The advantage of this approach is that it generalizes easily to prove Theorem 2.4.1, but in the complementary-dimensional case a more concrete picture is available, which may be useful in applications where estimates of C become desirable. We sketch this approach in Appendix B.

[7] Put more intuitively, both sides are just the set of $h \in \mathring{\mathcal{H}}$ such that $h(p) = q$ as a submanifold of $\mathcal{H}$ with the induced Riemannian metric.

2.4 Generalization

We now describe the generalization of Theorem 2.1.1 to the case when $\dim(V) + \dim(W) \geq \dim(M)$, where $\#(h(V) \cap W)$ is replaced by the appropriate-dimensional Hausdorff measure associated to the metric space structure on M.

For $r \geq 0$ and any Riemannian manifold X, let H^r_X denote the r-dimensional Hausdorff measure on X associated to its induced metric space structure. When $r = \dim(X)$, H^r_X agrees with the volume measure ν_X induced by the metric on X. More generally, if $Y \subset X$ is a k-dimensional submanifold with Riemannian metric inherited from X and induced volume measure ν_Y, we have

$$H^k_X\big|_Y = H^k_Y = \nu_Y.$$

In particular, the volume of a k-dimensional submanifold of X is equal to its k-dimensional Hausdorff measure in X. In this sense, for $k \in \mathbb{N}$ one can think of H^k_X as a generalization of k-dimensional volume to non-smooth subsets of X.

We now consider a fixed closed Riemannian manifold M of dimension n. For simplicity we write $H^r := H^r_M$.

Theorem 2.4.1. *Suppose $(\mathcal{H}, \psi)$ is compact C^1 family of diffeomorphisms of M such that*

(A1) *$ev : \mathcal{H} \times M \to M \times M$ defined by $ev(h, p) = (p, h(p))$ is a submersion;*

(A2) *$\mathring{ev}_k : \mathring{\mathcal{H}} \times \mathrm{Gr}_k(TM) \to \mathrm{Gr}_k(TM) \times \mathrm{Gr}_k(TM)$ defined by $ev_k(h, \sigma) = (\sigma, dh(\sigma))$ is surjective for all $0 \leq k \leq n-1$.*

Then there is a constant $C = C(\mathcal{H}, \psi) \geq 1$ such that for any submanifolds $V^k, W^\ell \subset M$ with $k + \ell \geq n$, we have

$$\frac{1}{C}\operatorname{vol}(V)\operatorname{vol}(W) \quad \leq \quad \int_{h \in \mathcal{H}} H^{k+\ell-n}(h(V) \cap W)\, d\mathcal{H} \quad \leq \quad C \operatorname{vol}(V)\operatorname{vol}(W). \tag{2.4}$$

The need to work with Hausdorff measures rather than volumes here derives from our insistence on assuming only C^1 regularity. In this context, when $\dim(V)+\dim(W) > \dim(M)$ we can no longer assume that $h(V) \pitchfork W$ for a.e. $h \in \mathcal{H}$; thus $\mathrm{vol}(h(V)\cap W)$ is not necessarily a.e. well-defined. However, if we let $h(V) \cap_T W$ denote the set of transversal intersection points between $h(V)$ and W (which when nonempty is always a $(k+\ell-n)$-dimensional submanifold of M), we will see below that indeed

$$H^{k+\ell-m}(h(V) \cap W) = H^{k+\ell-m}(h(V) \cap_T W) = \mathrm{vol}(h(V) \cap_T W)$$

for a.e. $h \in \mathcal{H}$.

Note that, since H^0 is the counting measure on M, Theorem 2.4.1 does indeed include Theorem 2.1.1 as a special case. Moreover, the proof method is essentially the same. Thus in this section we will content ourselves with some remarks on the modifications needed to adapt our proof of Theorem 2.1.1 to the general setting.

Let $V^k, W^\ell \subset M$ be submanifolds with $k+\ell \geq n$. The solution manifold $\widetilde{\mathcal{V}} \subset \mathcal{H} \times M \times M$ remains defined as before, and again (A1) implies that

$$\mathcal{V} := \left(\tilde{\pi}_2|_{\mathrm{int}(\widetilde{\mathcal{V}})}\right)^{-1} (V \times W)$$

is a C^1 submanifold of $\widetilde{\mathcal{V}}$. Now, however, we have

$$\dim(\mathcal{V}) = (d+n) - \mathrm{codim}(V \times W) = (d+n) - (2n-k-\ell) = d+k+\ell-n.$$

The basic observation motivating these definitions and underlying our entire proof strategy for Theorem 2.1.1 was that

$$\#(h(V) \cap W) = \#(\pi_1^{-1}(h)),$$

or equivalently,

$$H^0_M(h(V) \cap W) = H^0_{\mathcal{V}}(\pi_1^{-1}(h)).$$

The reason was that there is an obvious bijection between these sets (given explicitly by $\pi_3 : \mathcal{V} \to M$, defined below), and bijections between metric spaces preserve 0-dimensional Hausdorff measure. Obviously this fails for higher dimensional Hausdorff measures, so for the general case we must attend to how the maps $\pi_3^h := \pi_3|_{\pi_1^{-1}(h)} : \pi_1^{-1}(h) \to h(V) \cap W$ transform $(k+\ell-n)$-dimensional Hausdorff measures, where $\pi_3 : \mathcal{V} \to W$ is defined as one would expect, by $\pi_3(h,p,q) = q$.[8]

First suppose that h is a regular value of π_1; this is true iff $h(V) \pitchfork W$. In this case, $\pi_1^{-1}(h)$ and $h(V) \cap W$ are $(k+\ell-n)$-dimensional submanifolds of $\mathcal{V}$ and M, respectively, and π_3^h is a C^1 diffeomorphism. Their $(k+\ell-n)$-Hausdorff measures then agree with the volume measures induced by their Riemannian structures, and moreover we have

$$\begin{aligned} H_M^{k+\ell-n}(h(V)\cap W) = \mathrm{vol}(h(V)\cap W) &= \mathrm{vol}(\pi_3(\pi_1^{-1}(h))) \\ &= \int_{\pi_1^{-1}(h)} \left\| \bigwedge^{k+\ell-n} d\pi_3^h \right\| d\pi_1^{-1}(h) \\ &= \int_{\pi_1^{-1}(h)} \left\| \bigwedge^{k+\ell-n} d\pi_3^h \right\| dH_{\mathcal{V}}^{k+\ell-n}. \end{aligned}$$

Note also that

$$d\pi_3^h(h,p,q) = d\pi_3|_{T_{(h,p,q)}\pi_1^{-1}(h)} = d\pi_3|_{\ker(d\pi_1(h,p,q))},$$

and this last expression is well-defined even when (h,p,q) is not a regular point for π_1. Thus we can define

$$\rho : \mathcal{V} \longrightarrow \mathbb{R}_{\geq 0}$$

by

$$\rho(h,p,q) = \left\| \bigwedge^{k+\ell-n} d\pi_3 \big|_{\ker(d\pi_1(h,p,q))} \right\|,$$

[8] In fact this issue is largely an artifact of our original choice of set-up: it can be avoided by replacing $\widetilde{\mathcal{V}}$ with $\mathcal{H} \times M$ and $\widetilde{\pi}_2 : \widetilde{\mathcal{V}} \to M \times M$ by the map $\mathcal{H} \times M \to M \times M$, $(h,q) \mapsto (h^{-1}(q), q)$, since for these choices the resulting identification of $\pi_1^{-1}(h)$ with $h(V) \cap W$ is isometric. See [BC13] for an example of this approach.

and the above equation becomes

$$H_M^{k+\ell-n}(h(V) \cap W) = \int_{\pi_1^{-1}(h)} \rho \; dH_{\mathcal{V}}^{k+\ell-n}. \tag{2.5}$$

So far, (2.5) holds only when h is a regular value of π_1. In the complementary dimensional case this was enough, since when $\dim(\mathcal{H}) = \dim(\mathcal{V})$ Sard's theorem applies and thus (2.5) holds for a.e. $h \in \mathcal{H}$. When $\dim(V) + \dim(W) > \dim(M)$, however, we have $\dim(\mathcal{V}) > \dim(\mathcal{H})$, so Sard's no longer applies (since π_1 can only be assumed C^1). We appeal instead to the following generalization:

Proposition 2.4.2 ([Sim14], Theorem 6.4)**.** *Let $f : M^m \to N^n$ be a C^1 map between Riemannian manifolds, and let $C \subset M$ denote the set of critical points of f. Set $r = \max\{0, m-n\}$. Then*

$$H_M^r(f^{-1}(y) \cap C) = 0 \tag{2.6}$$

for Lebesgue a.e. $y \in N$.

Remark. The interesting case (and the one relevant here) is when $m > n$. As Simon notes in [Sim14], this proposition is itself a straightfoward consequence of Federer's general co-area formula; see [Fed69, §3.2] for details.

To apply this to the present setting, first write $C \subset \mathcal{V}$ for the set of critical points of π_1, and for $h \in \mathcal{H}$, let $h(V) \cap_T W$ denote the set of transversal intersection points between $h(V)$ and W, and $h(V) \cap_{NT} W = (h(V) \cap W) \setminus (h(V) \cap_T W)$. Then we have

$$\pi_3(\pi_1^{-1}(h) \setminus C) = h(V) \cap_T W,$$
$$\pi_3(\pi_1^{-1}(h) \cap C) = h(V) \cap_{NT} W,$$

and

$$\begin{aligned} H_M^{k+\ell-n}(h(V) \cap W) &= H_M^{k+\ell-n}(h(V) \cap_T W) + H_M^{k+\ell-n}(h(V) \cap_{NT} W) \\ &= \int_{\pi_1^{-1}(h) \setminus C} \rho \, dH_{\mathcal{V}}^{k+\ell-n} + H_M^{k+\ell-n}(\pi_3(\pi_1^{-1}(h) \cap C)). \end{aligned}$$

By Proposition 2.4.2,

$$\int_{\pi_1^{-1}(h) \setminus C} \rho \, dH_{\mathcal{V}}^{k+\ell-n} = \int_{\pi_1^{-1}(h)} \rho \, dH_{\mathcal{V}}^{k+\ell-n}$$

for a.e. $h \in \mathcal{H}$. For the second term, observe that $\pi_3 = \widetilde{\pi}_3|_{\pi_1^{-1}(h)}$, where $\widetilde{\pi}_3 : \widetilde{\pi}_1^{-1}(h) \to M$ is a distance non-increasing diffeomorphism (indeed $\widetilde{\pi}_1^{-1}(h)$ is canonically identified with $\Gamma(h) \subset M \times M$, with $\widetilde{\pi}_3$ corresponding to projection onto the second coordinate), so that

$$H_M^{k+\ell-n}(\pi_3(\pi_1^{-1}(h) \cap C)) \leq H_{\mathcal{V}}^{k+\ell-n}(\pi_1^{-1}(h) \cap C).$$

Since the right-hand side is zero for a.e. $h \in \mathcal{H}$ by Proposition 2.4.2, so is the left. We conclude that

$$\begin{aligned} H_M^{k+\ell-n}(h(V) \cap W) &= \int_{\pi_1^{-1}(h) \setminus C} \rho \, dH_{\mathcal{V}}^{k+\ell-n} + H_M^{k+\ell-n}(\pi_3(\pi_1^{-1}(h) \cap C)) \\ &= \int_{\pi_1^{-1}(h)} \rho \, dH_{\mathcal{V}}^{k+\ell-n} \end{aligned}$$

for Lebesgue a.e. $h \in \mathcal{H}$.

To apply this formula, we need the following slightly more general form of the co-area formula:

Proposition 2.4.3 (The co-area formula, C^1 case)**.** *Let $f : M^m \to N^n$ be a C^1 map between Riemannian manifolds, $m \geq n$. For $q \in N$, write $F_q = f^{-1}(q)$. Then for any Borel measurable function h on M we have*

$$\int_M h \cdot \mathrm{NJ}(f) \, dM = \int_{q \in N} \left(\int_{F_q} h \, dH_N^{m-n} \right) dN,$$

with both sides finite if one is.

We can now proceed as before. First, apply Proposition 2.4.3 twice to deduce

$$\begin{aligned}\int_{h\in\mathcal{H}} H^{k+\ell-n}(h(V)\cap W)\; d\mathcal{H} &= \int_{h\in\mathcal{H}} \left(\int_{\pi_1^{-1}(h)} \rho\; dH_{\mathcal{V}}^{k+\ell-n}\right)\; d\mathcal{H} \\ &= \int_{\mathcal{V}} \rho\cdot \mathrm{NJ}(\pi_1)\; d\mathcal{V} \\ &= \int_{(p,q)\in V\times W} \left(\int_{\mathcal{V}_{p,q}} \frac{\rho\cdot \mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\; d\mathcal{V}_{p,q}\right) dV\, dW,\end{aligned}$$

where $\mathcal{V}_{p,q} = \pi_2^{-1}(p,q)$ as before. The proof now reduces to showing that the integrand $\frac{\rho\cdot \mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}$ can be viewed as a continuous function on the appropriate Grassmannian bundle $\mathcal{G}_{k,\ell}$ over $\widetilde{\mathcal{V}}$, which is positive somewhere on each fiber of the submersion $\mathring{\mathcal{G}}_{k,\ell} \to \mathrm{Gr}_k(TM)\times \mathrm{Gr}_\ell(TM)$.

The space $\mathcal{G}_{k,\ell}\subset \mathcal{H}\times \mathrm{Gr}_k(TM)\times \mathrm{Gr}_\ell(TM)$, the bundle $E_{k,\ell}$, and the functions η_1, η_2, and η are defined exactly as in §2.2.4 (so e.g. in our original notation, $\mathcal{G}_k = \mathcal{G}_{k,n-k}$). Again, $\eta : \mathcal{G}_{k,\ell}\to \mathbb{R}_{\geq 0}$ is continuous, and positive at (h,σ_p,σ_q) iff $dh(\sigma_p)+\sigma_q = T_qM$. To extend ρ to a function on $\mathcal{G}_{k,\ell}$, simply define

$$\begin{aligned}\Pi_3(h,\sigma_p,\sigma_q) : E_{k,\ell}(h,\sigma_p,\sigma_q) &\longrightarrow T_qM \\ (\dot{h}, v, w) &\longmapsto w,\end{aligned}$$

and $\widehat{\rho}\colon \mathcal{G}_{k,\ell}\to \mathbb{R}_{\geq 0}$ by

$$\widehat{\rho}(h,\sigma_p,\sigma_q) = \left\| \bigwedge^{k+\ell-n} \Pi_3(h,\sigma_p,\sigma_q)\big|_{\ker(\Pi_1(h,\sigma_p,\sigma_q))} \right\|.$$

Clearly we have

$$\widehat{\rho}(h, T_pV, T_qW) = \rho(h,p,q).$$

Moreover, $\widehat{\rho}$ is continuous at all points where Π_1 has full rank, that is, away from the zeros of η. Since also $0<\widehat{\rho}\leq 1$, it follows that $\widehat{\rho}\cdot\eta$ is continuous on $\mathcal{G}_{k,\ell}$, with the same zero set

as η, and satisfies

$$(\widehat{\rho} \cdot \eta)(h, T_pV, T_qW) = \left(\rho \cdot \frac{NJ(\pi_1)}{NJ(\pi_2)}\right)(h, p, q).$$

One can now apply the results of Appendix A as before to complete the proof.

2.5 Construction of $(\mathcal{H}, \psi)$

In this section we show that Theorems 2.1.1 and 2.4.1 are non-vacuous, by constructing, for an arbitrary closed manifold M, a compact C^1 family of diffeomorphisms of M satisfying hypotheses (A1) and (A2). This proves Theorem 2.1.2.

Actually, since it is no more difficult (and arguably more natural) to work in the smooth category here, the family we construct will be C^∞. In addition, we will be unconcerned with compactness until the end. Specifically, we first obtain a C^∞ family $(\mathbb{R}^N, \psi)$ of the form

$$\psi : (\mathbb{R}^N, 0) \longrightarrow (\mathrm{Diff}^\infty(M), \mathrm{id})$$

which satisfies hypothesis (A1) and (A2), such that the surjectivity requirements of (A2) are satisfied by $(\mathbb{B}^N_R, \psi|_{\mathbb{B}^N_R})$ for all R sufficiently large.[9] The desired compact family is then obtained by restricting ψ to $\overline{\mathbb{B}}^N_R$ for some large R.

We begin with some general lemmas. Recall that for $1 \leq r \leq \infty$, a C^r *family of diffeomorphisms of* M is a pair $(\mathcal{H}, \psi)$, with $\mathcal{H}$ a smooth manifold, possibly with boundary, and $\psi : \mathcal{H} \to \mathrm{Diff}^r(M)$ such that the associated evaluation map $ev_\psi : \mathcal{H} \to M$ is C^r; this implies in particular that ψ is continuous. Any C^r family of diffeomorphisms of M is, of course, a C^s family of diffeomorphisms for all $s < r$.

One motivation for interpreting smoothness in this way—instead of via the strictly stronger requirement that $\psi \in C^r(\mathcal{H}, \mathrm{Diff}^r(M))$ for the standard Banach/Fréchet mani-

[9]Here and below we write $\mathbb{B}^N_R, \overline{\mathbb{B}}^N_R \subset \mathbb{R}^N$ for the open and closed balls, respectively, centered at 0 of radius R, with the subscript occasionally omitted in the special case $R = 1$, i.e. $\mathbb{B}^N := \mathbb{B}^N_1$.

fold structure on $\mathrm{Diff}^r(M)$—is the following elementary fact, which shows that the class of C^r families of diffeomorphisms in our weaker sense is in a natural way closed under composition.[10]

Lemma 2.5.1. *Let $(\mathcal{H}_1, \psi_1)$ and $(\mathcal{H}_2, \psi_2)$ be C^r families of diffeomorphisms of M, with $\partial\mathcal{H}_1 = \partial\mathcal{H}_2 = \varnothing$. Define*

$$\psi : \mathcal{H}_1 \times \mathcal{H}_2 \longrightarrow \mathrm{Diff}^r(M)$$

by $\psi(h_1, h_2) = \psi_2(h_2) \circ \psi_1(h_1)$, i.e. $\psi = \mathrm{comp} \circ (\psi_1 \times \psi_2)$.[11] *Then $(\mathcal{H}_1 \times \mathcal{H}_2, \psi)$ is a C^r family of diffeomorphisms of M.*

Proof. It suffices to check that the evaluation map

$$\begin{aligned} ev_\psi : (\mathcal{H}_1 \times \mathcal{H}_2) \times M &\longrightarrow M \\ (h_1, h_2, p) &\longmapsto \psi(h_1, h_2)(p) \end{aligned}$$

is C^r. We have

$$\begin{aligned} ev_\psi(h_1, h_2, p) &= \psi(h_1, h_2)(p) \\ &= \psi_1(h_1)(\psi_2(h_2)(p)) \\ &= ev_{\psi_1}(h_1, \psi_2(h_2)(p)) = ev_{\psi_1}(h_1, ev_{\psi_2}(h_2, p)). \end{aligned}$$

Since ev_{ψ_1} and ev_{ψ_2} are both C^r by assumption, so is ev_ψ. □

This lemma means that we can assemble our C^∞ family in the obvious way, namely by composing a finite collection of smaller C^∞ families supported in (and defined via) charts. Moreover, working within charts it is easy to produce families which *locally* satisfy the desired

[10] That this fails for the stronger notion is a consequence of the well-known fact that the composition map $[\mathrm{Diff}^r(M)]^2 \to \mathrm{Diff}^r(M)$ is only C^0 when $r < \infty$.

[11] For notational convenience we adopt the convention which defines the composition map $[\mathrm{Diff}^r(M)]^k \to \mathrm{Diff}^r(M)$ by $\mathrm{comp}(f_1, \ldots, f_k) = f_k \circ \cdots \circ f_1$.

properties (A1) and (A2); see Proposition 2.5.3 below. The next lemma gives us a criterion for ensuring that a composition of such families satisfies (A1) globally.

Lemma 2.5.2. *Let $(\mathcal{H}_1, \psi_1), \ldots, (\mathcal{H}_k, \psi_k)$ be C^r families of diffeomorphisms of M, with $\partial\mathcal{H}_i = \varnothing$ for all $1 \leq i \leq k$. Suppose that for all $i = 1, \ldots, k$, $h_i \in \mathcal{H}_i$, and $p \in M$, we have*

$$\psi_i(h_i)(p) \neq p \implies d(ev_{\psi_i,p})_{h_i} \text{ is surjective,}$$

where $ev_{\psi_i,p} := ev_{\psi_i}(\bullet, p) : \mathcal{H}_i \to M$. Finally, suppose that for all $p \in M$, there exists j such that $ev_{\psi_j,p}$ is a submersion, i.e. such that $d(ev_{\psi_j,p})_{h_j}$ is surjective for all $h_j \in \mathcal{H}_j$. Then letting $\mathcal{H} := \mathcal{H}_1 \times \cdots \mathcal{H}_k$, the map

$$\psi : \mathcal{H} \longrightarrow \mathrm{Diff}^r(M)$$
$$(h_1, \ldots, h_k) \longmapsto \psi_k(h_k) \circ \cdots \circ \psi_1(h_1)$$

defines a C^r family of diffeomorphisms of M, and $ev_{\psi,p} := ev_\psi(\cdot, p) : \mathcal{H} \to M$ is a submersion for all $p \in M$.

Proof. The fact that $(\mathcal{H}, \psi)$ is a C^r family of diffeomorphisms of M follows inductively from Lemma 2.5.1. Now let $p \in M$ and $h = (h_1, \ldots, h_k) \in \mathcal{H}$. We claim that

$$d(ev_{\psi,p})_h : T_h\mathcal{H} \longrightarrow T_qM$$

is surjective, where $q = ev_{\psi,p}(h) = \psi(h)(p) = \psi_k(h_k) \circ \cdots \circ \psi_1(h_1)(p)$. Clearly it suffices to show that

$$\partial_{h_j}(ev_{\psi,p})_h : T_{h_j}\mathcal{H}_j \longrightarrow T_qM$$

is surjective for some $1 \leq j \leq k$.

To compute these derivatives, write $p_0 := p$ and $p_i = \psi_i(h_i) \circ \cdots \circ \psi_1(h_1)(p)$ for $i \geq 1$; in

particular $p_k = q$. Setting $F_j = \psi_k(h_k) \circ \cdots \circ \psi_{j+1}(h_{j+1})$, we then have

$$\partial_{h_j}(ev_{\psi,p})_h = (dF_j)_{p_j} \circ d(ev_{\psi_j,p_{j-1}})_{h_j}.$$

Now set $\ell = \min\{1 \leq i \leq k \mid d(ev_{\psi_i,p})_{h_i} \text{ is surjective}\}$. Then by our assumption we must have $\psi_i(h_i)(p) = p$, and thus inductively $p_i = p$, for all $i < \ell$. Applying the above formula gives

$$\partial_{h_\ell}(ev_{\psi,p})_h = (dF_\ell)_{p_\ell} \circ d(ev_{\psi_\ell,p_{\ell-1}})_{h_j} = (dF_\ell)_{p_\ell} \circ d(ev_{\psi_\ell,p})_{h_\ell}.$$

By the definition of ℓ, $d(ev_{\psi_\ell,p})_{h_\ell}$ is surjective; since F_ℓ is a diffeomorphism, it follows that $\partial_{h_\ell}(ev_{\psi,p})_h : T_{h_\ell}\mathcal{H}_\ell \to T_{p_k}M$ is surjective as well, so we are done. □

We can now proceed with the construction of $\mathcal{H}$. Our approach is the "obvious" one, beginning with the observation that $\mathrm{Isom}^+(\mathbb{R}^n) \cong SO(n,\mathbb{R}) \ltimes \mathbb{R}^n$ acts smoothly and transitively on $\mathrm{Gr}_k(T\mathbb{R}^n) \cong \mathrm{Gr}_k(\mathbb{R}^n) \times \mathbb{R}^n$ for all $0 \leq k \leq n-1$. In particular, for every $\sigma \in \mathrm{Gr}_k(T\mathbb{R}^n)$, $ev_\sigma : SO(n,\mathbb{R}) \times \mathbb{R}^n \to \mathrm{Gr}_k(T\mathbb{R}^n)$ is a surjective submersion. To realize these properties inside an arbitrary manifold M, we first "cut off" this action to a smooth family of (uniformly) compactly supported diffeomorphisms of $\mathbb{R}^n$, which satisfies the submersion property (A1) on the interior of its support, and "acts transitively" on the restrictions of the Grassmann bundles to some slightly smaller ball. Using charts this family can be be realized inside any coordinate ball in M by a locally supported family of diffeomorphisms. The lemmas above then allow us to obtain the desired "action" of $\mathbb{R}^N$ on M by composing sufficiently many such families.

2.5.1 Step 1: Local construction.

Choose $\beta \in C^\infty(\mathbb{R}_{\geq 0}, [0,1])$ such that

- $\beta|_{[0,2]} \equiv 1$

- $\beta|_{[3,+\infty)} \equiv 0$
- $\beta'(x) < 0$ for all $x \in (2, 3)$.

Define

$$L : \mathfrak{so}_n \longrightarrow \mathrm{Diff}_c^\infty(\mathbb{R}^n, 0)$$

by

$$L(v)(p) = \exp(\beta(|p|)v)(p),$$

where $\exp : \mathfrak{so}_n \to SO(n, \mathbb{R})$ denotes the Lie exponential. Thus $L(v)$ preserves each sphere $\mathbb{S}_r^{n-1} := \{x \in \mathbb{R}^n \ : \ |x| = r\}$, and acts on $\mathbb{S}_r^{n-1}$ by $\exp(\beta(r)v) \in SO(n, \mathbb{R})$. In particular, $L(v)|_{\mathbb{B}_2^n} = \exp(v)$, and $\mathrm{supp}(L(v)) = \overline{\mathbb{B}}_3^n$ for all nonzero $v \in \mathfrak{so}_n$.

Define smooth vector fields $V_1, \ldots, V_n \in \Gamma_c^\infty(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n, \mathbb{R}^n)$ by

$$V_i(p) = \beta(|p|)\mathbf{e}_i, \ 1 \leq i \leq n.$$

For each i let h_i^t denote the flow associated to V_i. By composing these flows, we obtain a map

$$\tau : \mathbb{R}^n \longrightarrow \mathrm{Diff}_c^\infty(\mathbb{R}^n)$$
$$(t_1, \ldots, t_n) \longmapsto h_n^{t_n} \circ \cdots \circ h_1^{t_1}.$$

Finally, we define

$$\psi : \mathbb{R}^n \times \mathfrak{so}_n \longrightarrow \mathrm{Diff}_c^\infty(\mathbb{R}^n)$$

by

$$\psi(t, v) = \tau(t) \circ L(v).$$

Proposition 2.5.3. *The map*

$$ev_\psi : (\mathbb{R}^n \times \mathfrak{so}_n) \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$
$$(t, v, p) \longmapsto \psi(t, v)(p)$$

is C^∞, and for all $p \in \mathbb{B}^n_3$, the map $ev_{\psi,p} = ev_\psi(\cdot, p)$ is a submersion. Moreover, for all $0 \leq k \leq n-1$ and $\sigma_p, \sigma_q \in \mathrm{Gr}_k(T\mathbb{B}^n)$, we have

$$d(\psi(t, v))(\sigma_p) = \sigma_q.$$

for some $(t, v) \in \mathbb{B}^n_2 \times B_{\mathfrak{so}_n}(0, R)$, where $R > 0$ is such that $\exp(B_{\mathfrak{so}_n}(0, R)) = SO(n, \mathbb{R})$.

Proof. To see that ev_ψ is smooth, first observe that smoothness of

$$ev_L : \mathfrak{so}_n \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$
$$(v, p) \longmapsto L(v)(p) = \exp(\beta(|p|)v)(p)$$

follows from that of the Lie exponential exp and β. Also, since each V_i is smooth,

$$ev_i : \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$
$$(t, p) \longmapsto h_i^t(p)$$

is smooth for all $1 \leq i \leq n$. Smoothness of ev_ψ then follows inductively from Lemma 2.5.1.

Now let $p \in \mathbb{B}^n_3$ and $(t, v) \in \mathbb{R}^n \times \mathfrak{so}_n$. Writing $t = (t_1, \ldots, t_n)$, we set $q_0 = L(v)(p)$ and

$$q_j = (h_j^{t_j} \circ \cdots \circ h_1^{t_1})(q)$$

for $1 \leq j \leq n$; in particular, $q_n = \psi(t, v)(p)$. Observe that since every $L(v)$ and $h_j^{t_j}$ preserves $\mathbb{B}^n_3$, $q_i \in \mathbb{B}^n_3$ for all i.

We would like to show that that

$$d(ev_{\psi,p})_{(t,v)} : T_t\mathbb{R}^n \times T_v\mathfrak{so}_n \longrightarrow T_{q_n}\mathbb{R}^n$$

is surjective. Clearly it suffices to show that this is true for

$$\partial_t(ev_{\psi,p})_{(t,v)} : T_t\mathbb{R}^n \longrightarrow T_{q_n}\mathbb{R}^n.$$

Fix $1 \leq j \leq n$. We compute

$$\begin{aligned}
\partial_{t_j}(\psi(t,v)(p)) = \partial_{t_j}(\tau(t)(q_0)) &= \partial_{t_j}(h_n^{t_n} \circ \cdots \circ h_1^{t_1}(q_0)) \\
&= \partial_{t_j}(h_n^{t_n} \circ \cdots \circ h_j^{t_j}(q_{j-1})) \\
&= d(h_n^{t_n} \circ \cdots \circ h_{j+1}^{t_{j+1}}) \left(\partial_{t_j}(h_j^{t_j}(q_{j-1}))\right) \\
&= d(h_n^{t_n} \circ \cdots \circ h_{j+1}^{t_{j+1}}) \left(V_j(q_j)\right) \\
&= d(h_n^{t_n} \circ \cdots \circ h_{j+1}^{t_{j+1}}) \left(\beta(|q_j|)\mathbf{e}_j\right) \\
&= \beta(|q_j|) \cdot d(h_n^{t_n} \circ \cdots \circ h_{j+1}^{t_{j+1}}) \left(\mathbf{e}_j\right) \\
&= \beta(|q_j|) \left(\mathbf{e}_j + \alpha_{j+1,j}\mathbf{e}_{j+1} + \cdots + \alpha_{n,j}\mathbf{e}_n\right),
\end{aligned}$$

for some $\alpha_{i,j} \in \mathbb{R}$. In particular, $\partial_t(ev_{\psi,p})_{(t,v)}$ is lower triangular, with diagonal entries $\beta(|q_1|), \ldots, \beta(|q_n|)$. Since $|q_j| < 3$ and thus $\beta(|q_j|) > 0$ for all $1 \leq j \leq n$, it follows that $\partial_t(ev_{\psi,p})_{(t,v)}$ is surjective.

For the final statement, let $\sigma_p, \sigma_q \in \mathrm{Gr}_k(T\mathbb{B}^n) = \mathbb{B}^n \times \mathrm{Gr}_k(\mathbb{R}^n)$, say $\sigma_p = (p, \sigma_1), \sigma_q = (q, \sigma_2)$. Pick $A \in SO(n, \mathbb{R})$ such that $A\sigma_1 = \sigma_2$ and $v \in B_{\mathfrak{so}_n}(0, R)$ such that $\exp(v) = A$. Since $L(v)|_{\mathbb{B}^n} = A$ we have

$$d(L(v))(\sigma_p) = (Ap, A\sigma_1) = (Ap, \sigma_2).$$

Observe that, for all $1 \leq j \leq n$, if b and $h_j^{t_j}(b) \in \mathbb{B}_2^n$, then $h_j^{t_j} : x \mapsto x + t_j\mathbf{e}_j$ in a neighborhood of b, so that $d(h_j^{t_j})_b = \mathrm{id}$. Writing $q = (q_1, \ldots, q_n)$, $r = (r_1, \ldots, r_n) = Ap$, and $t = (t_1, \ldots, t_n) = q - r \in \mathbb{B}_2^n$, for all $0 \leq j \leq n$ we have

$$h_j^{t_j} \circ \cdots \circ h_1^{t_1}(r) = (q_1, \ldots, q_j, r_{j+1}, \ldots, r_n).$$

Thus $|h_j^{t_j} \circ \cdots \circ h_1^{t_1}(Ap)| \leq |q| + |r| < 2$. By the comment above, we inductively conclude that

$$d(h_n^{t_n} \circ \cdots \circ h_1^{t_1})_{Ap} = \mathrm{id}\,.$$

It follows that

$$\begin{aligned} d(\psi(t,v))(\sigma_p) &= d(h_n^{t_n} \circ \cdots \circ h_1^{t_1}) \circ d(L(v))(\sigma_p) \\ &= d(h_n^{t_n} \circ \cdots \circ h_1^{t_1})(Ap, \sigma_2) \\ &= (q, \sigma_2) = \sigma_q, \end{aligned}$$

so we are done. □

2.5.2 Step 2: Patching together.

Choose a finite collection $\{(U_i, \phi_i)\}_{i=1}^L$ of smooth charts $\phi_i : U_i \xrightarrow{\cong} \mathbb{B}_4^n$ for M such that

$$M = \bigcup_{i=1}^{L} \phi_i^{-1}(\mathbb{B}_1^n).$$

Set $W_i = \phi_i^{-1}(\mathbb{B}_1^n)$, and let D denote the diameter of the Čech 1-complex associated to the cover $\{W_i\}_{i=1}^L$ of M.[12] Since M is connected clearly we have $D \leq L - 1 < \infty$.

Now for each $i = 1, \ldots, L$ we can define

$$\psi_i : \mathbb{R}^n \times \mathfrak{so}_n \longrightarrow \mathrm{Diff}^\infty(M)$$

by

$$\psi_i(t,v) = \begin{cases} \phi_i^{-1} \circ \psi(t,v) \circ \phi_i & \text{on } U_i \\ \mathrm{id} & \text{on } M \setminus U_i. \end{cases}$$

[12]Explicitly, D is the smallest number such that for all $1 \leq k, \ell \leq L$, there exist $i_0, i_1, \ldots, i_{D-1}, i_D \in \{1, \ldots, L\}$ such that $i_0 = k$, $i_D = \ell$, and $W_{i_j} \cap W_{i_{j+1}} \neq \varnothing$ for all $0 \leq j \leq D - 1$.

Then define

$$\widehat{\psi} : (\mathbb{R}^n \times \mathfrak{so}_n)^L \longrightarrow \mathrm{Diff}^\infty(M)$$

by $\widehat{\psi} = \mathrm{comp} \circ (\psi_1 \times \cdots \times \psi_L)$, i.e.

$$\widehat{\psi}((t_1, v_1), \ldots, (t_L, v_L)) = \psi_L(t_L, v_L) \circ \cdots \circ \psi_1(t_1, v_1).$$

Finally, define

$$\Psi : ((\mathbb{R}^n \times \mathfrak{so}_n)^L)^{D+1} \to \mathrm{Diff}^\infty(M)$$

by

$$\Psi(q_1, \ldots, q_{D+1}) = \widehat{\psi}(q_{D+1}) \circ \cdots \circ \widehat{\psi}(q_1).$$

For convenience, let $N = \left(2n + \binom{n}{2}\right)(D+1)L$, and fix some isomorphism $((\mathbb{R}^{2n} \times \mathfrak{so}_n)^L)^{D+1} \cong \mathbb{R}^N$. Then we have

Proposition 2.5.4. *The evaluation map*

$$ev_\Psi : \mathbb{R}^N \times M \longrightarrow M \times M$$

associated to Ψ is a C^∞ submersion. Moreover, if for all $0 \leq k \leq n$ we define

$$\widehat{ev}^k_\Psi : \mathbb{R}^n \times \mathrm{Gr}_k(TM) \longrightarrow \mathrm{Gr}_k(TM) \times \mathrm{Gr}_k(TM)$$
$$(w, \sigma) \longmapsto (\sigma, d(\Psi(w))(\sigma)),$$

then $\widehat{ev}^k_\Psi\big|_{\mathbb{B}^N_R \times \mathrm{Gr}_k(TM)}$ is surjective for all R sufficiently large.

Proof. The fact that ev_Ψ is a smooth submersion follows immediately from Lemma 2.5.2 and Proposition 2.5.3. The surjectivity of $\widehat{ev}^k_\Psi\big|_{\mathbb{B}^N_R \times \mathrm{Gr}_k(TM)}$ for $R \gg 0$ follows from the second half of Proposition 2.5.3 and the definition of D. □

Corollary 2.5.5. *The hypotheses of Theorem 2.1.1 are satisfied by the family $(\overline{\mathbb{B}}^N_R, \Psi)$.*

2.6 Application 1: Stochastic Artin-Mazur

As usual write $\mathrm{End}^r(M) = C^r(M, M)$ for the space of C^r maps from M to itself M endowed with the C^r topology. With this topology, $\mathrm{End}^r(M)$ is a Baire space.

For $f \in \mathrm{End}^r(M)$ we define

$$\mathrm{Per}_n f = \mathrm{Fix}(f^n) = \{x \in M \ : \ f^n(x) = x\},$$

the set of periodic points for f with period dividing n. Note that $\#\mathrm{Per}_n f < \infty$ for all n iff the fixed point set of f^n consists of isolated points for every n. By the Kupka-Smale theorem, this property is C^r-generic.

We say that f has *at most exponential growth of periodic points* if all periodic points of f are isolated, and moreover there exists $\lambda > 1$ such that

$$\#\mathrm{Per}_n f \leq \lambda^n$$

for all $n \in \mathbb{N}$. The classical Artin-Mazur theorem says that this property is dense:

Theorem 2.6.1 ([AM65])**.** *The set of $f : M \to M$ with at most exponential growth of periodic points is dense in* $\mathrm{End}^r(M)$*.*

Though obvious one should note that no such bound can hold for *every* $f \in \mathrm{End}^r(M)$ (example: $f = \mathrm{id}$). More interestingly, deep work by a number of authors has shown that the *density* in the Artin-Mazur theorem cannot be replaced by *generic*: if $\dim(M) \geq 2$ and $2 \leq r \leq \infty$, then there exist open $U \subset \mathrm{Diff}^r(M)$ such that the generic $f \in U$ exhibits arbitrarily fast growth of periodic points [Ber21].

Here we apply our main estimate to show that the picture becomes much simpler in the stochastic setting. For $\mathbf{f} = (f_1, f_2, f_3, \dots) \in \mathrm{Diff}^r(M)^{\mathbb{N}}$ we write

$$\mathrm{Per}_n \mathbf{f} = \{x \in M \ : \ f_n \circ \cdots \circ f_1(x) = x\}.$$

Then we have

Theorem 2.6.2. *For any compact submersive family $\psi : \mathcal{H} \to \mathrm{Diff}^r(M)$ normalized so that $\mathrm{vol}(\mathcal{H}) = 1$, the random walk with law $\mu = \psi_*(\mathrm{vol}_{\mathcal{H}})$ has at most exponential growth of periodic points. More precisely, there exists $\beta > 1$ such that for $\mu^{\mathbb{N}}$-a.e. $\mathbf{f} \in \mathrm{Diff}^r(M)$ we have*

$$\#\operatorname{Per}_n \mathbf{f} \leq \beta^n$$

for all sufficiently large n.

Proof. First, consider a submersive family $\psi : \mathcal{H} \to \mathrm{Diff}^r(M)$ with $\mathrm{id} \in \mathrm{im}(\psi)$. Note that the family $(\mathcal{H}^2, \psi \times \psi)$ is submersive on $M \times M$. Moreover, the set of graphs $\Gamma_f \subset M \times M$ of C^r diffeomorphisms of M is invariant under this family, since for $f \in \mathrm{Diff}^r(M)$ we have

$$\begin{aligned}(h_1, h_2)\Gamma_f &= (h_1, h_2)\{(x, fx) \ : \ x \in M\} \\ &= \{(h_1 x, h_2 f x) : x \in M\} \\ &= \{(y, h_2 f h_1^{-1} y) : y \in M\} = \Gamma_{h_2 f h_1^{-1}}.\end{aligned}$$

Let $C > 1$ be an upper bound intersection constant for the family $(\mathcal{H}^2, \psi \times \psi)$ in the sense of (and guaranteed by) Theorem 2.1.1. Then

$$\int_{\mathcal{H}^2} \#\operatorname{Fix}(h_2 f h_1^{-1}) d\mathcal{H}^2 = \int_{\mathcal{H}^2} \#[(h_1, h_2)\Gamma_f \cap \Delta] \ d\mathcal{H}^2 \leq \widehat{C}\,\mathrm{vol}(\Gamma_f),$$

where $\widehat{C} := C\,\mathrm{vol}(\Delta)$.

This essentially means that, after averaging along a rich class of random perturbations (namely $\{h_2 f h_1^{-1}\}_{(h_1,h_2)\in\mathcal{H}^2}$ for $\mathcal{H}$ a small submersive family about id), the number of fixed points of a diffeomorphism is controlled from above by the volume of its graph. This fact, combined with the observation that graph volumes are bounded from above by a polynomial function of C^1-norm (which is sub-multiplicative over composition) will the desired result.

In detail: Fix $(\mathcal{H}, \psi)$ a submersive family for M with $\mathrm{vol}(\mathcal{H}) = 1$. We study the random

walk on $\mathrm{Diff}^r(M)$ with law $\mu = \psi_* \operatorname{vol}_{\mathcal{H}}$.

As above observe that the family $(\mathcal{H}^2, (\mathrm{inv} \circ \psi) \times \psi)$ is submersive on $M \times M$ and preserves the class of graphs. Again we write C for the upper bound constant for this family given by Theorem 2.1.1. Then

$$\begin{aligned}\int_{(h_1,\dots,h_k)\in\mathcal{H}^k} &\#\operatorname{Fix}(h_k \circ \cdots \circ h_1)\; d\mathcal{H}^k \\ &= \int_{\mathcal{H}^{k-2}} \left(\int \#\left[(h_1^{-1}, h_k)\Gamma_{h_{k-1}\circ\cdots\circ h_2} \cap \Delta\right] d\mathcal{H}^2 \right) d\mathcal{H}^{k-2} \\ &\leq \widetilde{C} \int_{\mathcal{H}^{k-2}} \operatorname{vol}(\Gamma_{h_{k-1}\circ\cdots\circ h_2})\; d\mathcal{H}^{k-2} \leq \widetilde{C}\lambda^{k-2},\end{aligned}$$

where $\widetilde{C} := C \operatorname{vol}(\Delta) \operatorname{vol}(M)$ and $\lambda = (1 + \max_{(h,x)\in\mathcal{H}\times M} \|D_x h\|)^{\dim(M)}$.

For each $k \in \mathbb{N}$ and any $\beta > \lambda$, define

$$E_k^\beta := \{(h_i) \in \mathcal{H}^{\mathbb{N}} \;:\; \#\operatorname{Fix}(h_k \circ \cdots \circ h_1) \geq \beta^k\}.$$

The above implies that

$$\mu^{\mathbb{N}}(E_k^\beta) \leq \frac{\widetilde{C}\lambda^{k-2}}{\beta^k} = \widetilde{C}\lambda^{-2}\left(\frac{\lambda}{\beta}\right)^k.$$

In particular $\sum_{k\geq 1} \mu^{\mathbb{N}}(E_k^\beta) < \infty$. By the Borel-Cantelli lemma, for $\mu^{\mathbb{N}}$-a.e. $(h_i) \in \mathcal{H}^{\mathbb{N}}$ we have $\#\operatorname{Fix}(h_k \circ \cdots \circ h_1) < \beta^k$ for all but finitely many k, as desired. □

Remark. If we choose the submersive family to be concentrated arbitrarily close to $g \in \mathrm{Diff}^1(M)$, we can view the measure $\mu = \psi_* \operatorname{vol}_{\mathcal{H}}$ as defining random perturbations of g. From this perspective one can interpret Theorem 2.6.2 as a kind of stochastic Artin-Mazur bound: in the presence of sufficiently rich random noise, *every* C^r diffeomorphism has at most exponential growth of periodic orbits. Moreover this fact is an essentially immediate consequence of Theorem 2.1.1 the fact that, for $\mathcal{H} \subset \mathrm{Diff}^r(M)$ any C^1-bounded set (a fortiori for a compact C^r family), there exists $\alpha > 1$ such that $\operatorname{vol}(\Gamma_{h_k\circ\cdots\circ h_1}) \leq \alpha^k$ for all $h_i \in \mathcal{H}$.

Remark. The proof shows that the constant β realizing

$$\#\operatorname{Per}_n \mathbf{f} \leq \beta^n$$

asymptotically for $\mu^{\mathbb{N}}$-a.e. $\mathbf{f}$ can be chosen arbitrarily close (from above) to

$$\lambda = \left(1 + \max_{(h,x)\in\mathcal{H}\times M} \|D_x h\|\right)^{\dim(M)}.$$

This latter quantity, meanwhile, converges to

$$\lambda_0 = (1 + \|Df\|_\infty)^{\dim(M)}$$

as $\operatorname{diam}_{C^1}(\mathcal{H}) \to 0$ within the class of families containing f. Thus our result also gives a quantitative (though far from sharp) upper bound for the exponential periodic point growth rate for any diffeomorphism in the presence of sufficiently rich (i.e. submersive) noise as above: up to a small error that goes to zero with the perturbation size, this exponential rate is bounded from above by a polynomial function of $\|f\|_{C^1}$ of degree $\dim(M)$.

2.7 Application 2: Uniformly expanding measures

The notion of a *uniformly expanding* probability measure μ on $\operatorname{Diff}^r_{\text{vol}}(M)$ was introduced in the study of measure rigidity for smooth actions on manifolds. In brief it requires uniform $\mu^{(n)}$-average expansion along all tangent directions (or more generally tangent planes) for some finite n. This hypothesis turns out to have strong consequences for the classification of μ-stationary measures on M ([Chu20], [BEFH25], [Fil]).

Definition 2.7.1 ([Chu20], [Pot21], [ES23])**.** A Borel probability measure on $\operatorname{Diff}^r(M)$ is called *uniformly expanding* if there exists $N \in \mathbb{N}$ such that for all $v \in T^1M$ we have

$$\int \log \|D_x f(v)\| \, d\mu^{(N)}(f) > 2.$$

More generally, we say μ is *uniformly expanding in dimension* k ($k \in \{1, \dots, d-1\}$) if there exists $N \in \mathbb{N}$ such that

$$\int \log \left\| \Lambda^k Df \big|_\sigma \right\| d\mu^{(N)}(f) > 2$$

for all $\sigma \in \mathrm{Gr}_k(TM)$. Finally μ is called *uniformly expanding in all dimensions* if it is uniformly expanding in dimension k for all $1 \leq k \leq d-1$.

In other words, μ is *uniformly expanding in all dimensions* if the associated random walk on M expands length/volume of all tangent vectors/subspaces *on average* by a definite uniform amount in some uniform finite time.

The qualifier *on average* here is obviously essential, since no specific sample path can expand all tangent vectors at once. Relatedly we emphasize that this definition only becomes interesting/non-vacuous in the genuinely random setting, when $\#(\mathrm{supp}(\mu)) > 1$. In the deterministic case $\#(\mathrm{supp}(\mu)) = 1$, i.e. when studying the dynamics of a single map, this definition reduces to the classical one of *expanding maps* [Shu69], which cannot be diffeomorphisms.

Given the well-known topological rigidity of their deterministic cousins (see e.g. [Shu69], [Fra70], [Gro81]), it is perhaps surprising that uniformly expanding measures exist in abundance, on all manifolds M. Specifically, Potrie in [Pot21] showed that every neighborhood $U \subset \mathrm{Diff}^\infty_{\mathrm{vol}}(\mathbb{T}^2)$ contains the support of some uniformly expanding and finitely supported measure, and later Elliott Smith [ES23] generalized this result to all dimensions/manifolds, moreover obtaining measures with the stronger property of uniform expansion in all dimensions.

The rather modest goal of this section is to relate the class of C^1 families considered earlier in this chapter to these realization results, by showing that compact, volume-preserving families $\mathcal{H}$ as in Theorem 2.1.1 (with a mild but essential additional assumption) naturally produce uniformly expanding measures on $\mathrm{Diff}^\infty_{\mathrm{vol}}(M)$.

Special thanks are due to Rafael Potrie for pointing out the similarities between our construction and Elliott Smith's, and encouraging us to explore the relevance of our class of families to the study of uniform expansion.

Theorem 2.7.2. *Suppose $\psi : \mathcal{H} \to \mathrm{Diff}^1_{\mathrm{vol}}(M)$ is a compact C^1 family of diffeomorphisms of M such that*

(UE1) *$ev : \mathcal{H} \times M \to M \times M$ defined by $ev(h, p) = (p, h(p))$ is a submersion;*

(UE2) *ev is surjective, and for all $1 \leq k \leq d-1$, $\sigma \in \mathrm{Gr}_k(TM)$, and $y \in M$, $\{Dh(\sigma) \ : \ h \in \mathcal{H}\}$ intersects $\mathrm{Gr}_k(T_yM)$ in a Zariski dense set;*

(UE3) *For some $h \in \mathcal{H}$ and $x \in M$ we have $h(x) = x$ and $\mathrm{sp}(Dh_x) > 1$.*

Then the measure $\mu = \psi_*(\mathrm{vol}_{\mathcal{H}})$ is uniformly expanding in all dimensions.

Remark. The reference to Zariski density in (UE2) is well-defined via the Plücker embedding, which realizes every $\mathrm{Gr}_k(T_xM)$ as a subvariety of $\mathbb{P}(\Lambda^k T_xM)$. With minimal sacrifice in generality one could replace this with the condition that for all k, $\sigma \in \mathrm{Gr}_k(TM)$, and $y \in M$, $\mathcal{H}(\sigma_x) \cap \mathrm{Gr}_k(T_yM)$ has non-empty interior in $\mathrm{Gr}_k(T_yM)$. However, the sharper algebraic condition is most natural from the perspective of Proposition 2.7.3, on which the proof depends.

Remark. An obvious but important point is that μ is uniformly expanding in all dimensions if and only if $\mu^{(n)}$ is uniformly expanding in all dimensions for some $n \in \mathbb{N}$. Then if $(\mathcal{H}, \psi)$ is a compact, volume-preserving family such that the family $(\mathcal{H}^n, \mathrm{comp} \circ \psi^n)$ satisfies (UE1-3), by Theorem 2.7.2 the measure

$$(\mathrm{comp} \circ \psi^n)_*(\mathrm{vol}_{\mathcal{H}^n}) = (\mathrm{comp} \circ \psi^n)_*(\mathrm{vol}^n_{\mathcal{H}}) = \mathrm{comp}_*(\mu \times \cdots \times \mu) = \mu * \cdots * \mu =: \mu^{(n)}$$

is uniformly expanding in all dimensions, and thus μ is as well. In other words, the theorem remains true requiring only that (UE1-3) hold for $\mathcal{H}$ possibly after finitely many compositions with itself.

This point is significant for the realization question because, while (UE2) (specifically its surjectivity requirement) forces $\psi(\mathcal{H})$ to have large diameter in $\mathrm{Diff}^1_{\mathrm{vol}}(M)$ and is thus unachievable for perturbative families, it is easy to construct submersive, perturbative families $(\mathcal{H}, \psi)$ concentrated arbitrarily close to any $g \in \mathrm{Diff}^1_{\mathrm{vol}}(M)$ which satisfy (UE2) after finitely many iterations, i.e. such that $(\mathcal{H}^n, \mathrm{comp} \circ \psi^n)$ satisfies (UE2) for some n.

Finally we note that our preference for assumptions realizable by compositions of perturbative families near arbitrary g also accounts for the replacement of (A2) from Theorem 2.1.1 by (UE2) here. While (A2) clearly implies (UE2) and is thus (more than) sufficient for this Theorem 2.7.2, it is not always realizable by compositions of perturbative families, i.e. there exist $g \in \mathrm{Diff}^1_{\mathrm{vol}}(M)$ such that no $\mathcal{H}$ concentrated close to g can satisfy (A2) even up to arbitrary iterations. For example if g is Anosov, then there is a proper cone field $C \subset TM$ containing E^u_g which is forward-invariant under Df for all f C^1-close to g, in the sense that $Df(C(x)) \subset C(fx)$ for nearby f. It follows immediately that no family concentrated near g can (generate a semigroup) act(ing) transitively on $\mathbb{P}(TM) = \mathrm{Gr}_1(TM)$.

Remark. Though we omit the proof, it is not hard to see using a modification of our construction in §2.5 to the volume-preserving setting that families satisfying (UE1-3) after finitely many compositions exist and moreover can be concentrated arbitrarily close to any $g \in \mathrm{Diff}^1_{\mathrm{vol}}(M)$ (see also the construction in [ES23]).

The proof of Theorem 2.7.2 follows the same basic strategy for proving uniform expansion established in [Pot21] and [ES23]. The key result is the following, which relates uniform expansion to the non-existence of certain invariant, measurable tangent bundle structures.

Proposition 2.7.3 ([ES23], Theorem 3.2)**.** *If a probability measure μ on $\mathrm{Diff}^1_{\mathrm{vol}}(M)$ is not*

uniformly expanding in all dimensions, then there is an ergodic μ-stationary measure ν on M which admits either *a proper μ-a.e. invariant ν-measurable algebraic structure in $\mathrm{Gr}_k(TM)$, for some $1 \leq k \leq d-1$,* or *a μ-a.e. invariant ν-measurable conformal structure on TM.*

Remark. This result follows from the *invariance principle* of Ledrappier and later Avila and Viana. In [ES23] the result is stated and proved in the C^∞ setting, but the proof goes through unchanged for C^1 measures.

In short, Proposition 2.7.3 means that one can establish uniform expansion for μ by ruling out the existence of μ-stationary measures supporting certain invariant tangent structures. Moreover, by the following lemmas, assuming (UE1) alone it suffices to rule out such invariant structures for Leb_M. First, almost trivially, we have

Lemma 2.7.4. *If $\psi : \mathcal{H} \to \mathrm{Diff}^1(M)$ is a submersive family and $\mu = \psi_*(\mathrm{vol}_{\mathcal{H}})$, then for ν any Borel probability measure on M we have $\mu * \nu \ll \mathrm{Leb}_M$. In particular, all μ-stationary measures ν are absolutely continuous with respect to Leb_M.*

And moreover in the conservative case:

Lemma 2.7.5. *If $\psi : \mathcal{H} \to \mathrm{Diff}^1_{\mathrm{vol}}(M)$ is a submersive family and $\mu = \psi_*(\mathrm{vol}_{\mathcal{H}})$, then every μ-stationary measure is equivalent to Leb_M.*

Proof of Lemma 2.7.4. Obvious. □

Proof of Lemma 2.7.5. By Lemma 2.7.4, $\nu \ll \mathrm{Leb}_M$, so it suffices to show $\mathrm{Leb}_M \ll \nu$. Suppose not, so there exist Borel $B \subset M$ such that $\nu(B) = 0$ but $\mathrm{Leb}_M(B) > 0$. Let $E \subset M$ be the "largest" such subset, in the sense that $\mathrm{Leb}_M(E) > 0$ and $\nu(E) = 0$, *and* if $\widehat{E} \subset M$ also satisfies these properties then $\mathrm{Leb}_M(\widehat{E} \setminus E) = 0$. It is easy to check that E is uniquely defined up to sets of Lebesgue measure zero, and that its maximality property—combined with μ-stationarity of ν and the μ-invariance of Leb_M—implies that E is μ-a.e.

Leb_M-essentially invariant, in the sense that $\mathrm{Leb}_M(fE \,\Delta\, E) = 0$ for μ-a.e. $f \in \mathrm{Diff}^1_{\mathrm{vol}}(M)$. By Fubini one gets that $\mathrm{ev}(V) = E$ mod Leb_M-measure 0 for V the full-measure subset of $\mathcal{H} \times E$ defined $V = \{(h, x) \ : \ \mathrm{Leb}_M(hE\Delta E) = 0,\ x \in h^{-1}(E) \cap E\}$. On the other hand, basic properties of submersions imply that $\mathrm{ev}(V)$ also has full Lebesgue measure in $N_\varepsilon(E)$ for some $\varepsilon > 0$. Since $\mathrm{Leb}(E^c) > 0$ (proof: ν is a positive measure and equivalent to Leb_M on E^c), contradiction. Thus $\mathrm{Leb}_M \ll \nu$ everywhere on M, as desired. □

Proof of Theorem 2.7.2. Let $(\mathcal{H}, \psi)$ be a compact C^1 volume-preserving family satisfying (UE1-3). We want to show that $\mu = \psi_*(\mathrm{vol}_{\mathcal{H}})$ is uniformly expanding in all dimensions. By Proposition 2.7.3 and Lemma 2.7.5, it suffices to check that no proper Leb-measurable algebraic structures in $\mathrm{Gr}_k(TM)$ or Leb-measurable conformal structures on TM are μ-a.e. invariant.

Fix $1 \leq k \leq d-1$, and consider E a proper Leb-measurable algebraic structure in $\mathrm{Gr}_k(TM)$, which for our purposes amounts to a Leb-measurable bundle section $E : M \to \mathrm{Haus}(\mathrm{Gr}_k(TM))$ such that $E(x)$ is a proper (thus positive codimension) algebraic subset of $\mathrm{Gr}_k(T_xM)$ for Leb_M-a.e. $x \in M$. We show that E cannot be μ-a.e. invariant, or equivalently that the set of $(f, x) \in \mathrm{Diff}^1_{\mathrm{vol}}(M) \times M$ such that

$$Df(E(x)) \neq E(f(x))$$

has $(\mu \times \mathrm{Leb}_M)$-measure positive.[13]

Indeed, by Lusin's theorem we can pick compact $K \subset M$ with $\mathrm{Leb}(K) > 0$ such that $E|_K$ is continuous. Fix $x \in K$ a Lebesgue density point for K and pick $\sigma_0 \in E(x)$. By (UE2), we can find $h \in \mathcal{H}$ such that $h(x) = x$ and $Dh(\sigma_0) \notin E(x)$. In particular, for this x and choice of h we have

$$Dh(E(x)) \neq E(hx).$$

[13]In fact algebraicity here is irrelevant; we mention it only for coherence with the literature. Our proof gives the stronger claim that no proper Leb-measurable bundle section $E : M \to \mathrm{Haus}(\mathrm{Gr}_k(TM))$ (i.e. $E(x) \subsetneq \mathrm{Gr}_k(T_xM)$ for Leb-a.e. $x \in M$) can be μ-a.e. invariant.

This is an open condition, so by continuity of $E|_K$ we have

$$Dg(E(y)) \neq E(gy)$$

for all g close to h and $y \in K \cap g^{-1}K \cap B_\delta(x)$ with δ small. On the other hand, since h is a diffeomorphism, $x = h^{-1}(x)$ is also a density point for $h^{-1}K$, and thus for $K \cap h^{-1}K$. It follows that $\mathrm{Leb}_M(K \cap g^{-1}K \cap B_\delta(x)) > 0$ for all $g \in U$ a small neighborhood of h, and thus

$$(\mathrm{vol}_{\mathcal{H}} \times \mathrm{Leb}_M)\left(\{(g,y) \ : \ g \in U, y \in K \cap g^{-1}K \cap B_\delta(x)\}\right) > 0.$$

Since $Dg(E(y)) \neq E(g(y))$ for all (g,y) in this set, we are done.

Finally, consider $\mathcal{C} : M \to \mathrm{CS}(TM)$ a Leb-measurable conformal structure on M. It suffices to show that $\mathcal{C}$ cannot be μ-a.e. invariant, equivalently that for some $n \in \mathbb{N}$ we have

$$(\mu^{(n)} \times \mathrm{Leb}_M)\left(\{(f,x) \in \mathcal{H} \times M \ : \ df(\mathcal{C}(x)) \neq \mathcal{C}(fx)\}\right) > 0.$$

Again we pick $K \subset M$ compact with $\mathrm{Leb}_M(K) > 0$ and $\mathcal{C}|_K$ continuous. As in the preceding paragraph it suffices to find x a density point of K and $f = h_1 \circ \cdots \circ h_n$, $h_i \in \mathcal{H}$, such that $Df(\mathcal{C}(x)) \neq \mathcal{C}(fx)$, since then $Dg(\mathcal{C}(y)) \neq \mathcal{C}(gy)$ for all g near f and $y \in K \cap g^{-1}K \cap B_\delta(x)$, and it is easy to see that this set has positive $(\mu^{(n)} \times \mathrm{Leb}_M)$-measure.

So let x be a Lebesgue density point for K. By (UE3) we can find $h \in \mathcal{H}$ and $y \in M$ such that $h(y) = y$ with $\mathrm{sp}(Dh_y) > 1$, and by (UE2) there also exist $f, g \in \mathcal{H}$ such that $fx = y$ and $gy = x$. The condition on $\mathrm{sp}(Dh_x)$ implies that $\|D(g \circ h^m \circ f)_x\| \to \infty$ as $m \to \infty$; since $\det(D(g \circ h^m \circ f)_x) = 1$ this forces also $m(D(g \circ h^m \circ f)_x) = \|D(g \circ h^m \circ f)_x^{-1}\|^{-1} \to 0$, so for some $m \in \mathbb{N}$ we must have

$$D(g \circ h^m \circ f)(\mathcal{C}(x)) \neq \mathcal{C}(g \circ h^m \circ f(x))$$

for some $m \in \mathbb{N}$. Applying the argument above we conclude

$$(\mu^{(m+2)} \times \mathrm{Leb}_M)\left(\left\{(f,x) \in \mathrm{Diff}^1_{\mathrm{vol}}(M) \times M \ : \ df(\mathcal{C}(x)) \neq \mathcal{C}(fx)\right\}\right) > 0,$$

as desired. □

Chapter 3

Domination and Bowen disks

In this chapter we consider the following question of Shub's:

Question 3.0.1. *Let M be a closed Riemannian manifold of dimension d, $f \in \mathrm{Diff}^2(M)$, and $1 \leq k < d$. Is there a sequence $(\varepsilon_n) \subset \mathbb{R}_{>0}$, with $\lim_{n\to\infty} \frac{\log(\varepsilon_n)}{n} = 0$, such that for all $x \in M$, there exists a k-disk*

$$D_x = \mathrm{graph}(\sigma)$$

with $\sigma : E^{u,k}_{x,n}(\varepsilon_n) \to [E^{u,k}_{x,n}(\varepsilon_n)]^{\perp}, \sigma(0) = 0$, and $\mathrm{Lip}(\sigma) \leq 1$, for $E^{u,k}_{x,n}$ the k-unstable subspace for some singular value decomposition of Df^n_x, such that for all $1 \leq j \leq n$ we have

$$f^j(D_x \cap B_f(x, n, \varepsilon_n)) \subset \mathrm{graph}(\sigma_j)$$

for some $\sigma_j : Df^j_x(E^{u,k}_{x,n}) \to [Df^j_x(E^{u,k}_{x,n})]^{\perp}$ satisfying $\mathrm{Lip}(\sigma_j) \leq 1$?

3.1 Background

We define an (n, ε)-*Bowen k-disk* to be any subset of the form

$$D_{n,\varepsilon} = D \cap B_f(x, n, \varepsilon)$$

where $D \subset M$ is an embedded k-disk and $x \in D$. Very generally we are interested in finding assumptions which allow one to control the geometry of $f^n D_{n,\varepsilon}$ for an (n,ε)-Bowen disk $D_{n,\varepsilon}$. Classically, such control is obtainable using some degree of hyperbolicity, or domination more generally. The following toy result illustrates this method and its consequences.

Proposition 3.1.1. *Let $f \in \mathrm{Diff}^1(M)$, and suppose f admits a global dominated splitting $M = E^u \oplus E^s$, with $\dim(E_u) = k$. Then for every embedded k-disk $D \subset M$ transverse to E^s we have*

$$h(f) \geq \limsup_{n\to\infty} \frac{1}{n} \log \mathrm{vol}(f^n D).$$

Proof. Using the graph transform and uniform domination it is easy to check that, for $\varepsilon > 0$ sufficiently small, all $n \in \mathbb{N}$ large, and all $x \in D$, we have

$$f^n(D \cap B_f(x,n,\varepsilon)) \subset \mathrm{graph}(\sigma_n)$$

for some $\sigma_n : E^u_{f^n x}(\varepsilon) \to E^s_{f^n x}$ with $\sigma_n(0) = 0$ and $\mathrm{Lip}(\sigma_n) \leq 1$. In particular there exists $C = C(\varepsilon) > 0$ such that for all $x \in D$ and $n \in \mathbb{N}$,

$$\mathrm{vol}\left(f^n(D \cap B_f(x,n,\varepsilon))\right) \leq C\varepsilon^k.$$

Then if $X^D_{n,\varepsilon} \subset D$ is some minimal (n,ε)-spanning set we get

$$\begin{aligned}
\mathrm{vol}(f^n D) &= \mathrm{vol}\left(\bigcup_{x \in X^D_{n,\varepsilon}} f^n(D \cap B_f(x,n,\varepsilon))\right) \\
&\leq \sum_{x \in X^D_{n,\varepsilon}} \mathrm{vol}\left(f^n(D \cap B_f(x,n\varepsilon))\right) \\
&\leq \#(X^D_{n,\varepsilon}) \cdot C\varepsilon^k,
\end{aligned}$$

so that

$$\limsup_{n\to\infty} \frac{1}{n} \log \mathrm{vol}(f^n D) \leq \lim_{\varepsilon\to 0} \limsup_{n\to\infty} \frac{1}{n} \log \#(X^D_{n,\varepsilon}) = h(f,D,\varepsilon) \leq h(f,D) \leq h(f),$$

as desired. □

Very roughly, Question 3.0.1 is motivated by a desire to generalize this method to non-hyperbolic settings.

3.2 Local version

We fix a dimension $d > 1$, $1 \leq k < d$, and any (real) constant $K > 1$. Given this data we will write $B_1(0) \subset \mathbb{R}^d$ for the open unit ball centered at 0 and

$$\mathbb{R}^d = \mathbb{R}^k \oplus \mathbb{R}^{d-k} =: E \oplus F.$$

The class of maps whose compositions we consider is given by

$$\begin{aligned}\mathcal{D}_K = \{f : B_1(0) \to \mathbb{R}^d \ : f \text{ is a diffeomorphism onto its image, } f(0) = 0,\\ Df_0 \text{ preserves the splitting } E \oplus F, \text{ and } \|f\|_{C^2} \leq K\},\end{aligned}$$

where

$$\|f\|_{C^2} = \sup_{x \in B_1(0)} \max\{\|Df_x\|, \left\|D(f^{-1})_{f(x)}\right\|, \left\|D^2 f_x\right\|, \left\|D^2(f^{-1})_{f(x)}\right\|\}.$$

In the natural initial approach to Question 3.0.1, the key assumption on M, compactness, is used only to ensure that the local coordinate expressions of f have uniformly bounded C^2 norm. Thus one is led to consider the following purely local version of this question. Indeed here we consider an even weaker statement, which assumes that each Df_x^j preserves the orthogonality of the splitting $E^{u,k}_{x,n} \oplus [E^{u,k}_{x,n}]^\perp$, and hopes for flatness only at times 0 and n, not necessarily at intermediate j. Namely we ask

Question 3.2.1. *Is there a sequence* $(\varepsilon_n) \in (0,1)^{\mathbb{N}}$ *with* $\lim_{n\to\infty} \dfrac{\log(\varepsilon_n)}{n} = 0$ *such that for all*

$\mathbf{f} = (f_1, \ldots, f_n) \in (\mathcal{D}_K)^n$ *satisfying*

$$\frac{m(D(f_n \circ \cdots \circ f_1)_0|_E)}{\|D(f_n \circ \cdots \circ f_1)_0|_F\|} \geq 1, \tag{3.1}$$

there is a k*-disk*

$$D = \text{graph}(\sigma : E \to F)$$

with $\sigma(0) = 0, \text{Lip}(\sigma) \leq 1$*, so that*

$$f_n \circ \cdots \circ f_1(D \cap B_{\mathbf{f}}(0, n, \varepsilon_n)) \subset \text{graph}(\widehat{\sigma})$$

for some $\widehat{\sigma} : E \to F$ *with* $\text{Lip}(\widehat{\sigma}) \leq 1$*?*

The set $B_{\mathbf{f}}(0, n, \delta)$ is defined in the obvious way, namely $x \in B_{\mathbf{f}}(0, n, \delta)$ iff $d(f_j \circ \cdots \circ f_1(x), 0) < \delta$ for all $0 \leq j \leq n-1$. Our definition of $\mathcal{D}_K$ implies that the composition $f_n \circ \cdots \circ f_1$ is only well-defined on $B_{\mathbf{f}}(0, n, 1)$; this accounts for the (unproblematic) additional assumption that $0 < \varepsilon_n < 1$ for all n.

If $\mathbf{f} = (f_1, \ldots, f_n) \in (\mathcal{D}_K)^n$ satisfies Condition (3.1) we will say that the splitting $E \oplus F = T_0\mathbb{R}^2$ is *weakly dominated* for $\mathbf{f}$. Such a sequence will be called *forward Pliss* if moreover $E \oplus F$ is weakly dominated for $(f_1, \ldots, f_j)$ for all $1 \leq j \leq n$, and *backward Pliss* if $E \oplus F$ is weakly dominated for $(f_j, \ldots, f_n)$ for all $1 \leq j \leq n$. In various attempts to answer Question 3.2.1 positively it has seemed useful to add the assumption that $\mathbf{f}$ is forward or backward Pliss (or both).

Our main result gives a negative answer to Question 3.2.1, even under the additional assumption that all sequences considered are both forward- and backward- Pliss.

The construction we give moreover appears flexible enough to bypass most of the natural extra assumptions one might attempt to impose (for instance, replacing weak domination with α-domination, $1 < \alpha < K^2$; see Section 3.5), and it appears unlikely that any extra assumptions weaker than a global dominated splitting are sufficient to imply a positive

answer to Question 3.0.1. Indeed, in the final section we substantiate this claim with a result characterizing controlled behavior of graphs under compositions of C^2 perturbations of linear maps in terms of uniform domination of the linear part.

3.3 Main result

From now on we assume $d = 2$ (so $k = 1$). It is straightforward to adapt this example to arbitrary $d, k \in \mathbb{N}$ with $1 \leq k < d$.

In Question 3.2.1, we are trying to bound the rate of decay, as a function of n, of

$$\begin{aligned}\varepsilon_n := \sup\{\varepsilon > 0 \ : &\forall f_1, \dots, f_n \in \mathcal{D}_K \text{ s.t. } E \oplus F \text{ is weakly dominated for } f_n \circ \cdots \circ f_1, \\ &\exists\ \sigma, \widehat{\sigma} : E \to F \text{ with } \max(\mathrm{Lip}(\sigma), \mathrm{Lip}(\widehat{\sigma})) \leq 1 \text{ and } \sigma(0) = \widehat{\sigma}(0) = 0, \\ &\text{s.t. } f_n \circ \cdots \circ f_1(\mathrm{graph}(\sigma) \cap B_{\mathbf{f}}(0, n, \varepsilon)) \subset \mathrm{graph}(\widehat{\sigma})\}.\end{aligned}$$

Specifically, the claim of the conjecture is that ε_n decays subexponentially in n.

To estimate the decay rate of ε_n, it turns out to be easier to bound the *growth* rate of a sort of "inverse" quantity. Namely, for $0 < \varepsilon < 1$ and $K > 1$ fixed as before, define

$$\begin{aligned}N(\varepsilon) = \min\{n \in \mathbb{N} \ : &\exists f_1, \dots, f_n \in \mathcal{D}_K \text{ s.t. } E \oplus F \text{ is weakly dominated for } f_n \circ \cdots \circ f_1, \\ &\text{and } \forall\ \sigma : E \to F \text{ with } \mathrm{Lip}(\sigma) \leq 1 \text{ and } \sigma(0) = 0, \\ &f_n \circ \cdots \circ f_1(\mathrm{graph}(\sigma) \cap B_{\mathbf{f}}(0, n, \varepsilon)) \text{ is not a graph over } E\}.\end{aligned}$$

Then our main result is

Proposition 3.3.1. *There are constants $C_1, C_2 > 0$, depending only on K, such that for all $0 < \varepsilon < 1$ we have*

$$N(\varepsilon) \leq C_1 \log\left(\frac{1}{\varepsilon}\right) + C_2.$$

This immediately implies

Corollary 3.3.2. *For any $K > 1$ fixed and ε_n defined as above we have*

$$\limsup_{n\to\infty} \frac{\log(\varepsilon_n)}{n} < 0.$$

Proof of corollary. It is immediate from the definitions that

$$N(\varepsilon_n) > n.$$

Then by Proposition 3.3.1 we have

$$n < -C_1 \log(\varepsilon_n) + C_2$$

or

$$\frac{\log(\varepsilon_n)}{n} < -\frac{1}{C_1} + \frac{C_2}{nC_1}$$

for all $n \in \mathbb{N}$. Taking $n \to \infty$, the result follows. □

3.4 Construction of example

Fix $0 < \varepsilon < 1$. We will use the maps

- $A = \begin{bmatrix} K^{-1} & 0 \\ 0 & 1 \end{bmatrix}$
- $B = \begin{bmatrix} 1 & 0 \\ 0 & K^{-1} \end{bmatrix}$
- $f(x, y) = (x, y + Cx^2)$
- $g(x, y) = (x - Cy^2, y)$.

Here we take $0 < C < 1$ sufficiently small so that $\left\|f|_{B_1(0)}\right\|_{C^2} = \left\|g|_{B_1(0)}\right\|_{C^2} \leq K$, say $C = \min\{1/2, (\sqrt{K}-1)/2\}$. Thus $f, g, A, B \in \mathcal{D}_K$. We will study the behavior of 1-Lipschitz graphs under compositions of the form

$$B^m g A^m f B^m,$$

with $m \in \mathbb{N}$ to be determined later. For notational convenience, we write $\mathbf{f} = (f_1, \ldots, f_{3m+2})$ where $f_1 = \cdots = f_m = B$, $f_{m+1} = f$, $f_{m+2} = \cdots = f_{2m+1} = A$, $f_{2m+2} = g$, and $f_{2m+3} = \cdots = f_{3m+2} = B$.

We first consider the special case when $\sigma = \sigma_0 : E \to F$ is the constant zero function. It is straightforward to check that $(t, 0) \in B_{\mathbf{f}}(0, 3m+2, \varepsilon)$ provided $|t| \leq \frac{\varepsilon}{2}$. Define

$$\gamma = \{(t, 0) : 0 \leq t \leq \varepsilon/2\} \subset \operatorname{graph}(\sigma_0) \cap B_{\mathbf{f}}(0, 3m+2, \varepsilon),$$

and $p = (0, \varepsilon/2)$ the right endpoint of γ. Clearly

$$B^m(p) = p, \;\; fB^m(p) = \left(\frac{\varepsilon}{2}, \frac{C\varepsilon^2}{4}\right), \;\; A^m f B^m(p) = \left(\frac{K^{-m}\varepsilon}{2}, \frac{C\varepsilon^2}{4}\right),$$
$$\text{and } gA^m f B^m(p) = \left(\frac{K^{-m}\varepsilon}{2} - \frac{C^3\varepsilon^4}{16}, \frac{C\varepsilon^2}{4}\right).$$

Then it is straightforward to check that

$$\frac{K^{-m}\varepsilon}{2} - \frac{C^3\varepsilon^4}{16} < 0$$

provided

$$m > \frac{3}{\log K} \log\left(\frac{1}{\varepsilon}\right) + \frac{3\log(2) - 3\log(C)}{\log K}. \tag{3.2}$$

Moreover, since γ is tangent to the x-axis at 0, and $D(gA^m f B^m)_0(\mathbf{e}_1) = K^{-m}\mathbf{e}_1$, some initial segment of $gA^m f B^m(\gamma)$ must lie in the plane $\{x > 0\}$. Thus if m satisfies (3.2), $gA^m f B^m(\gamma)$ must cross the y-axis at some point above the origin. Since B^m preserves the positive y-axis

and the origin, clearly the same is true of

$$\begin{aligned} B^m g A^m f B^m(\gamma) &\subset B^m g A^m f B^m(\text{graph}(\sigma_0) \cap B_{\mathbf{f}}(0, 3m+2, \varepsilon)) \\ &= f_{3m+2} \circ \cdots f_1(\text{graph}(\sigma_0) \cap B_{\mathbf{f}}(0, 3m+2, \varepsilon)). \end{aligned}$$

Since this curve also contains the origin, it cannot be a graph over E.

For a general 1-Lipschitz function $\sigma : E \to F$ with $\sigma(0) = 0$, one repeats this argument with γ replaced by the curve

$$\widehat{\gamma} = \{(t, \sigma(t)) : 0 \leq t \leq \varepsilon/4\}.$$

Again it is not difficult to check that $\widehat{\gamma} \subset B_{\mathbf{f}}(0, 3m+2, \varepsilon)$, and the problem again reduces to tracking the orbit of the right endpoint $q = (\varepsilon/4, \sigma(\varepsilon/4))$. The main difference here is that the initial factor B^m is required to move q close enough to the x-axis that f can push $B^m(q)$ a distance proportional to ε^2 above. To see as before that some initial segment of $gA^m f B^m(\widehat{\gamma})$ lies in the half plane $R = \{x > 0\}$, it suffices to observe that $A^m f B^m(\widehat{\gamma})$ is contained in some open cone strictly contained in R, and $Dg_0 = I$. These modifications in mind, it is straightforward to check that, in general, the bound

$$m > \frac{3}{\log K} \log\left(\frac{1}{\varepsilon}\right) + \frac{8\log(2) - 3\log(C)}{\log K}. \tag{3.3}$$

suffices to ensure that $B^m g A^m f B^m(\widehat{\gamma})$ (and thus $f_{3m+2} \circ \cdots f_1(\text{graph}(\sigma) \cap B_{\mathbf{f}}(0, 3m+2, \varepsilon)))$ is not a graph over E.

Putting everything together, this example implies that

$$\begin{aligned} N(\varepsilon) &\leq 3\left(\frac{3}{\log K} \log\left(\frac{1}{\varepsilon}\right) + \frac{8\log(2) - 3\log(C)}{\log K} + 1\right) + 2 \\ &= \frac{9}{\log K} \log\left(\frac{1}{\varepsilon}\right) + \frac{24\log 2 - 9\log C + 5\log K}{\log K}. \end{aligned}$$

Since $C = \min\{1/2, (\sqrt{K} - 1)/2\}$ is a function of K, this proves Proposition 3.3.1.

3.5 Pliss domination

It is easy to check that the sequence $\mathbf{f}$ constructed above is both forward and backward Pliss. Moreover, it is not hard to see that Proposition 3.3.1 remains true even if weak domination in the definition of $N(\varepsilon)$ is replaced by some stricter domination condition on the composition. For instance, for $\alpha > 1$ we say that $E \oplus F$ is *α-dominated* for $\mathbf{f} = (f_1, \dots, f_n) \in (\mathcal{D}_K)^n$ if

$$\frac{m(D(f_n \circ \cdots \circ f_1)_0|_E)}{\|D(f_n \circ \cdots \circ f_1)_0|_F\|} \geq \alpha^n.$$

Such an $\mathbf{f}$ will be called *α-forward Pliss* if $E \oplus F$ is α-dominated for $(f_1, \dots, f_j)$ for all $1 \leq j \leq n$, and *α-backward Pliss* if $E \oplus F$ is α-dominated for $(f_j, \dots, f_n)$ for all $1 \leq j \leq n$. Finally, we say that $E \oplus F$ is *strongly α-dominated* for $\mathbf{f}$ if it is *α-dominated*, and $\mathbf{f}$ is both α-forward and α-backward Pliss.

Now for any $1 < \alpha < K^2$ we can define

$$\begin{aligned} N_\alpha(\varepsilon) := \min\{n \in \mathbb{N} \ : \exists f_1, \dots, f_n \in \mathcal{D}_K \text{ s.t. } & E \oplus F \text{ is strongly } \alpha\text{-dominated for } \mathbf{f} = (f_1, \dots, f_n), \\ & \text{and } \forall\, \sigma : E \to F \text{ with } \operatorname{Lip}(\sigma) \leq 1 \text{ and } \sigma(0) = 0, \\ & f_n \circ \cdots \circ f_1(\operatorname{graph}(\sigma) \cap B_{\mathbf{f}}(0, n, \varepsilon)) \text{ is not a graph over } E\}. \end{aligned}$$

Then we have

Proposition 3.5.1. *There are constants $C_{1,\alpha}, C_{2,\alpha} > 0$, depending only on K and α, such that for all $0 < \varepsilon < 1$ we have*

$$N_\alpha(\varepsilon) \leq C_{1,\alpha} \log\left(\frac{1}{\varepsilon}\right) + C_{2,\alpha}.$$

This can be proved using a modification of the earlier construction, namely by considering products of the form

$$\widehat{B}^{m+\ell} g A^m f \widehat{B}^{m+\ell},$$

where A, f, g are defined as above and

$$\widehat{B} = \begin{bmatrix} K & 0 \\ 0 & K^{-1} \end{bmatrix}.$$

One first shows as before that the non-graph condition can be obtained using m bounded by a logarithmic function of $1/\varepsilon$, then checks that strong α-domination can be achieved by taking ℓ a linear function of m, with constants depending only on K and α.

The assumption here that $\alpha < K^2$ is essential, because

$$\frac{m(D(f_n \circ \cdots \circ f_1)_0|_E)}{\|D(f_n \circ \cdots \circ f_1)_0|_F\|} \leq K^{2n}$$

for all $f_1, \ldots, f_n \in \mathcal{D}_K$, with equality iff

$$D(f_i)_0 = \begin{bmatrix} K & 0 \\ 0 & K^{-1} \end{bmatrix}$$

for all $1 \leq i \leq K$. In particular, in this case the splitting $E \oplus F$ is strictly dominated for *every* f_i, and in this restrictive setting, a positive answer to Question 3.2.1 follows immediately from the classical fact that the graph transform preserves the class of 1-Lipschitz graphs near hyperbolic fixed points.

3.6 Characterizing uniform domination

We conclude this chapter by sketching a slightly more systematic framework for our result. Suppose we have a sequence $\mathcal{A} = (A_i) \in GL(2, \mathbb{R})^{\mathbb{N}}$ of diagonal matrices

$$\begin{bmatrix} a_i & 0 \\ 0 & b_i \end{bmatrix}$$

with $\|A_i\|, \left\|A_i^{-1}\right\| \leq K$ which is non-uniformly hyperbolic in the sense that

$$\lim_{n\to\infty} \frac{1}{n} \log \|A_n \cdots A_1\| = \lambda^+ > 0, \tag{3.4}$$

$$\lim_{n\to\infty} \frac{1}{n} \log m(A_n \cdots A_1) = \lambda^- < 0. \tag{3.5}$$

Fix $C > 0$. Then we consider the class

$$\mathcal{D}(\mathcal{A}) = \{\mathbf{f} = (f_i) \in \mathrm{Diff}^2(\mathbb{R}^2, 0)^{\mathbb{N}} \ : \ \forall i \in \mathbb{N} \ D_0 f_i = A_i, \text{ and } \|A_i - f_i\|_{C^2} \leq C\}$$

of C^2 perturbations of the sequence (A_i) preserving the derivative at 0. The question we consider is: For which sequences $\mathcal{A}$ can we guarantee that all perturbations $\mathbf{f} \in \mathcal{D}(\mathcal{A})$ admit flat Bowen disks of a definite size $\varepsilon > 0$ at all times?

More precisely, for $\mathbf{f} \in \mathcal{D}(\mathcal{A})$ define

$$\varepsilon_n(\mathbf{f}) = \sup\{\varepsilon > 0 \ : \exists \ \sigma : E^u \to E^s, \mathrm{Lip}(\sigma) \leq 1, \text{ s.t.}$$
$$\mathbf{f}^n(\mathrm{graph}(\sigma) \cap B_{\mathbf{f}}(0, n, \varepsilon)) \subset \mathrm{graph}(\widehat{\sigma}), \widehat{\sigma} : E^u \to E^s, \mathrm{Lip}(\widehat{\sigma}) \leq 1\},$$

where $\mathbf{f}^n := f_n \circ \cdots \circ f_1$. We say that $\mathcal{A}$ is *uniformly dominated* if there exists $k_0 \in \mathbb{N}$ and $\beta > 1$ such that for all $j \in \mathbb{N}$,

$$\frac{\|A_{j+k_0} \cdots A_{j+1}\|}{m(A_{j+k_0} \cdots A_{j+1})} \geq \beta.$$

Then we have

Proposition 3.6.1. *The sequence $\mathcal{A}$ is uniformly dominated if and only if for every $\mathbf{f} \in \mathcal{D}(\mathcal{A})$ there exists $\varepsilon_0 > 0$ such that $\varepsilon_n(\mathbf{f}) \geq \varepsilon_0$ for all n.*

The left-to-right implication is obvious by standard graph transform techniques. For the other direction, one assumes that $\mathcal{A}$ is *not* uniformly dominated. This means that there are arbitrarily long subsequences of $\mathcal{A}$ with domination arbitrarily close to (or even less than) one. In such stretches one can then use perturbations of the f_i as in the example above to the force every 1-Lipschitz to eventually have arbitrarily large slope, arbitrarily near 0.

Not addressed here is the interesting question of whether, in the case when $\mathcal{A}$ is non-uniformly dominated, one can always find $\mathbf{f} \in \mathcal{D}(\mathcal{A})$ such that $\varepsilon_n(\mathbf{f}) \to 0$ exponentially.

Chapter 4

Density subshifts and generalized entropy

Let X be a compact metric space and $f : X \to X$ a continuous map. In [CP23], Correa & Pujals define the *generalized entropy* (or *order of growth*) of f by

$$o(f) := \sup_{\varepsilon>0} [r_f(n,\varepsilon)],$$

where $r_f(n,\varepsilon)$ is the minimum number of (n,ε)-Bowen balls for f needed to cover X, and $[r_f(n,\varepsilon)]$ denotes the *order of growth* of the sequence $(r_f(n,\varepsilon))_{n=1}^{\infty}$. The space $\mathbb{O}$ of orders of growth is the set of equivalence classes of non-decreasing sequences $f : \mathbb{N} \to \mathbb{R}_{>0}$, where $f \sim g$ iff $1/C \leq f/g \leq C$ for some $C \geq 1$. The set $\mathbb{O}$ is naturally a poset, under the relation $f \leq g$ iff $f \leq Cg$ for some constant C; in order to consider sups/infs, we take its lattice completion, $\overline{\mathbb{O}}$. We occasionally call elements of $\overline{\mathbb{O}} \setminus \mathbb{O}$ *abstract* orders of growth—as opposed to the *concrete* orders of growth, $\mathbb{O}$—when the distinction requires emphasis. Note that, in general, $o(f) \in \overline{\mathbb{O}}$.

Now define

$$\mathbb{P} = \{[n^t] : t > 0\} \subset \mathbb{O}$$

and

$$\mathbb{E} = \{[e^{\alpha n}] : \alpha > 0\} \subset \mathbb{O},$$

the sets of polynomial and exponential orders of growth, respectively. It is not hard to see that, for all $\varepsilon > 0$, there exists $a = a(\varepsilon)$ such that $[r_f(n, \varepsilon)] \leq [e^{an}]$; thus we always have $o(f) \leq \sup \mathbb{E}$. Moreover, $o(f) = \sup \mathbb{E}$ iff $h_{\text{top}}(f) = \infty$ and $o(f) \leq \inf \mathbb{E}$ iff $h_{\text{top}}(f) = 0$, where $h_{\text{top}}(f)$ denotes the usual topological entropy of f.

Correa & Pujals give several examples of zero entropy dynamical systems with polynomial order of growth (by later work of Correa & de Paula, this class also includes Morse-Smale diffeomorphisms [Cd23]), and ask [CP23, p. 1591] whether systems f exist such that

$$\sup \mathbb{P} < o(f) < \inf \mathbb{E}. \tag{4.1}$$

The simple goal of this chapter is to answer this question in the affirmative, by describing a simple family of subshifts satisfying (4.1).

Remark. We became aware of Stanley's paper [Sta13] investigating a more general class of subshifts only at the end of the thesis-writing process. Though we have not had a chance to read this paper in detail, it presumably contains many of our results.

4.1 Density subshifts

Write $\Sigma = \Sigma_2 = \{0, 1\}^{\mathbb{Z}}$. We equip Σ with the product topology; this can be metrized by e.g.

$$d(\mathbf{x}, \mathbf{y}) = 2^{-\min\{|n| : x_n \neq y_n\}}.$$

Let $\sigma : \Sigma \to \Sigma$ denote the left shift map, i.e. $(\sigma(\mathbf{x}))_i = x_{i+1}$. A subshift is a closed, σ-invariant subset Λ of Σ.

Given a subshift Λ and $n \in \mathbb{N}$ we denote by $[\Lambda]_n$ the set of strings $a_1 a_2 \dots a_n \in \{0, 1\}^n$

which appear as substrings of elements of Λ. An elementary classical result says that

$$h_{\text{top}}(\sigma|_\Lambda) = \lim_{n\to\infty} \frac{1}{n} \log \#[\Lambda]_n.$$

In the context of generalized entropy, it is similarly straightforward to show that

$$o(\sigma|_\Lambda) = [\#[\Lambda]_n].$$

Let $\mathcal{G}$ denote the set of non-decreasing, subadditive functions $g : \mathbb{N}_0 \to \mathbb{N}_0$ such that $g(n) \leq n$.[1] For any $g \in \mathcal{G}$ we define

$$\Lambda(g) = \{\mathbf{x} \in \Sigma : \text{ Every substring of } \mathbf{x} \text{ of length } j \text{ has at most } g(j) \text{ 1s.}\}$$

For any $g \in \mathcal{G}$, $\Lambda(g) \subset \Sigma$ is closed (since defined by a closed condition on finite-length strings) and σ-invariant (obvious). Thus it is a subshift.

Example 4.1.1. $\Lambda(0) = \{\mathbf{0}\}$. $\Lambda(n) = \Sigma$.

We call $\mathcal{D} = \{\Lambda(g) : g \in \mathcal{G}\}$ the class of *density subshifts*, and denote $\widehat{\sigma}_g = \sigma|_{\Lambda(g)}$. We will estimate the generalized entropy of density subshifts in terms of (the growth rates of) their defining functions g.

Given $g \in \mathcal{G}$ and $n \in \mathbb{N} \cup \{\infty\}$, we say $\mathbf{a} \in \{0,1\}^n$ is *g-admissible* if every substring of $\mathbf{a}$ of length j has at most $g(j)$ 1s, $1 \leq j \leq n$. Equivalently, $\mathbf{a}$ is g-admissible iff $\mathbf{a} \in [\Lambda(g)]_n$.

Lemma 4.1.2. *For any $g \in \mathcal{G}$, there is an infinite g-admissible sequence $\mathbf{a} \in \Lambda(g)$ such that for all $n \in \mathbb{N}$, the truncation $[\mathbf{a}]_n = a_1 a_2 \dots a_n$ contains precisely $g(n)$ 1s.*

Proof. We define $\mathbf{a} = (a_i)_{i\in\mathbb{Z}} \in \Sigma$ by $a_j = 0$ if $j \leq 0$ and

$$a_k = g(k) - g(k-1)$$

[1] In particular, $g(0) = 0$ and $g(1) \leq 1$, with $g(1) = 0$ iff $g \equiv 0$.

for $k \geq 1$. This is well-defined (i.e. $a_k \in \{0,1\}$ for all k) because g is non-decreasing and subadditive with $g(1) \leq 1$. Then for all $n \in \mathbb{N}$ the number of 1s in the substring $[\mathbf{a}]_n = a_1 a_2 \dots a_n$ is precisely

$$\sum_{j=1}^{n} a_j = \sum_{j=1}^{n} (g(j) - g(j-1)) = g(n) - g(0) = g(n).$$

It remains to show that $\mathbf{a}$ is g-admissible, i.e. $\mathbf{a} \in \Lambda(g)$. This is true iff

$$\sum_{j=m}^{m+n-1} a_j \leq g(n)$$

for all $m \in \mathbb{Z}, n \in \mathbb{N}$, and it clearly suffices to check the case when $m \geq 1$. But subadditivity of g implies

$$\sum_{j=m}^{m+n-1} a_j = g(m+n-1) - g(m-1) \leq g(n) + g(m-1) - g(m-1) = g(n),$$

so we are done. □

Remark. This lemma implies that if $g, h \in \mathcal{G}$ with $g \neq h$, then $\Lambda(g) \neq \Lambda(h)$. In particular, the class $\mathcal{D}$ of density subshifts is in one-to-one correspondence with the space $\mathcal{G}$ of non-decreasing, subadditive functions $g : \mathbb{N}_0 \to \mathbb{N}_0$ such that $g(n) \leq n$.

Now write

$$A_n = A_g(n) = \#[\Lambda(g)]_n = \text{ the number of } g\text{-admissible strings of length } n,$$

so that $o(\widehat{\sigma}_g) = [A_n]$. We have the following simple estimate:

Proposition 4.1.3. *For all $n \in \mathbb{N}$ such that $2 \leq g(n) \leq n/2$ we have*

$$2^{g(n)} \leq A_n \leq g(n) \cdot \binom{n}{g(n)}. \tag{4.2}$$

Proof. For the lower bound, choose $\mathbf{a} \in \Lambda(g)$ as given by Lemma 4.1.2, so that $[\mathbf{a}]_n$ contains precisely $g(n)$ 1s for all n. Since $[\mathbf{a}]_n$ is g-admissible, any string obtained from $[\mathbf{a}]_n$ by

changing some subset of its 1s to 0s is also g-admissible (i.e. density subshifts are *hereditary*). Since there are $2^{g(n)}$ such subsets, we get $2^{g(n)} \leq A_n$.

For the upper bound, note that A_n is clearly dominated by the number of subsets $S \subset \{1, \ldots, n\}$ with $\#S \leq g(n)$. Then

$$A_n \leq \sum_{j=0}^{g(n)} \binom{n}{j} = 1 + \sum_{j=1}^{g(n)} \binom{n}{j} < 1 + g(n) \cdot \binom{n}{g(n)},$$

where the last inequality follows from the assumption $2 \leq g(n) \leq n/2$. □

Proposition 4.1.4. *For any metric space X and continuous $f : X \to X$,*

$$h(f) = 0 \implies o(f) < \inf \mathbb{E}.$$

Proof. Clearly $h(f) = 0$ implies $o(f) \leq \inf \mathbb{E}$. Strict inequality then follows from abstract (i.e. dynamically contentless) properties of the lattice $\overline{\mathbb{O}}$. Specifically we show that $\inf \mathbb{E}$ cannot be realized as a countable supremum of concrete orders of growth.

First note that $\inf \mathbb{E} \in \overline{\mathbb{O}} \setminus \mathbb{O}$, i.e. $\inf \mathbb{E}$ is not itself a concrete order of growth. If it were, say $\inf \mathbb{E} = [a_n]$, we would have $\inf \mathbb{E} = [a_n] < [na_n] \leq \mathbb{E}$, a contradiction. Thus for all $\varepsilon > 0$ we have

$$h(f) = 0 \implies [r_f(n, \varepsilon)] \leq \inf \mathbb{E} \implies [r_f(n, \varepsilon)] < \inf \mathbb{E}.$$

Now consider a countable collection $A = \{\alpha_i\}_{i \in \mathbb{N}} \subset \mathbb{O}$ with $\alpha_i < \inf \mathbb{E}$ for all i. We claim that A admits a concrete, subexponential upper bound, i.e. there exists $\beta \in \mathbb{O}$ with $\beta \leq \inf \mathbb{E}$ (thus $\beta < \inf \mathbb{E}$, by concreteness) with $\alpha_i \leq \beta$ for all i. Thus $\sup A < \inf \mathbb{E}$.

We construct β via a straightforward "stretched" diagonalization argument. Write

$$\alpha_i = [a_n^{(i)}].$$

For any non-decreasing function $f : \mathbb{N} \to \mathbb{N}$ we define $\beta_f = [b_n]$ by setting $b_n = \max\limits_{1 \leq i \leq f(n)} a_n^{(i)}$.

For f unbounded, β_f eventually dominates every α_i, thus $\alpha_i \leq \beta$. Using the fact that

$\frac{1}{n}\log a_n^{(i)} \to 0$ for all i, it is easy to see that we can pick f such that $f(n) \to \infty$, but slowly enough that $\frac{1}{n}\log b_n \to 0$. Since $\beta_f \in \mathbb{O}$, we get

$$\sup \alpha_i \leq \beta_f < \inf \mathbb{O},$$

as desired.

Finally we observe that every generalized entropy $o(f)$ is a countable supremum of concrete orders of growth:

$$o(f) = \sup_{\varepsilon>0}[r_f(n,\varepsilon)] = \sup_{m\in\mathbb{N}}[r_f(n,1/m)] =: \sup_{m\in\mathbb{N}} \alpha_m.$$

If $h(f) = 0$ it is easy to see that $\alpha_m < \inf \mathbb{E}$, so by the above $o(f) < \inf \mathbb{E}$, completing the proof. □

Proposition 4.1.5. *If $g \in \mathcal{G}$ satisfies*

$$\frac{g(n)\log n}{n} \xrightarrow{n\to\infty} 0, \tag{4.3}$$

then $o(\widehat{\sigma}_g) < \inf \mathbb{E}$.

Proof. By Lemma 4.1.4, it suffices to show that

$$\frac{\log A_n}{n} \to 0.$$

If $g(n) \leq 1$ for all n this result holds trivially (since then $A_n \leq n$). Otherwise we have $g(n) \geq 2$ and (by (4.3)) $g(n) \leq n/2$ for all $n \gg 0$, whence Proposition 4.1.3 gives

$$\frac{\log A_n}{n} \leq \frac{\log g(n)}{n} + \frac{1}{n}\log \binom{n}{g(n)}. \tag{4.4}$$

Since $g(n) \leq n$ for all $n \in \mathbb{N}$, the first summand goes to zero. The fact that

$$\lim_{n\to\infty} \frac{1}{n}\log \binom{n}{g(n)} = 0$$

follows (after a straightforward but tedious calculation) from Stirling's approximation and (4.3). □

Lemma 4.1.6. *Suppose $(b_n) \subset \mathbb{R}_{>0}$ is a non-decreasing sequence. Then $\sup \mathbb{P} < [b_n]$ if and only if for all $t > 0$ we have*

$$\lim_{n\to\infty} \frac{n^t}{b_n} = 0. \tag{4.5}$$

Proof. It is immediate that (4.5) implies $[b_n] \geq \sup \mathbb{P}$. Moreover it is easy to see (as in the proof of Lemma 4.1.4) that $\sup \mathbb{P} \in \overline{\mathbb{O}} \setminus \mathbb{O}$, forcing strict inequality, $[b_n] > \sup \mathbb{P}$.

For the converse, suppose $\sup \mathbb{P} < [b_n]$. Then for all $t > 0$, the sequence (n^t/b_n) is bounded above, say by $C_t < \infty$. It follows that

$$\frac{n^t}{b_n} = \frac{1}{n} \cdot \frac{n^{t+1}}{b_n} \leq \frac{C_{t+1}}{n} \xrightarrow{n\to\infty} 0,$$

as desired. □

We can now state our main theorem, giving a criterion for a density subshift to have strictly superpolynomial and subexponential order of growth.

Theorem 4.1.7. *Suppose $g \in \mathcal{G}$ satisfies*

1. $\sup\limits_{t>0} \lim\limits_{n\to\infty} \dfrac{n^t}{2^{g(n)}} = 0$, *and*

2. $\lim\limits_{n\to\infty} \dfrac{g(n) \log n}{n} = 0.$

Then

$$\sup \mathbb{P} \;<\; [2^{g(n)}] \;\leq\; o(\widehat{\sigma}_g) \;\leq\; \left[g(n) \cdot \binom{n}{g(n)} \right] \;<\; \sup \mathbb{E}.$$

Proof. This is immediate from Proposition 4.1.3, Proposition 4.1.5, and Lemma 4.1.6. □

Example 4.1.8. Define $g : \mathbb{N} \to \mathbb{N}$ by

$$g(n) = \lfloor \sqrt{n} \rfloor.$$

Then $g \in \mathcal{G}$ and satisfies conditions (1) and (2) of Theorem 4.1.7, so that

$$\sup \mathbb{P} < [2^{\lfloor \sqrt{n} \rfloor}] \leq o(\widehat{\sigma}_g) < \inf \mathbb{E}.$$

More generally setting $g_t(n) = \lfloor n^t \rfloor$ for any $0 < t < 1$ we have

$$\sup \mathbb{P} < [2^{\lfloor n^t \rfloor}] \leq o(\widehat{\sigma}_g) < \inf \mathbb{E}.$$

This gives a one-parameter family of subshifts $\Lambda_t := \Lambda(g_t)$, $0 < t < 1$, increasing in the sense that $\Lambda_t \subset \Lambda_s$ when $t < s$, such that

$$\sup \mathbb{P} < o(\sigma|_{\Lambda_t}) < \inf \mathbb{E}$$

for all $0 < t < 1$.

We end this section with some observations on the topological and measurable dynamics of the density subshifts considered here.

Proposition 4.1.9. *Suppose $g \in \mathcal{G}$ grows sublinearly, i.e. $g(n)/n \to 0$. Then*

$$\mathrm{Per}(\widehat{\sigma}_g) = \mathrm{Fix}(\widehat{\sigma}_g) = \{\mathbf{0}\}.$$

If in addition $g(n) \to \infty$, then $\widehat{\sigma}_g$ is transitive, and in fact topologically mixing.

Remark. Both conditions here (sublinear growth and unboundedness) are implied by the assumptions of Theorem 4.1.7.

Proof. For any σ-periodic sequence $\mathbf{0} \neq \mathbf{a} \in \Sigma$, the set of $i \in \mathbb{N}$ such that $a_i = 1$ has positive asymptotic density. This is ruled out by sublinear growth of g. The fact that $g(n) \to \infty$ implies $\widehat{\sigma}_g$ mixing follows from standard criteria for subshifts in terms of the concatenation of admissible words. □

In particular, when $g \in \mathcal{G}$ and $g(n) \to \infty$, $\overline{\mathrm{Rec}(\widehat{\sigma}_g)} = \Omega(\widehat{\sigma}_g) = \Lambda(g)$. This shows that density subshifts can have non-trivial topological dynamics even when the defining function

$g \in \mathcal{G}$ grows sublinearly and (therefore) $\widehat{\sigma}_g$ has no non-trivial periodic points. In contrast, the ergodic theory of such subshifts (defined by slowly growing g) is always trivial:

Proposition 4.1.10. *If $g \in \mathcal{G}$ grows sublinearly, then $\widehat{\sigma}_g$ has only one invariant measure, $\delta_{\mathbf{0}}$.*

Proof. Fix $\mathbf{x} \in \Lambda(g)$ and define

$$\mu_n = \frac{1}{n} \sum_{j=0}^{n-1} \delta_{\sigma^j \mathbf{x}}$$

For any $m \in \mathbb{N}$, let

$$U_m = \{0\}^m \times \Sigma.$$

We show that

$$\lim_{n\to\infty} \mu_n(U_m) = 1. \tag{4.6}$$

Since $\partial U_m = \varnothing$, it follows that $\mu(U_m) = 1$ for any weak-$*$ limit point μ of (μ_n), and thus (since $\bigcap_m U_m = \{\mathbf{0}\}$) that $\mu(\{\mathbf{0}\}) = 1$. Since $\mathbf{x} \in \Lambda(g)$ was arbitrary, the ergodic theorem implies that $\delta_{\mathbf{0}}$ is the only σ-invariant measure supported on $\Lambda(g)$.

To show (4.6), write $\mathbf{x} = (x_1, x_2, \dots)$ and observe that

$$\begin{aligned}
\mu_n(U_m) &= \frac{\#\{0 \le j \le n-1 : \sigma^j \mathbf{x} \in U_m\}}{n} \\
&= \frac{\#\left\{1 \le j \le n : \sum_{k=j}^{j+m-1} x_k = 0\right\}}{n} \\
&= 1 - \frac{\#\left\{1 \le j \le n : \sum_{k=j}^{j+m-1} x_k \ge 1\right\}}{n} =: 1 - \frac{B_m(n)}{n}.
\end{aligned}$$

Now

$$B_m(n) \le \sum_{j=1}^{n} \sum_{k=j}^{j+m-1} x_k \le m \cdot \sum_{j=1}^{n+m-1} x_j \le m(m-1) + m \cdot \sum_{j=1}^{n} x_j,$$

so the fact that g grows sublinearly and $\mathbf{x} \in \Lambda(g)$ implies

$$\frac{B_m(n)}{n} \leq \frac{m(m-1)}{n} + m\left(\frac{1}{n}\sum_{j=1}^{n} x_j\right) \xrightarrow{n\to\infty} 0,$$

completing the proof. □

By the variational principle, this shows that $h_{\text{top}}(\widehat{\sigma}_g) = 0$ whenever g grows sublinearly; by Lemma 4.1.4 it follows that $o(\widehat{\sigma}_g) < \inf \mathbb{E}$. This improves Proposition 4.1.5, though without an explicit expression for the upper bound when g grows too fast for our earlier Stirling's approximation-based estimate to work.

Corollary 4.1.11. *If $g \in \mathcal{G}$ grows sublinearly and satisfies $\lim_{n\to\infty} \frac{n^t}{2^{g(n)}} = 0$ for all $t > 0$, then $\sup \mathbb{P} < o(\widehat{\sigma}_g) < \inf \mathbb{E}$.*

4.2 Smooth realization

We now show how to realize any density subshift σ_g as a maximal compact invariant set Λ for a smooth diffeomorphism of $\mathbb{S}^3$. Our straightforward construction produces $f \in \text{Diff}^\infty(\mathbb{S}^3)$ such that $\Omega(f) = \Lambda \cup \{N\} \cup \{S\}$, so that for the classical topological entropy we have

$$h(f) = h(f|_{\Omega(f)}) = h(f|_\Lambda) = h(\sigma_g).$$

In the context of generalized entropy we obviously still have $o(f|_{\Omega(f)}) \leq o(f)$, so if σ_g is chosen so that $\sup \mathbb{P} < o(\sigma_g) < \inf \mathbb{E}$ we get $\sup \mathbb{P} < o(f)$, and since $h(f) = h(\sigma_g) = 0$, similarly have $o(f) < \inf \mathbb{E}$ by remarks above. However we should note that, since there are many cases where the inequality $o(f|_{\Omega(f)}) \leq o(f)$ is strict (Morse-Smale diffeomorphisms are an instructrive example [Cd23]), it does not immediately follow that a sharp quantitative upper bound for a subshift's order of growth will carry through to its smooth realization.

In fact the construction here works for any subshift $\Lambda \subset \{0,1\}^{\mathbb{Z}}$. We begin by choosing

a C^∞ diffeomorphism $h : \mathbb{S}^2 \to \mathbb{S}^2$ realizing the standard Smale horseshoe; more precisely, we assume h is Axiom A with $\Omega(h) = \{P\} \cup K$, K a maximal invariant set such that $h|_K$ is conjugate to the full shift on 2 symbols, say via a homeomorphism $\eta : \{0,1\}^{\mathbb{Z}} \to K$. Define $\Lambda' = \eta(\Lambda)$. Since Λ' is a compact subset of $\mathbb{S}^2$, by Whitney's extension theorem there is a C^∞ function $\varphi : \mathbb{S}^2 \to \mathbb{R}$ such that $\varphi \geq 0$ and $\varphi^{-1}(0) = \Lambda'$.

To define $f : \mathbb{S}^3 \to \mathbb{S}^3$, we view $\mathbb{S}^3$ as the suspension of $\mathbb{S}^2$,

$$\mathbb{S}^3 = S\mathbb{S}^2 = (\mathbb{S}^2 \times I)/\sim,$$

and use coordinates $(x,t) \in \mathbb{S}^2 \times I$ (here $I := [-1,1]$). Pick $q : I \to I$ any diffeomorphism satisfying $q(x) \geq x$ and $\mathrm{Fix}(q) = \{-1,0,1\}$, and let $\psi : I \to \mathbb{R}_{\geq 0}$ be a bump function supported in a small neighborhood of 0 with $\psi(0) > 0$. After scaling ψ if necessary, we can assume that

$$\|\psi'\|_\infty \cdot \|\varphi\|_\infty < \min_{t \in I} q'(t).$$

It follows that $f : \mathbb{S}^3 \to \mathbb{S}^3$ defined by

$$f(x,t) = (h(x), q(t) + \varphi(x)\psi(t)) =: (h(x), f_2(x,t))$$

is a C^∞ diffeomorphism of $\mathbb{S}^3$.

Now assume that Λ is a density subshift with $\sup \mathbb{P} < o(\sigma|_\Lambda) < \inf \mathbb{E}$. By the above, to conclude that $\sup \mathbb{P} < o(\sigma|_\Lambda) \leq o(f) < \inf \mathbb{E}$ it suffices to show that $\Omega(f) = \{N\} \cup \{S\} \cup (\Lambda' \times \{0\})$.

But this is obvious. If $s \notin \{-1,0,1\}$, then any sufficiently small open interval J containing s is wandering for q (specifically satisfies $q(J) > J$), and consequently $\mathbb{S}^2 \times J$ gives an f-wandering neighborhood of (x,s) for any $x \in \mathbb{S}^2$. Finally consider $(x,0)$ for $x \notin \Lambda'$. Write $\varepsilon = \varphi(x)\psi(0) > 0$, and pick δ such that $d_{\mathbb{S}^3}((x,0),(y,s)) < \delta$ implies $\varphi(y)\psi(s) > \varepsilon/2$. Then

if $\eta < \min\{\delta, \varepsilon/4\}$ the neighborhood $U := B_\eta(x) \times (-\eta, \eta)$ is wandering for f, since

$$(y,s) \in U \implies f_2(y,s) = q(s) + \varphi(y)\psi(s) > -\eta + \frac{\varepsilon}{2} > \frac{\varepsilon}{4} > \eta \implies f(y,s) \notin U,$$

and the second coordinate is non-decreasing along orbits.

Chapter 5

How (not) to prove the Entropy Conjecture

We conclude with some remarks on the Entropy Conjecture. Recall from the Introduction:

Conjecture 5.0.1 ([Shu74])**.** *Let M be a closed manifold. For all $1 \leq k \leq \dim(M)$ and $f \in \mathrm{Diff}^1(M)$ one has*

$$\log \mathrm{sp}(f_{*,k}) \leq h(f) \tag{5.1}$$

where $f_* : H_*(M, \mathbb{R}) \to H_*(M, \mathbb{R})$ denotes the induced map on real homology, and $\mathrm{sp}(f_*)$ its spectral radius.

Conjecture 5.0.1 has many partial results. Our main touchstone here is Yomdin's proof that the Entropy Conjecture holds in the C^∞ category:

Theorem 5.0.2 ([Yom87])**.** *For all closed manifolds M and $f \in \mathrm{End}^\infty(M)$,*

$$\log \mathrm{sp}(f_*) \leq h(f).$$

5.1 Stable volume growth

Yomdin's proof reduces to showing that the following quantity vanishes for all $f \in \mathrm{Diff}^\infty(M)$ and C^∞ submanifolds $S \subset M$:

Definition 5.1.1. For $S \subset M$ a compact submanifold we define the *local volume growth of* S *under* f to be

$$\mathrm{LV}(f, S) = \lim_{\varepsilon \to 0} \liminf_{n \to \infty} \frac{1}{n} \log \left(\sup_{x \in S} \ \mathrm{vol}\, f^n(S \cap B_f(x, n, \varepsilon)) \right).$$

Then define the *local volume growth of* f to be

$$\mathrm{LV}^r(f) = \sup\{\mathrm{LV}(f, S) \ : \ S \subset M \text{ a } C^r \text{ compact submanifold}\}.$$

For convenience we will drop the superscript when restricting to smooth submanifolds: $\mathrm{LV}(f) := \mathrm{LV}^\infty(f)$.

This quantity can be nonzero in finite regularity:

Example 5.1.2 ([Yom87], attributed to Margulis)**.** The curve in γ_r in $\mathbb{R}^2$ defined by $y = x^{2r+1} \sin(1/x)$ is C^r. Consider a hyperbolic linear linear map

$$A = \begin{bmatrix} 1/\lambda & 0 \\ 0 & \lambda \end{bmatrix} : \mathbb{R}^2 \to \mathbb{R}^2.$$

Then a straightforward estimate shows that for all $\varepsilon > 0$ fixed, there are constants $C_1 = C_1(\varepsilon) > 0$ and $C_2 = C_2(\varepsilon) > 0$ such that

$$\text{length } f^n(\gamma_r \cap B_A(0, n, \varepsilon)) \geq C_1 \lambda^{\frac{n}{2r+1}} - C_2$$

for all n. Since this construction is clearly local and easily generalizable to higher dimensions, it shows that for any f with a hyperbolic fixed point we have $\mathrm{LV}^r(f) > 0$ when $r < \infty$.

More to the point, in this example it is easy to see how to pass regularity between

submanifold and map. Consider a C^∞ surface diffeomorphism with a hyperbolic fixed point, p. Take $q \in W^s_{\mathrm{loc}}(p)\backslash\{p\}$. For a sufficiently small neighborhood U of q we have $f(U)\cap U = \varnothing$. Now by a C^r change of f in U (appropriately cut off to allow C^r gluing) one can obtain $\widehat{f}$ which agrees with f outside U but sends $W^s_{\mathrm{loc}}(p) \cap U$ to a version of the curve γ_r in $f(U)$ still "skewered" by the stable manifold of p for f (which is by construction the same as its $\widehat{f}$ stable manifold in $f(U)$). Though the non-smooth point of this curve is no longer fixed, it lies on the stable manifold of p, and it is easy to see that the same asymptotic estimate as above holds. Thus one obtains a C^r diffeomorphism f with $\mathrm{LV}(f) > 0$.

Our first rather obvious observation is that the examples described here are not *stable*: one can destroy the pathological local volume growth phenomenon via arbitrarily small perturbations of the curves. On the other hand, we have seen that volume growth coming from homology growth must be stable not only under C^1 perturbations of a representing submanifold, but over its entire homotopy class.

This leads us to define the following quantities by analogy with the local volume growth defined above.

Definition 5.1.3. For $S \subset M$ a compact submanifold we define the *homotopy volume growth of S under f* to be

$$\mathrm{HV}(f, S) = \liminf_{n\to\infty} \frac{1}{n} \log \inf_{\hat{S}\sim S} \mathrm{vol}\, f^n(\hat{S}),$$

where the relation $\hat{S} \sim S$ means that $\hat{S}$ is homotopic to S, rel boundary if $\partial S \neq \varnothing$ or freely if not. Then we define the *homotopy local volume growth of S under f* to be

$$\mathrm{HLV}(f, S) = \lim_{\varepsilon\to 0} \liminf_{n\to\infty} \frac{1}{n} \log \left(\inf_{\hat{S}\sim S} \sup_{x\in\hat{S}} \mathrm{vol}\, f^n(S \cap B_f(x, n, \varepsilon)) \right),$$

and finally the *homotopy local volume growth of f* by

$$\mathrm{HLV}(f) = \sup\{\mathrm{HLV}(f, S) \ : \ S \subset M \text{ a compact submanifold}\}.$$

Obviously we have

Fact 5.1.4. *$HLV(f) \leq LV(f)$.*

In addition, the same argument showing that $\log \mathrm{sp}(f_*) \leq h(f) + LV(f)$ easily shows

Proposition 5.1.5. *For all $f \in \mathrm{Diff}^1(M)$ one has*

$$\log \mathrm{sp}(f_*) \leq h(f) + \mathrm{HLV}(f).$$

This suggests the following question, which as far as we can tell remains entirely open.

Question 5.1.6. *Do there exist $f \in \mathrm{Diff}^r(M)$, $1 \leq r < \infty$, for which $\mathrm{HLV}(f) > 0$?*

A negative answer to this question in any regularity would imply the Entropy Conjecture in that regularity. Thus, one natural indirect approach to the EC would be to ask: What are the consequences of $\mathrm{HLV}(f) > 0$?

5.2 Finding separated intersection sets

Suppose $V^k, W^{d-k} \subset M$ are closed, homologically non-trivial submanifolds of M such that

$$\iota([f^n V], [W]) \sim \rho^n$$

where ι denotes the intersection form $\iota : H_k(M, \mathbb{R}) \times H_{d-k}(M, \mathbb{R}) \to \mathbb{R}$. Up to a perturbation of V or W one can assume that $f^n(V) \pitchfork W$ for all $n \in \mathbb{N}$, in which case we have $\#(f^n(V) \cap W) \geq \iota([f^n V], [W]) \geq C\rho^n$. Moreover the bound $\#(f^n(\widehat{V}) \cap \widehat{W}) \geq C\rho^n$ holds for all $\widehat{V}, \widehat{W}$ homotopic to V, W, respectively, such that $f^n(\widehat{V}) \pitchfork \widehat{W}$ for all n, thus for generic such $\widehat{V}$ and $\widehat{W}$.

In particular, the Entropy Conjecture would follow from the claim that there exists $\varepsilon > 0$ such that for every $n \in \mathbb{N}$ we can find $\widehat{V}_n \sim V$ such that $\widehat{V}_n \pitchfork f^n W$ and $\widehat{V}_n \cap f^n W$ is an (n, ε)-separated set.

5.2.1 A modified H_1 proof

We illustrate the use of this strategy via a new proof of the one-dimensional case.

Theorem 2 ([Man75]). *For all C^0 $f : M \to M$ we have* $\log \mathrm{sp}(f_{*,1}) \leq h(f)$.

The main motivation for re-formulating the H_1 proof in this way is the search for proof types potentially more amenable to higher-dimensional generalizations. Attempting such a generalization of the proof we give here, however, one encounters new topological obstructions in terms of the linking of disk boundaries around intersection points.

The appearance of some new obstructions is not surprising given that the almost purely topological H_1 argument reflects the fact that the Entropy Conjecture for H_1 holds in the C^0 category. Since this is no longer true for higher homology,[1] when generalizing this argument one should *expect* novel obstructions. The general hope is that these obstructions can eventually be overcome by appeal to differentiability.

We note that the proof we give here, since it appeals to transversality, appears to require f at least C^1.

For essentially all versions of the H_1 proof, the key technical lemma is

Lemma 5.2.1 ([Man75], [Kat86]). *There exists $\varepsilon > 0$ such that if $\gamma : [0,1] \to M$ is an embedded curve with* $\mathrm{diam}(\gamma) < \varepsilon$ *and endpoints $x_0 = \gamma(0)$, $x_1 = \gamma(1)$ satisfying $d_n^f(x,y) < \varepsilon$, then for all $0 \leq k \leq n-1$, γ is homotopic rel boundary to γ_k such that*

$$\mathrm{length}(f^k \gamma_k) < \varepsilon.$$

Proof. First note that there exists $\delta > 0$ such that any curve of diameter less than δ is homotopic rel boundary to a minimizing geodesic segment (namely any $\delta < \mathrm{injrad}(M)$ works). Now by uniform continuity we can choose $0 \leq \varepsilon \ll \delta$ small enough that for any

[1] Though the stock of examples remains surprisingly limited, a classical construction shows that C^1 regularity is at least necessary for EC to hold on $H_k(M)$ with $k \geq 2$ and $\dim(M) \geq 8$ [Pug75].

curve γ with $\operatorname{diam}(\gamma) < \varepsilon$, $\operatorname{diam}(f\gamma) < \delta$. Now clearly γ is homotopic rel boundary to γ_1 such that $f\gamma_1$ is a minimizing geodesic. If moreover we have $d(fx, fy) < \varepsilon$ (where x, y denote the endpoints of the curve), then $\operatorname{diam}(f\gamma_1) < \varepsilon$. Repeating inductively we see that γ is homotopic rel boundary to γ_k such that $\operatorname{diam}(f^k\gamma_k) < \varepsilon$ provided $d(f^\ell x, f^\ell y) < \varepsilon$ for all $0 \leq \ell \leq k$, i.e. $d^f_{k+1}(x, y) < \varepsilon$. □

The next lemma follows from standard representation arguments for homology in dimension or codimension 1.

Lemma 5.2.2. *Suppose* $\operatorname{sp}(f_{*,1}) = \rho > 1$. *Then there exist closed embedded submanifolds* $V, W \subset M$, *with* $\dim(V) = 1$ *and* $\dim(W) = d - 1$, *and* $C > 0$ *such that*

$$\iota([f^n V], [W]) \geq C\rho^n$$

for all $n \in \mathbb{N}$.

Proposition 5.2.3. *Suppose* $V^1, W^{d-1} \subset M$ *are closed submanifolds. Then there exists* $\delta > 0$ *such that for all* n, V *is homotopic to* $\widehat{V}_n$ *such that* $\widehat{V}_n \pitchfork f^{-n}(W)$, *and* $\widehat{V}_n \cap f^{-n}(W)$ *is an* (n, δ)*-separated set.*

Proof. After perturbing V and W we can assume that $f^n V \pitchfork W$ for all n. Fix $\mathcal{T}_W$ a tubular neighborhood of W. Let $\varepsilon > 0$ be given by Lemma 5.2.1 and let $0 < \delta \ll \varepsilon$ be small enough that for any $x, y \in V$ with $d(x, y) < \delta$, if $V(x, y)$ denotes the short segment of $V \setminus \{x, y\}$ then we have $\operatorname{diam}(V(x, y)) < \varepsilon$. We also assume δ small enough that $B_\delta(x) \subset \mathcal{T}_W$ for all $x \in W$.

Now suppose $x, y \in f^{-n}V \cap W$ with $d^f_n(x, y) < \delta$. By Lemma 5.2.1 we may homotope $V(x, y)$ rel $\{x, y\}$ to $\widehat{V}(x, y)$ such that $f^n\widehat{V}(x, y)$ is the minimizing geodesic connecting $f^n x$ and $f^n y$, which then lies in the tubular neighborhood of W. By a further homotopy concentrated in an arbitrarily small neighborhood of this geodesic in $\widehat{V}$ we can either remove all

intersections in this neighborhood, or all except x. In this latter case we can ensure that x is fixed by this homotopy.

This argument shows that if $f^{-n}V \cap W$ is not (n, δ)-separated, we can homotope V to $\widehat{V}$ such that $\#f^{-n}\widehat{V} \cap W < \#f^{-n}V \cap W$. Moreover the construction shows that the homotopy can be taken rel $\#f^{-n}\widehat{V} \cap W$. This latter fact implies that one can apply this argument inductively, since at each stage the curve connecting two intersection points is homotopic to the corresponding curve in V. When the process terminates, we are done. □

This result combined with Lemma 5.2.2 immediately gives

Corollary 5.2.4. *For all closed manifolds M and $f \in \mathrm{Diff}^1(M)$ we have* $\log \mathrm{sp}(f_{*,1}) \leq h(f)$.

5.3 Nerves of dynamical covers

For the discussion in this section it is convenient to assume that M^d is smoothly embedded in some $\mathbb{R}^N$, and that f extends to $F : \mathcal{T}_M \to \mathbb{R}^N$ for some tubular neighborhood $\mathcal{T}_M$ of M. Fixing some smooth identification of $\mathcal{T}_M$ with N_M the normal bundle of M in $\mathbb{R}^N$, one can assume (possibly after shrinking the tube width and composing with sufficiently strong contraction in the normal direction) that $\overline{F(\mathcal{T}_M)} \subset \mathcal{T}_M$. We write $\pi : \mathcal{T}_M \to M$ for the smooth projection map given by our fixed identification $\mathcal{T}_M \cong N_M$.

Any $(k+1)$-tuple $\mathbf{x} = (x_0, \ldots, x_k) \in (\mathbb{R}^N)^{k+1}$ determines an affine map $\sigma_{\mathbf{x}} : \Delta^k \to \mathbb{R}^N$ from the standard k-simplex $\Delta^k = \{(t_0, \ldots, t_{k+1}) \in (\mathbb{R}_{\geq})^{k+1} \ : \ \sum t_i = 1\}$, given by

$$\sigma_{\mathbf{x}}(t_0, \ldots, t_k) = t_0 x_0 + \ldots t_k x_k.$$

Since $M \subset \mathbb{R}^N$ is smooth, there exists $\varepsilon_0 > 0$ such that if $\mathbf{x} = (x_0, \ldots, x_k) \in M^{k+1}$ satisfies $d(x_i, x_j) < \varepsilon_0$ for all i, j, then $\mathrm{im}(\sigma_{\mathbf{x}}) \subset \mathcal{T}_M$. Then composing $\sigma_{\mathbf{x}}$ with the projection π one gets a smooth map $\widehat{\sigma}_{\mathbf{x}} = \pi \circ \sigma_{\mathbf{x}} : \Delta^k \to M$.

Now consider $\mathcal{U} = \{U_i\}_{i\in\mathcal{I}}$ any finite open cover of M. Then the *nerve* of $\mathcal{U}$ is the abstract simplicial complex defined by

$$\operatorname{nerve}(\mathcal{U}) = \{\{i_0, \ldots, i_k\} \subset \mathcal{I} \ : \ U_{i_0} \cap \cdots \cap U_{i_k} \neq \varnothing\}.$$

We write $|\operatorname{nerve}(\mathcal{U})|$ for its geometric realization, which we can view concretely as a subcomplex of the standard simplex $\Delta^{|\mathcal{I}|-1}$ in $\mathbb{R}^{\mathcal{I}}$, namely

$$|\operatorname{nerve}(\mathcal{U})| = \left\{ (t_i) \in \mathbb{R}^{\mathcal{I}} \ : \ t_i \geq 0, \ \sum_{i\in\mathcal{I}} t_i = 1, \text{ and } \bigcap_{\{i\in\mathcal{I}:t_i>0\}} U_i \neq \varnothing \right\}.$$

Any partition of unity $\Phi = \{\phi_i\}_{i\in\mathcal{I}}$ subordinate to $\mathcal{U}$ specifies a map

$$G_\Phi : M \to |\operatorname{nerve}(\mathcal{U})|$$

by $G_\Phi(x) = (\phi_i(x))_{i\in\mathcal{I}}$.

Define $\operatorname{diam}(\mathcal{U}) = \sup_{i\in\mathcal{I}} \operatorname{diam}(U_i)$. If $\operatorname{diam}(\mathcal{U}) < \varepsilon_0/2$ one can define a map

$$H : |\operatorname{nerve}(\mathcal{U})| \to M$$

by first choosing $x_i \in U_i$ for all $i \in \mathcal{I}$ and then defining

$$H((t_i)) = \pi\left(\sum_{i\in\mathcal{I}} t_i x_i\right).$$

In other words, the map H is defined first on vertices by mapping the vertex corresponding to U_i to x_i, then extended simplicially via the maps $\widehat{\sigma}_{\mathbf{x}}$. Though the definition of H depends on the choice of the x_i, it is easy to see that it is independent of these choices up to homotopy provided $\operatorname{diam}(\mathcal{U})$ is sufficiently small (namely, small enough that the straight-line homotopy in $\mathbb{R}^N$ from H to $\widehat{H}$ defined using some other choice $\widehat{x}_i \in U_i$ is always contained in $\mathcal{T}_M$). Moreover, we have the following:

Lemma 5.3.1. *There exists $\varepsilon > 0$, depending only on M and its embedding in $\mathbb{R}^N$, such*

that for any open cover $\mathcal{U}$ *of* M *with* $\operatorname{diam}(\mathcal{U}) < \varepsilon$ *and partition of unity* Φ *subordinate to* $\mathcal{U}$ *we have*

$$H \circ G_\Phi \simeq \mathrm{id}_M .$$

In particular, for all sufficiently small-diameter open covers, H *surjects on homology.*

Now consider $f \in \mathrm{Diff}^1(M)$ and $\mathcal{U}_{n,\varepsilon}$ a minimal (n,ε)-cover of M for f. Consider $\alpha \in H_k(|\operatorname{nerve}(\mathcal{U}_{n,\varepsilon})|, \mathbb{R})$ and a representation

$$\alpha = \left[\sum a_i \sigma_i\right]$$

with the σ_i all k-simplices of the nerve. Using an easy generalization of Lemma 5.2.1 (see also [Gro07, §5.43]), it is easy to see that for $\varepsilon > 0$ sufficiently small and some constant $C > 0$ we have

$$\|f_*^n H_* \alpha\| = \left\| \left[\sum a_i (f^n \circ H \circ \sigma_i)\right] \right\| \leq \left(\sum |a_i|\right) \cdot C\varepsilon^k.$$

In particular, if $\beta \in H_k(M, \mathbb{R})$ is such that $\|f_*^n \beta\| \sim \lambda^n$, then for some constant $\widehat{C} > 0$ we have $\widehat{C}\lambda^n$ for all n, and for Φ any partition of unity subordinate to $\mathcal{U}_{n,\varepsilon}$ we get

$$\widehat{C}\lambda^n \leq \|f_*^n \beta\| = \|f_*^n H_*(G_\Phi)_* \beta\| = \|(G_\Phi)_* \beta\|_{\ell^1} \cdot C\varepsilon^k,$$

where $\|\cdot\|_{\ell^1}$ denotes Gromov's simplicial norm, i.e.

$$\|\alpha\|_{\ell^1} = \inf \left\{ \sum |a_i| \ : \ \alpha = \left[\sum a_i \sigma_i\right], \sigma_i \in [\operatorname{nerve}(\mathcal{U}_{n,\varepsilon})]^{(k)} \right\}$$

for $[\operatorname{nerve}(\mathcal{U}_{n,\varepsilon})]^{(k)}$ the k-skeleton of the nerve. This discussion shows that the following would imply the Entropy Conjecture:

Conjecture 5.3.2. *For* $f \in \mathrm{Diff}^1(M)$ *with* $\operatorname{sp}(f_*) = \lambda > 1$*, we can find* $\beta \in H_*(M, \mathbb{R})$ *with* $\|f_*^n \beta\| \sim \lambda^n$ *such that for all* $\varepsilon > 0$ *sufficiently small there exists a constant* $C > 0$ *such that for all* n *we can find a minimal* (n,ε)*-cover* $\mathcal{U}_{n,\varepsilon}$ *and partition of unity* Φ *subordinate to* $\mathcal{U}_{n,\varepsilon}$

such that

$$\|(G_\Phi)_*\beta\| \leq Cr_f(n,\varepsilon).$$

We have relatively little evidence for this conjecture, which could indeed turn out to be stronger than the Entropy Conjecture, and/or false. Since the Entropy Conjecture is generally false in the C^0 setting, it is instructive to look at the nerves of dynamical covers for the standard C^0 counterexamples to see what goes wrong. Careful study of the example $f(z) = 2z^2/|z|$ on $\widehat{\mathbb{C}}$ suggests that the unboundedness of G_Φ on H_2 originates at the non-differentiable fixed point $z = 0$, and in particular depends on the unboundedness of $\mathrm{ind}_{f^n}(0)$. Shub-Sullivan's result in [SS74] then suggests the following sub-conjecture, which seems a natural place for understanding the interplay of regularity and the structure of dynamical nerves.

Conjecture 5.3.3. *For x a fixed point of a C^1 map and all $\varepsilon > 0$ sufficiently small, the local degree of G_Φ at x is bounded as a function of n.*

Appendix A

Submersions from manifolds with boundary

Our allowance of the case $\partial\mathcal{H} \neq \varnothing$ in Theorem 2.1.1 necessitates a discussion of submersions from compact Riemannian manifolds with boundary. In this appendix we establish some elementary geometric properties of such maps which are used in proving the main result.

These maps are messier than submersions from compact manifolds without boundary, essentially because their fibers can intersect the boundary badly. Most dramatically, the fibers need not be manifolds (even with boundary) in general. For similar reasons, submersions from manifolds with boundary need not be open maps at boundary points, and (most relevant for our purposes) the locally trivial fibration property enjoyed by proper submersions between manifolds without boundary (known as Ehresmann's theorem) fails completely in this setting.[1]

Here we circumvent these issues rather naively, by restricting our maps to the interiors of the manifolds in question. An obvious drawback of this approach is the loss of compactness.

[1] All these pathologies are easily illustrated by maps of the form $\pi : D \to \mathbb{R}$ where $D \subset \mathbb{R}^2$ is a compact planar domain with smooth boundary curves and π denotes (the restriction of) projection onto some one-dimensional subspace. One can of course also obtain examples with closed codomain by post-composing π with a smooth covering map $\mathbb{R} \to \mathbb{S}^1$.

Nevertheless, this trick turns out to be sufficient for our purposes because the geometry of such maps is still tamer than that of submersions from general open manifolds. The purpose of this section is to state and prove the needed constraints.

Let $f : M^m \to N^n$ be a C^1 map between smooth manifolds without boundary. Recall that a *box chart* (or *submersion chart*) adapted to f on M is a triple (B, ϕ, ψ) consisting of an open set $B \subset M$ and a pair of C^1 diffeomorphisms $\phi : B \to (-r, r)^m$ and $\psi : f(B) \to (-r, r)^n$ such that $\psi \circ f \circ \phi^{-1} : (-r, r)^m \to (-r, r)^n$ is given by projection onto the first n coordinates. For convenience we will use the following nominal refinement of this notion:

Definition A.0.1. We say that a box chart (B, ϕ_B, ψ_B) is *nice* if B is precompact, and there is another box chart $(B', \phi_{B'}, \psi_{B'})$ such that $\overline{B} \subset B'$, $\phi_B = \phi_{B'}|_B$, and $\psi_B = \psi_{B'}|_{f(B)}$. In this case we say that the box chart B' *cocoons* B.

The fundamental fact about submersions from manifolds without boundary is that their domains can be covered by box charts. Observe that nothing is lost in this statement by restricting our attention to nice box charts, since for $f : M \to N$ a smooth map with $\partial M = \varnothing$ it is immediate that

$$\begin{aligned} f \text{ is a submersion } &\iff f \text{ can be covered by box charts} \\ &\iff f \text{ can be covered by nice box charts.} \end{aligned}$$

On the other hand, we have the following straightforward fact:

Lemma A.0.2. *Let $f : M^m \to N^n$ be a C^1 submersion between smooth Riemannian manifolds without boundary and (B, ϕ_B, ψ_B) a nice box chart adapted to f. Then there are constants $0 < C_L^B < C_U^B$ such that*

$$C_L^B \leq \mathrm{vol}_{m-n}(f^{-1}(q) \cap B) \leq C_U^B$$

for all $q \in f(B)$.

Proof. Such an inequality holds trivially for the standard projection $\pi_n : \mathbb{R}^m \to \mathbb{R}^n$ and any (precompact) box $(-r, r)^m \subset \mathbb{R}^m$, with $C_L = C_U = r^{m-n}$. For the general case, let B be a nice box chart for f and $(B', \phi_{B'}, \psi_{B'})$ a box chart cocooning B. By definition $\overline{B} \subset B'$ is compact and ϕ'_B is C^1; it follows that $\phi_{B'}\big|_B : B \hookrightarrow \mathbb{R}^m$ distorts Riemannian metrics by a uniformly bounded amount, i.e. for some $C > 1$ we have $C^{-1}g \leq (\phi_{B'})^*\eta \leq Cg$ on B, where g denotes the metric on M and η the (Euclidean) metric on $\mathbb{R}^m$. Moreover, for all $q \in f(B)$ we have

$$f^{-1}(q) \cap B = \phi_{B'}^{-1}\big(\pi_n^{-1}(\psi_{B'}(q)) \cap \phi_{B'}(B)\big).$$

The general result then follows trivially from the Euclidean case. □

Lemma A.0.3. *Let $f : M^m \to N^n$ be a C^1 submersion between compact Riemannian manifolds, with $\partial M \neq \varnothing$ and $\partial N = \varnothing$. Then $\mathrm{vol}_{m-n}(f^{-1}(q) \cap \mathring{M})$ is finite for all $q \in N$, and in fact uniformly bounded above. If $f\big|_{\mathring{M}}$ is also surjective, then $\mathrm{vol}_{m-n}(f^{-1}(q) \cap \mathring{M})$ is bounded away from zero.*

Proof. Let $\mathcal{F}_q := f^{-1}(q) \cap \mathring{M} = (f|_{\mathring{M}})^{-1}(q)$, and note that, since $\mathring{M}$ is a manifold without boundary and $f\big|_{\mathring{M}}$ is a submersion, each $\mathcal{F}_q$ is either empty or an embedded $(m-n)$-dimensional submanifold of $\mathring{M}$. Now define

$$\nu(q) := \mathrm{vol}_{m-n}(\mathcal{F}_q) = \int_{\mathcal{F}_q} d\mathcal{F}_q.$$

A priori we have $\nu : N \to [0, +\infty]$. We first show that in fact $\sup \nu < +\infty$.

Upper bound: By attaching a collar $C_\epsilon \cong \partial M \times [0, \epsilon)$ to M and extending the metric, we obtain an open Riemannian manifold without boundary $\widehat{M}$ containing M as an isometrically embedded compact submanifold. Choose an extension of f to a C^1 map $F : \widehat{M} \to N$; since the set $\{p \in \widehat{M} \ : \ dF_p \text{ is surjective}\}$ is an open set containing M, by shrinking the collar width ϵ we may assume that F is a submersion as well.

Now pick a cover $\mathcal{B}$ of $\widehat{M}$ by nice box charts adapted to F; by compactness of M we can

find a finite subset $\{B_1, \ldots, B_k\} \subset \mathcal{B}$ such that $M \subset B_1 \cup \cdots \cup B_k$. By Lemma A.0.2 there are $C_U^i < +\infty$, $1 \leq i \leq k$, such that $\mathrm{vol}(F^{-1}(q) \cap B_i) < C_U^i$ for all $q \in N$. Since clearly every fiber $F^{-1}(q)$ intersects each B_j in at most one plaque, for all q we have

$$\begin{aligned}\nu(q) = \mathrm{vol}(f^{-1}(q) \cap \mathring{M}) &\leq \mathrm{vol}\left(F^{-1}(q) \cap \left(\bigcup_{j=1}^{k} B_j\right)\right) \\ &\leq \sum_{j=1}^{k} \mathrm{vol}\left(F^{-1}(q) \cap B_j\right) \\ &\leq k \cdot \max_{1 \leq j \leq k} C_U^j < +\infty.\end{aligned}$$

Lower bound: It remains to show that $\inf \nu > 0$ if $\mathring{f} := f|_{\mathring{M}}$ is surjective. This assumption implies that for all $q \in N$, there is a nice box chart $E_q \subset \mathring{M}$ adapted to $\mathring{f}$ such that $E_q \cap \mathring{f}^{-1}(q) \neq \varnothing$. For each such E_q, $\mathring{f}(E_q) = f(E_q) \subset N$ is an open neighborhood of q, and

$$N = \bigcup_{q \in N} f(E_q).$$

Since N is compact we may extract a finite subcover $\{f(E_1), \ldots, f(E_\ell)\}$; in particular, every fiber $\mathcal{F}_q = \mathring{f}^{-1}(q)$ intersects and thus contains a fiber plaque in some E_j, $1 \leq j \leq \ell$. On the other hand, the lower bound in Lemma A.0.2 implies that every plaque in any E_j has volume at least $C_L := \min\{C_L^{E_1}, \ldots, C_L^{E_\ell}\} > 0$, so

$$\nu(q) = \mathrm{vol}(\mathring{f}^{-1}(q)) \geq C_L > 0$$

for all $q \in N$, as desired. □

Lemma A.0.4. *Let f, M, N be as in Lemma A.0.3. Suppose $\phi \in C^0(M, \mathbb{R}_{\geq 0})$ is such that for all $q \in N$, we have $\phi(p) > 0$ for some $p \in \mathcal{F}_q = f^{-1}(q) \cap \mathring{M}$. Then there exists $C_\phi > 1$ such that*

$$\frac{1}{C_\phi} \leq \int_{\mathcal{F}_q} \phi \, d\mathcal{F}_q \leq C_\phi$$

for all $q \in N$.

Proof. Let $C_1 = \sup_{p \in M} \phi < +\infty$. Then for all $q \in N$ we have

$$\mathcal{I}_\phi(q) := \int_{\mathcal{F}_q} \phi \, d\mathcal{F}_q \leq C_1 \operatorname{vol}(\mathcal{F}_q) \leq CC_1,$$

where C is the upper bound on fiber volumes guaranteed by Lemma A.0.3.

Thus it remains only to show that $\mathcal{I}_\phi(q)$ is uniformly bounded away from 0. This follows from an easy adaptation of the proof of the fiber volume lower bound. Consider the set $\mathcal{B}$ of all nice box charts B_i for $f|_{\overset{\circ}{M}}$ such that

$$\inf_{p \in B_i} \phi(p) > 0.$$

Since ϕ is continuous and positive somewhere on each fiber of $f|_{\overset{\circ}{M}}$, the image of $\mathcal{B}$ under f forms an open cover of N. Extract a finite subcover $\{f(B_1), \ldots, f(B_k)\} \subset f(\mathcal{B})$. Then for some $\delta > 0$ we have $\phi(p) > \delta$ for all $p \in B_1 \cup \cdots \cup B_k$, and by Lemma A.0.2 there exists $C_2 > 0$ such that the volumes of the plaques of each B_i are uniformly bounded below by C_2. Since $\{f(B_i)\}_{i=1}^k$ covers N, each $\mathcal{F}_q$ contains at least one full plaque in some B_i; thus

$$\int_{\mathcal{F}_q} \phi \, d\mathcal{F}_q \geq \delta C_2$$

for all $q \in N$. Setting

$$C_\phi = \max\left\{CC_1, \frac{1}{\delta C_2}\right\}$$

we are done. □

Appendix B

Normal Jacobians formula

Recall that our proof of Theorem 1.1 hinges on the existence of uniform bounds for the fiber integrals

$$\int_{\mathcal{V}_{p,q}} \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}\, d\mathcal{V}_{p,q}.$$

The key point is that these integrals can be identified with fiber integrals of a *universally defined* function η on a slightly larger compact manifold with boundary which submerses onto $\mathrm{Gr}_k(TM) \times \mathrm{Gr}_{n-k}(TM)$. By the results of Appendix A, it then suffices to show that η is (i) continuous, and (ii) positive somewhere on the interior of each fiber.

There are two main benefits to this approach. First, by using only what is required to apply Lemma A.0.4, it avoids obscuring the logic of the argument. Second, and largely as a consequence, it generalizes easily to the case when $\dim(V) + \dim(W) > M$.

The downside is that the constant C remains somewhat opaque. In particular, this method gives no further analysis of the function η, on which C depends, or equivalently of the function(s)

$$\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)} : \mathcal{V} \longrightarrow \mathbb{R}_{\geq 0}.$$

This is unsatisfying, especially given the fairly straightforward geometric intuition motivating the bounds. In addition, the mere existence of some constant is clearly insufficient for

applications where more precise control over the constant is needed.

In the complementary dimensional case, a more transparent expression for $\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}$ is available, which in particular allows us to decompose C into more tractable factors:

Proposition B.0.1. *For $(\mathcal{H}, \psi)$ satisfying the hypotheses of Theorem 2.1.1 and any complementary dimensional submanifolds $V, W \subset M$, we have*

$$\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h, p, q) = \frac{|\det(J_{(h,T_pV,T_qW)})|}{\mathrm{NJ}(d(ev_p)_h)}$$

for all $(h, p, q) \in \mathcal{V}$. Here J is defined as in Claim 2.2.1, and ev_p as usual denotes the map

$$ev_p := ev_\psi(\cdot, p) : \mathcal{H} \longrightarrow M$$

given by $ev_p(h) = h(p)$.

The purpose of this section is to prove this statement. We begin with

Lemma B.0.2. *Let V, W be finite dimensional inner product spaces and suppose $S \subset V \times W$ is a linear subspace such that $\pi_1|_S$ is an isomorphism. Then if*

$$G := \pi_2 \circ (\pi_1|_S)^{-1} : V \to W,$$

we have $S = \Gamma(G)$. If in addition $\pi_2|_S$ is surjective and S is endowed with the inner product inherited from the product structure on $V \times W$, then

$$\det(GG^*)^{-1/2} = \frac{|\det(\pi_1|_S)|}{\det((\pi_2|_S) \cdot (\pi_2|_S)^*)^{1/2}}. \tag{B.1}$$

Proof. The first statement is obvious. The second follows from some elementary linear algebra computations; see [BCSS98, pp. 241-243] for details. □

Remark. In terms of normal Jacobians, this result specifies a context in which the innocent-looking formula

$$\frac{1}{\mathrm{NJ}(AB^{-1})} = \frac{\mathrm{NJ}(B)}{\mathrm{NJ}(A)} \tag{B.2}$$

holds. Note however that this formula fails in general: normal Jacobians of general (not necessarily invertible) linear maps between inner product spaces are not typically multiplicative under composition. In particular, the fact that (B.2) holds in the setting of Lemma B.0.2 depends on the assumption that the metric on $V \times W$ (and thus the metric on S) is the product metric induced by the metrics on V and W.

Proof of Proposition B.0.1. As in the discussion following Claim 2.2.1, observe that

$$\mathrm{NJ}(\pi_1)(h,p,q) = |\det((d\pi_1)_{(h,p,q)})| = 0 \iff \det(J_{(h,T_pV,T_qW)}) = 0.$$

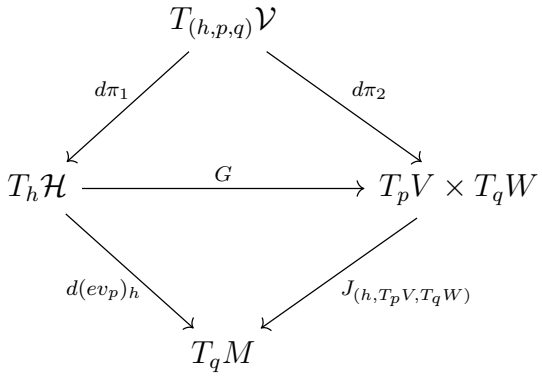
Thus it suffices to consider the case when both $d\pi_1$ and J are invertible. Let (h,p,q) be a regular point for π_1. Then we can define

$$G := (d\pi_2)_{(h,p,q)} \circ (d\pi_1)^{-1}_{(h,p,q)} : T_h\mathcal{H} \longrightarrow T_pV \times T_qW.$$

Since $\Gamma(G) = T_{(h,p,q)}\mathcal{V}$, with metric induced by the product metric on $T_h\mathcal{H} \times T_pV \times T_qW$, Lemma B.0.2 applies to give

$$\det(GG^*)^{-1/2} = \frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h,p,q). \tag{B.3}$$

Moreover, the description of $T_{(h,p,q)}\mathcal{V}$ given in Claim 2.2.1 immediately implies that the following diagram commutes:



In particular,

$$G = J^{-1}_{(h,T_pV,T_qW)} \circ d(ev_p)_h,$$

so by (B.3) we compute

$$\begin{aligned}
\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h,p,q) &= \det(GG^*)^{-1/2} \\
&= \det((J^{-1}\circ d(ev_p)_h)\cdot(J^{-1}\circ d(ev_p)_h)^*)^{-1/2} \\
&= \det(J^{-1}\cdot d(ev_p)_h\cdot d(ev_p)_h^*\cdot(J^{-1})^*)^{-1/2} \qquad \text{(B.4)} \\
&= \left[\det(J^{-1})\cdot\det(d(ev_p)_h\cdot d(ev_p)_h^*)\cdot\det((J^{-1})^*)\right]^{-1/2} \\
&= \left[\det(J^{-1})^2\cdot\det(d(ev_p)_h)\cdot d(ev_p)_h^*)\right]^{-1/2} \\
&= \frac{|\det(J)|}{\det(d(ev_p)_h)\cdot d(ev_p)_h^*)^{1/2}}. \quad \square
\end{aligned}$$

Remark. Up to a uniformly bounded factor, we have

$$|\det\left(J_{(h,\sigma_p,\sigma_q)}\right)| \approx \sin\measuredangle(dh(\sigma_p),\sigma_q).$$

The multiplicative error involved is at most η^k, where η is the supremum of $\max\left\{\|dh_q\|, \left\|dh_q^{-1}\right\|\right\}$ over all $h\in\mathcal{H}$ and $q\in M$.

Similarly, the fact that ev is a C^1 submersion implies that $\det(d(ev_p)_h)\cdot d(ev_p)_h^*)^{1/2}$ is uniformly bounded above and away from 0, i.e. constant up to a uniform factor.

Combining both errors we conclude that

$$\frac{\mathrm{NJ}(\pi_1)}{\mathrm{NJ}(\pi_2)}(h,p,q) \approx \sin\measuredangle(dh(\sigma_p),\sigma_q)$$

up to a bounded multiplicative constant (which is itself a function of the constants bounding $\mathrm{NJ}(ev_p)_h$, $\|dh_p\|$, and $\left\|dh_p^{-1}\right\|$ for all $(h,p)\in\mathcal{H}\times M$).

This means that one can view Theorem 2.1.1 as reducing to the following claim:

Proposition B.0.3. *For all $0\le k\le n$ there is a constant $C_k>1$ such that for all $(\sigma_p,\sigma_q)\in \mathrm{Gr}_k(TM)\times\mathrm{Gr}_{n-k}(TM)$,*

$$\frac{1}{C_k}\le\int_{h\in\mathcal{H}_{p,q}}\sin\measuredangle(dh(\sigma_p),\sigma_q)\ |d\mathcal{H}_{p,q}|\le C_k.$$

Here $\mathcal{H}_{p,q} = \{h \in \mathring{\mathcal{H}} : h(p) = q\}$; this is a submanifold of $\mathring{\mathcal{H}}$, with Riemannian metric induced from $\mathcal{H}$, which can be canonically (and isometrically) identified with $\mathcal{V}_{p,q}$ (indeed $\mathcal{V}_{p,q} = \mathcal{H}_{p,q} \times \{p\} \times \{q\}$).

Using the results above, our original constant C can be estimated as a function of the constants C_k and the bounds on $\mathrm{NJ}(ev_p)_h$, $\|dh_p\|$, and $\left\|dh_p^{-1}\right\|$.

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